SELFIE: AI Dental Assistant Application

IBEHS 5P06A

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Team 9

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Summary

Oral diseases are among the most prevalent global health conditions and impact the quality of life of nearly 3.7 billion people [1]. Despite being largely preventable, many individuals (particularly those from underserved populations) delay seeking dental care due to financial constraints, limited access to services, low oral health awareness, and dental anxiety. These delays often result in disease progression, leading to more invasive treatments and increased healthcare costs. Therefore, there is a critical need for an accessible and affordable solution that supports early awareness and preventive care.

To address this problem, we propose the *Selfie Dental Assistant*, a mobile application designed as a non-diagnostic screening and educational tool for oral health. The application detects four common oral conditions (dental caries, gingivitis, calculus, and tooth discoloration) by integrating two complementary systems. First, a multi-label convolutional neural network (CNN), based on the lightweight architecture MobileNetV3 Small, analyzes user-submitted images to identify visible signs of these conditions. Second, a symptom-based questionnaire uses rule-based clinical logic to evaluate user-reported symptoms and behavioral risk factors.

The outputs from both systems are combined into a fusion-based triage model that generates personalized risk assessments, preventive recommendations, and guidance on whether professional dental care should be sought. The system is intentionally designed to emphasize education and awareness rather than diagnosis, aligning with stakeholder feedback and ethical considerations. Additional design priorities include user-friendly interaction, data privacy through secure handling, and accessibility across diverse populations.

The expected impact of the Selfie Dental Assistant spans multiple domains. From a health perspective, it promotes early detection and preventive behaviors, potentially reducing the severity and progression of oral diseases. Socially and economically, it lowers barriers to oral health awareness and may reduce long-term treatment costs, particularly for individuals with limited access to dental care. Technologically, it demonstrates the application of mobile artificial intelligence in preventive healthcare. Overall, the solution contributes to improving oral health literacy and encouraging proactive self-care on a global scale.

Capstone Expo Description

SELFIE Dental Assistant is a mobile application that uses AI image analysis and a symptom-based questionnaire to screen for common oral conditions and provide personalized preventive guidance, improving access to early oral health awareness and supporting regular preventive care.

Updated Need Statement

There is a need for an easy-to-use, affordable, accessible educational platform that helps users understand and track their oral health, encouraging early intervention and regular preventive care, particularly for populations with limited access to dental services.

Declaration of Academic Achievement

As a future member of the engineering profession, the student is responsible for honestly performing the required work without plagiarism and cheating. If generative AI tools were used in the preparation of this report, I have clearly indicated how and where they were used and confirm that their use complies with the guidelines provided for this course.

Submitting this work with my name and student number is a statement of understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University and the Code of Conduct of the Professional Engineers of Ontario.

Submitted by [Ruidi Liu, liu127,400296859]

As a future member of the engineering profession, the student is responsible for honestly performing the required work without plagiarism and cheating. If generative AI tools were used in the preparation of this report, I have clearly indicated how and where they were used and confirm that their use complies with the guidelines provided for this course.

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Submitted by [Jamine Wang, wangj500, 400294807]

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List of Abbreviations and Symbols

Abbreviations

AES — Advanced Encryption Standard

ADA — America Dental Association

AI — Artificial Intelligence

API — Application Programming Interface

APK — Android Package Kit

AUC — Area Under the Curve

CAM — Class Activation Mapping

CASL — Canada's Anti-Spam Legislation

CDA — Canadian Dental Association

CDHA — Canadian Dental Hygienists Association

CI — Confidence Interval

CNN — Convolution Neural Network

CRC — Class Responsibility Collaboration

CPU — Central Processing Unit

DO — Design Output

DSIT — Department for Science, Innovation and Technology

FC — Fully Connected

FM — Failure Mode

FMEA — Failure Modes and Effects Analysis

FN — False Negatives

FP — False Positives

FPS — Frames per second

FR — Functional Requirements

GPU — Graphics Processing Unit

HIPAA — Health Insurance Portability and Accountability Act

HSV — Hue, Saturation, Value

HTA — Hierarchical Task Analysis

HTTPS — Hypertext Transfer Protocol Secure

IEC — International Electrotechnical Commission

ISO — International Organization for Standardization

LS — List of Services

MB — megabyte

ML — Machine Learning

NCSC-UK — United Kingdom's National Cyber Security Center

NFR — Non-functional Requirements

PIPEDA — Personal Information Protection and Electronic Documents Act

PHI — Personal Health Information

PHIPA — Ontario's Personal Health Information Protection Act

POC — Proof Of Concept

RGB — Red, Blue, Green

ReLU — Rectified Linear Unit

RPN — Risk Priority Number

ROC — Receiver Operating Characteristic

SaMD — Software as a Medical Device

SFWRENG — Software Engineering

SRS — Software Requirements Specification

TLS — Transport Layer Security

TN — True Negatives

TP — True Positives

UI — User Interface

UX — User Experience

USC — Uniform System of Coding

V&V — Verification & Validation

WCAG — Web Content Accessibility Guidelines

WHO — World Health Organization

Symbols

% percentage

° degree (s)

± plus, or minus

< less than

= equal

≤ less than or equal to

≥ greater than or equal to

$\boldsymbol{x}$ input oral image after preprocessing (RGB tensor of size $224 \times 224 \times 3$)

$\boldsymbol{f}(\cdot)$ convolutional neural network mapping learned during training

$\hat{\boldsymbol{y}}$ raw model output (logits)

$\hat{\boldsymbol{y}}_{\boldsymbol{i}}$ logit corresponding to condition i

$\mathbb{R}^{\mathbf{5}}$ 5-dimensional real-valued output space

$\boldsymbol{i} \in \{\mathbf{1}, \dots, \mathbf{5}\}$ disease index

$\boldsymbol{p}_{\boldsymbol{i}}$ predicted probability that condition $i$is present

$\boldsymbol{\sigma}(\cdot)$ sigmoid activation function

$\boldsymbol{e}$ Euler's constant

$\boldsymbol{L}$ total training loss

$\boldsymbol{y}_{\boldsymbol{i}} \in \{\mathbf{0}, \mathbf{1}\}$ ground-truth label for condition i

$\boldsymbol{p}_{\boldsymbol{i}}$ predicted probability

$\boldsymbol{i}$ condition index

$\boldsymbol{s}_{\boldsymbol{i}}$ cumulative symptom score for condition i

$\boldsymbol{P_i}$ normalized questionnaire probability for condition i

$\boldsymbol{j}$ summation index over all conditions

$\boldsymbol{q_k}$ binary or ordinal answer to symptom question k

$\boldsymbol{w_{ik}}$ clinical weight linking symptom $k$to disease i

$\boldsymbol{m}$ number of questionnaire items

$\boldsymbol{R_i}$ final fused risk score for condition i

$\boldsymbol{p_i}$ CNN subsystem probability

$\boldsymbol{P_i}$ questionnaire subsystem probability

$\boldsymbol{\alpha} \in [\mathbf{0}, \mathbf{1}]$ weighting factor controlling contribution of image vs questionnaire

$\alpha = 1$ (image only)

$\alpha = 0$ (questionnaire only)

$0 < \alpha < 1$ (hybrid system using probabilities from both)

* For equation and their relative symbol details please refer to **Appendix C: Equations**

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Chapter 1

*NO TABLES in Chapter 2,3,4 and 5.

Chapter 1

1.1 User Needs

Background

Oral diseases affect approximately 3.7 billion people worldwide [1]. They are extremely common and can have a significant impact on quality of life. Several reports have highlighted the importance of oral health and its relationship to systemic conditions, particularly cardiovascular diseases and diabetes [2]. Tooth loss is associated with greater illness and earlier than average death due to its ability to affect multiple oral functions such as speaking, chewing, swallowing, and smiling [3]. Periodontal disease (gum and bone inflammations) affects more than a billion people globally, destroying alveolar bone and leading to potential tooth loss [4]. Such limitations in oral function adversely affect an individual's emotional state and social interactions [3].

Although most oral health conditions are largely preventable and can be treated in their early stages, many individuals delay seeking care, which worsens the problem. This issue is illustrated by the Global Burden of Disease 2021: untreated dental caries in permanent teeth is the most common health condition, with approximately 2.37 billion cases of caries of permanent teeth [1][5]. Socioeconomic disparities, including healthcare systems, income, and educational levels, are the primary factors contributing to barriers that prevent many from accessing dental services. These barriers include dental cost burden, limited access, lack of awareness, and fear of dentists [6]. Out of all types of health care, dental care is the most unaffordable among the general population [7]. Low- and middle-income countries often bear a disproportionately higher burden due to limited access to oral healthcare services, poor oral hygiene practices, and unhealthy diets [5][8].

There are additional scientific supports for how early prevention and detection of oral diseases benefit patients: less invasive treatments, improved treatment outcomes, and prevention of complications [9]. Studies have shown that early dental visits correlate with better treatment experience for dental caries [10]. Early identification of enamel lesions also enables non-invasive intervention of dental caries. [11] Due to all the factors stated above, it is believed that there is a need for an accessible tool that helps with early detection of oral diseases.

Several studies have demonstrated significant improvements in health outcomes through mobile health (mHealth) interventions. For example, digital oral health education tools and reminder-based mobile applications have been shown to reduce plaque index and gingival inflammation by up to 20–30% in controlled studies [12]. Similarly, telehealth-supported screening programs have been associated with earlier detection of dental caries and reduced rates of advanced disease requiring invasive treatment [13]. These findings suggest that accessible, technology-assisted early screening and education tools can significantly reduce adverse oral health outcomes. Our stakeholder Zhuo, a dental assistant, confirmed this trend with her own experiences using similar applications in China.

Preliminary Need Statement:

There is a need for an easy-to-use, affordable platform that helps users understand and track their oral health, encouraging early intervention and regular preventive care, particularly for populations with limited access to dental services.

Impact

The designed solution should have an impact on health, social, economic, technological, environmental, cultural, and ethical areas.

Health Impact

The application should focus on prevention, education, encouragement of preventive care, and self-awareness to support WHO's oral health strategy [14]. Its personal benefit lies in the early detection of oral diseases which prevents the development of severe oral conditions.

Social & Economic Impact

There are disparities in oral health care among low-income, uninsured, rural, minority, and immigrant populations [15]. Children aged 6 to 9 from lower-income households were more than twice as likely (25%) to have untreated cavities than children from higher-income households [16]. According to the 2000 Surgeon General's report on oral health, the oral health inequities well-documented in children are also prevalent among working-age adults. Low-income adults have higher rates of untreated caries and missing teeth while tooth loss and dental caries affect most elderly [16][3]. Additionally, socioeconomic and literacy barriers create greater health inequities for aging adults, especially those aged 75 years and older [17].

A study in Egypt has shown smartphone-based solutions can bridge geographical and resource gaps by providing oral health care services to isolated rural areas and remote regions [18]. Hence, this application will not only reduce treatment costs but also bridge the gap between inequality, benefiting underserved, low-income, and rural populations while promoting equitable access to oral healthcare.

Technological & Environmental Impact

Radiological imaging instruments such as X-rays contribute greatly to energy consumption, exposure to ionizing radiation, and waste generation [19]. In comparison, smartphone-based applications are a more cost-saving, environmentally friendly, and sustainable alternative. Since it is a scalable smartphone application, it can be easily accessible and adaptable across languages/cultures.

Cultural & Ethical Impact

Attitudes towards sharing personal health data with applications vary widely across cultures, so the solution needs to respect cultural differences in health engagement. This would lower the barriers in remote areas where access to dental services is limited, providing a basic level of oral health screening.

Many studies that have used mobile applications focused on promoting oral health while other areas such as diagnostic and remote consultations remained using teledentistry [20]. This is because there are too many ethical factors that need to be taken into consideration carefully when working with diagnostic applications. Therefore, the application should serve only as an educational tool that emphasizes privacy and guidance instead of self-diagnosis.

1.2 Design Inputs

Based on stakeholder discussions and background research, the project aims to create a digital platform for oral-health self-monitoring and education. Overall, the design must be accessible, affordable, and user-friendly while being flexible enough to integrate future diagnostic technologies.

*Table 1. Functional Requirements categories, requirements and supporting rationale***.**

Function	Requirement	Rationale
Track Oral Health	The platform should allow users to log oral-health status (symptoms, discomfort, photos, or self-reports) and visualize historic trends.	Promotes continuous self-monitoring and awareness. According to Research, early tracking correlates with improved preventive behavior [21]
Identify Oral Health Issues	The Platform should highlight potential issues (caries, gums, sensitivity, misalignment) and encourage users to visit dentists when appropriate.	Supports early detection and reduces delay in care. This aligns with dentist stakeholder who recommends the app to be used for “screening, not diagnosis.”
User Education	Platform should provide preventive-care tips (brushing, flossing, diet) and educate users about oral health	Increase Health Literacy. Oral-health education significantly improves outcomes and reduces disease incidence [22].
Connectivity (Optional)	The Platform should have an optional function for users to share data or images with dentists for progress monitoring (orthodontics) or remote dental visits.	Dentist stakeholder indicated that similar apps in China have this function to improve continuity of care and dentist-patient engagement.
Emergency Guidance	The platform should provide immediate response helping users to seek emergency help.	Timely action is critical for emergencies so clear guidance will support user safety and reduce anxiety [21].
Reminder Notifications	The platform should send reminders for brushing, flossing, dental visits, or re-assessments.	Research indicates that behavioral reminders improve adherence to hygiene routines [23].
Personalized Feedback	The platform should tailor advice based on user input (symptoms, diet, etc.).	Personalized plans enhance engagement and compliance [23].
Security and Backup	The platform should make sure data should be backed up securely and retrievable on reinstall or device change.	Protects user data integrity and enhances user trust. This is also a standard Mobile Health expectation [23].

Table 2. ***Non- Functional Requirements categories, requirements and supporting rationale.***

Category	Requirement	Rationale

Appearance Requirements	The platform should have a clean, minimalist design with dental-health–related visuals.	Promotes a calm, clinical, and trustworthy aesthetic.
	The platform should use a consistent color palette, icon style, and typography across all screens.	Consistency throughout the platform enhances professionalism, trust, and ease of navigation.
	Platform should provide a progress indicator or loading animation when analysis or uploads take longer than 2 seconds [24].	Prevents user frustration and reassures them that the platform is functioning properly.
	The platform should offer both light and dark mode.	Improves accessibility, reduces eye strain, and conserves battery life [25].
	Health indicators on the platform (i.e. green/yellow/red icons) should visually communicate oral-health status.	Supports rapid understanding and engagement, especially for users with low health literacy.
Style Requirements (ISO 9241-210 (Human-Centred Design for Interactive Systems))	Text and messaging tone on the platform should be friendly, informative, and non-alarming.	Avoids anxiety, builds trust, and aligns with ethical guidance on self-monitoring health tools.
	Icons and visuals should be universally recognizable and culturally neutral.	Ensure usability across diverse user populations.
	The platform should adapt to multiple screen sizes and orientations (portrait and landscape).	Supports a wide range of devices for accessibility.
	All texts should be legible, with a minimum font size of 14pt and strong contrast.	Ensures accessibility for users with visual impairments.
	The platform's branding (logo, colors, typography) should reflect oral-health themes and remain consistent.	Builds trust and recognition.
	The app should provide smooth transitions and animations between screens.	Makes navigation on the platform intuitive.
	Users should access any core feature (capture, log, education, or history) within three taps.	Reduces cognitive load and supports usability for non-technical users.
	Users should be able to undo or edit an entry within 2 minutes.	Prevents frustration due to mistakes in data entry or photos for users.
	The platform should provide step-by-step guidance for taking good oral photos.	Ensures consistent input quality for tracking historical trends. Also improves reliability for future AI integration [26].

Usability and Humanity Requirements (Nielsen's Usability Heuristics)	New users should complete basic onboarding and perform a sample check within 5 minutes.	Ensures quick adoption of whatever technique is used to collect data.
	Provide in-app feedback or rating option to collect user input.	Enables continuous improvement based on real user experience.
	The platform should allow users to change text size and contrast.	Improves readability for users with visual limitations.
	Include dyslexia-friendly fonts and support for color-blind modes.	Ensures inclusivity for diverse users. (wcag 2.1 (web content accessibility guidelines))
	Include multilingual support (English, Chinese) at minimum.	Addresses language barriers, especially for immigrant and multicultural populations.
Performance Requirements	Average interaction response < 1 second [27].	Ensures smooth user experience.
	Platform startup time < 3 seconds [27].	Prevents user drop-off and frustration.
	Platform must clearly state that it is a screening and educational tool, not a diagnostic device.	Reduces liability and aligns with ethical health app standards.
	Emergency-guidance features must be clearly labeled and easily accessible.	Helps users seek urgent care for severe pain, swelling, or trauma.
	Crash-free rate ≥ 99.5% in testing [28].	Ensures stable user experience.
	Platform must back up logs and data daily.	Prevents data loss and improves reliability.
Operational and Environmental Requirements	The platform must maintain functionality under variable network conditions (3G–5G, Wi-Fi).	Oral health tracking should work for users with limited connectivity.
	The platform must operate efficiently on mid-range smartphones with ≤ 50 MB install size [29].	Ensures accessibility and sustainability (low data/battery use).
	The platform must comply with App Store / Google Play policies [30].	Required for public deployment.
Maintainability and Support Requirements	The platform must include an in-app option for accessing customer support.	Users need a built-in way to request help with navigation issues, general questions, or bug reports.
	The system must support the latest mobile devices as well as several previous generations.	Ensure broad user accessibility.
	All data (images, logs) are encrypted using AES-256; all	Protect user confidentiality.

	communications via HTTPS/TLS 1.2+.	
Security Requirements	Platform must request explicit consent before storing or sharing user data.	Meets PIPEDA/HIPAA privacy standards.
	No identifiable facial data stored unless consented.	Ensures anonymity and user trust.
	Platform must allow users to delete all personal data on demand.	Complies with ethical and legal privacy expectations.
	Educational content should be culturally sensitive and use inclusive imagery.	Prevents alienating or offending users.
Cultural and Political Requirements	Avoid alarming or judgmental language in results.	Encourages positive engagement and reduces anxiety.
	Respect local healthcare norms and provide dentist referral information appropriate to user's region.	Aligns with ethical guidelines and enhances trust.

Table 3. Core Evaluation Criteria with criterion and metric.

Criterion	Metric
App Size	Maximum compressed download size of APKs < 200 MB [31]
Processing Time	Processing time < 10 seconds [27]
Usability	System Usability Scale < 70 [32]
Accuracy	Sensitivity > 80% [33]
Privacy & Ethics	PIPEDA / App Store compliant

Constraints and Assumptions:

The Capstone Project is subject to several key constraints and assumptions. Firstly, the platform will rely solely on user input or visual data captured through smartphone cameras. Unfortunately, the inclusion of X-ray or intraoral scanning technologies is beyond the project's current scope. Additionally, due to the difficulty of collecting proprietary dental image data, the system will depend primarily on publicly available open-source datasets for development and testing, the prototype's accuracy is expected to be moderate and primarily suited for screening and educational purposes rather than diagnostic use. Ethical considerations also prevent the system from providing medical diagnoses, ensuring that its role remains supportive and informational. This is also in accordance with a dentist stakeholder's opinions. Lastly, the design and deployment of the platform must comply with relevant privacy and data protection regulations, including PIPEDA, as well as App Store and Google Play policies governing health-related applications.

1.3 Design Process

The design process began by revisiting the stakeholder's needs and design inputs defined in Sections 1.1 and 1.2. The team wanted to explore different solutions that could satisfy the need for an accessible, user-friendly platform for oral-health self-monitoring and education. Four preliminary design ideas were evaluated against the criteria of accessibility, usability, technical feasibility, scalability, and ethical compliance.

Symptom-Based Questionnaire App:

This concept mainly focuses on education and self-assessment through an interactive questionnaire. Users would answer questions about common oral symptoms (pain, bleeding gums, tooth sensitivity, discoloration, etc.), and the app would provide personalized preventive guidance and recommend when to seek professional care.

This concept was inspired by clinical decision-tree logic and rule-based expert systems, commonly used in early telemedicine applications [34]. The logic would be derived from published diagnostic criteria (probably from ADA symptom guidelines or stakeholders) translated into simple if-then rules [35]. Decision trees approximate how clinicians perform differential diagnosis by sequentially narrowing possibilities based on key indicators. Finally arriving at a prediction of disease for the user.

<u>Strengths:</u>

- High Accessibility: Low computational demand means this app will run on any mobile device.
- Avoids data-privacy issues tied to image processing.
- Supports strong educational and behavioral change goals, which have been proven to improve oral health habits.
- High Feasibility: Very easy to implement and code.

<u>Weaknesses:</u>

- Relies on subjective user input; accuracy is affected.
- Cannot visually detect structural or aesthetic conditions that the users can't see. This limits the usability of the platform.
- Limited engagement without interactive visuals, relying on text to convey ideas.

Possible Improvements:

Train a lightweight model (logistic regression, gradient boosting, or Naive Bayes) on questionnaire responses (symptoms, habits, history) for risk scoring. Can also utilize Rule-AI hybrid decision support by adding a small ML layer that adjusts thresholds to refine outputs and improve accuracy [36]. We would utilize tiny models to run on-device, and minimal data is required.

Traditional Image-Analysis Approach:

This concept uses classical computer-vision techniques (color segmentation, edge detection, contour analysis) to analyze photos of the user's teeth and gums. The system would highlight discoloration, plaque, or gum redness based on pixel thresholds. Based on image-processing fundamentals such as color-space transformations (HSV/RGB), Canny edge detection, and region segmentation (Otsu's method) [37] [38]. The underlying principle is that oral conditions exhibit distinct visual features, such as color variation (gum redness in gingivitis), texture irregularities, and edge discontinuities. These features can be mathematically extracted using image gradients, histogram distributions, and segmentation algorithms and used to identify disease.

Strengths:

- High Accessibility: Lightweight and fast, which means the platform is suitable for low-end devices.
- Transparent processing logic (no "black box" AI). Variables and decisions can be tailored and manipulated easily.
- Easier to validate and maintain [38].

Weaknesses:

- Highly sensitive to lighting and photo quality, it is difficult for users to maintain consistency which will affect accuracy [39].
- Limited ability to detect subtle or complex lesions.
- Requires preprocessing and calibration which needs to be tailored to individual users.
- Limited research applied to oral health field; methodology must be transferred from other areas.

AI-Based Image Analysis (Transfer Learning):

This concept includes a machine-learning component that utilizes transfer learning with a pre-trained convolutional neural network (MobileNetV3 or EfficientNet pretrained on ImageNet). Users will take a photo (or a series of photos), and the model identifies potential issues (caries, inflammation, discoloration). We don't have to train a complex model. Large-scale image models already trained for feature extraction exist and transfer learning allows us to just fine-tune the last couple layers on a smaller dental dataset, significantly reducing computational cost and increasing feasibility [40]. Visual analysis models train on disease vs healthy images to look for clinically relevant patterns.

Strengths:

- Can detect subtle patterns that are difficult to detect by rule-based methods.
- Adaptive and improve with more data, can also be personalized for individual users to be more accurate.
- Align current trends in digital dentistry and tele-diagnostics.
- Plenty of research in the field of oral health is available which can be helpful.

Weaknesses:

- Requires a large number of high-quality labeled datasets (currently limited, though our stakeholder is willing to help).
- Potential ethical concerns with the images taken revealing the user's identity.
- Computationally heavier and less explainable. It might be difficult to run on older models and might take significantly longer to process results.

Possible Improvements:

Incorporate subjective user feedback (dropdown menus) asking about pain, sensitivity or other features that cannot be detected from images. This addition would greatly increase the accuracy of the model. Can speak to stakeholders to determine which specific features are important to ask about in the future.

Dentist-Connected Platform:

This concept emphasizes professional connectivity and long-term engagement. Users could track progress (whitening, orthodontics, gum improvement) and share images or symptom logs with dentists for feedback or follow-up care. Dentists could review trends remotely, like telemonitoring systems already used in China [41]. Mainly used to educate users and enhance continuity of care. By staying connected to their dentists in day-to-day life, users are more likely to take care of their oral health and any issues can be caught quickly.

<u>Strengths</u>:

- Encourages preventive care and education for users about oral health.
- Scalable for public health programs and has real-world clinical linkage.
- Build trust by involving professionals (dentists and orthodontists)
    - Professionals can promote the platform with their current patients.

<u>Weaknesses:</u>

- Requires secure data handling.
- Dependent on user consent and clinician participation.
    - Ineffective for target populations who have limited access to regular dental care to begin with.
- It has dentist-side infrastructure which means higher development complexity.

Table 4. Overall Comparisons of Each Concept.

Criterion	Symptom-Based Questionnaire	Traditional Image-Analysis	AI-Based Image Analysis	Dentist-Connected Platform
App Size	A lightweight text-based app requires minimal resources and should run smoothly on most devices.	Basic image-processing algorithms need less memory and should run efficiently on most devices.	Incorporating any sort of pre-trained CNN models increases memory and CPU/GPU load significantly more than other concepts.	Moderate resources needed since client–server communication and storage increase memory and network overhead
Processing Time	Processing time is short since rule-based evaluation is simple.	Feature extraction and threshold detection typically complete <10 seconds.	Image inference can take >10 seconds depending on hardware optimization.	Typically < 5 seconds for message exchange but requires dentist feedback which greatly increases processing time.
Usability	Simple questionnaires should be intuitive and accessible to all literacy levels.	Users need to capture a clear photo under specific lighting requirements.	Users must align and capture images correctly, which may challenge some users.	Interface involves multiple users, making it more complex.
Accuracy	Relies on user-reported symptoms, giving moderate accuracy.	Detects visible discoloration, caries and gum redness but	Models can be trained to identify subtle issues with higher potential accuracy.	Dentist input ensures the highest diagnostic accuracy.

		limited against subtle issues.		
Privacy & Ethics	Handles only text data, minimizing privacy risk and simplifying compliance.	App will store and analyze images locally, posing limited privacy exposure	Collects and processes identifiable dental photos on Cloud, requiring strong consent and encryption.	Manages personal health records and communication, requires strict data-controls.

1.4 Preliminary Risk Assessment

Health and Safety

The safety risks primarily fall within the scope of software security. It is important to protect end-users' privacy and personal data from unauthorized access. Therefore, robust measures must be implemented to maintain data confidentiality and integrity. Some Performance risks still exist. For example, the application may fail to correctly identify oral conditions due to poor image quality, insufficient lighting, occlusion, or limitations in the training dataset. Resulting false negatives may delay users from seeking necessary care, while false positives may cause unnecessary anxiety. To mitigate this, the system will include confidence scores, clear disclaimers, and recommendations to seek professional evaluation when uncertainty is high.

Ethical Concerns

Algorithmic bias

The use of algorithms, ML training, and AIs may induce algorithmic bias into the system. Biases in training data, algorithm design, proxy data, and evaluation are the root causes of algorithmic bias. Common biases in health care include the underrepresentation of minority groups in data [42]. Biased algorithmic decisions reinforce existing societal disparities, creating discrimination and inequality, leading to potentially harmful outcomes. Furthermore, it may lead to non-compliance, such as legal and reputational damage.

Questionable personal data ownership

Personal Data could be seen as a valuable currency, especially one's health specifics. Many health websites, apps and wearable devices may operate outside of traditional healthcare privacy regulations, creating a "grey" area of questionable data ownership [43]. Personal health data may be sold to third parties or even claimed by developers for machine learning or algorithm training purposes. Violation of data privacy may expose individuals to identity theft and financial fraud, leading to significant financial loss. lead to hefty regulatory fines or even criminal charges. Unethical data use may further destroy the trust of the public and damage consumer relationships.

Standards and Codes

Risks of non-compliance, including legal fines and penalties, financial losses and cybersecurity risks, do exist within the scope of the designed solution. It is important to follow the following dental and software standards set by national, provincial, and territorial regulatory authorities.

The software application designed will follow Ontario Standards for dental procedures, where dental procedure codes are standardized, 5-digit alphanumeric codes based on the **CDA** [44]. Additional **CDHA** Codes are also included since many other professional organizations, and regulatory bodies refer to them as a standard.

- **CDA Dental Codes [45]**
    - The general classification system groups codes into the following categories:
        - Diagnostic (00000-09999) [46]
        - Preventive (10000-19999)
        - Restorative (20000-29999)
- **CDHA National Lists of Service Codes 2022 Edition [47]**
    - The Canadian Dental Hygienists Association CDHA is a national professional organization that advocates for dental hygienists and promotes the profession. It includes a brief description of each CDHA Code and the corresponding USC&LS.
- **CDHA Code of ethics [48]**
    - A document that guides dental professionals on expected behaviors and serves as a standard for ethical decision-making within CDHA.

Software development in Canada is required to follow various standards and code, including but not limited to the following:

- **PIPEDA [49]**
    - The federal law enforces how private sector organizations handle personal information during commercial activities.
- **CASL [50][51]:**
    - A federal law that protects users and businesses from the misuse of digital technology. It includes requirements for software developers to adhere to, i.e. developers need to obtain consent before installing software on a user's device.
- **Software Security Code of Practice [52]:**
    - The Canadian Centre for Cyber Security offers joint guidance with the NCSC-UK and DSIT, outlining 14 principles for building resilient and secure software.
- **Ontario's PHIPA [51]**
    - Provincial law establishes a framework to protect privacy, confidentiality, use, and disclosure of individuals' health information while ensuring that the data can be shared effectively for the provision of healthcare.

Chapter 2

2.1 Design Outputs

There is a total of 7 design outputs, they are:

- **DO1 – Software Architecture**
- **DO2 – Functional Design Outputs**
- **DO3 – UI/UX Design Outputs**
- **DO4 – Data Schema & Encryption**
- **DO5 — Privacy, Consent, and Access Control**
- **DO6 – Risk Management File**
- **DO7 – Verification & Validation Plan Output**

Table A1. Design Output Category Table in Appendix A shows what category each design output falls under.

The design outputs are derived directly from the functional and non-functional requirements defined in Chapter 1 and are expressed in measurable, quantitative terms to enable verification and validation. Each output ensures that system performance, usability, and safety targets are met. For example, the CNN model output must achieve ≥80% sensitivity (Design Input: Accuracy), processing time <10 seconds (Performance Requirement), and operate on-device within ≤50 MB of size limit (Operational Constraint). Similarly, the questionnaire module must produce risk scores within defined clinical thresholds aligned with standardized frameworks. These outputs ensure traceability between design inputs and system implementation.

DO1 Software Architecture

A complete Software Architecture Document is required to ensure all stakeholders and developers are aligned. It serves as a guide to software structure and helps developers to **design and implement** an effective system. It should define all modules and interface and ensure the system is **traceable, modular and testable**. This entire section should follow the general guidelines and templates provided in the course Software Design III – Large System Design (SFWRENG 3A04) [53].

SRS

The SRS document formalizes agreements and provides a road map for development and testing. It contains all design inputs required by the system and illustrates the business logic behind core system functionalities such as identifying oral health issues.

It should include the following sections:

1. Introduction: Provide an overview of the entire document, including subsections:
    - 1.1 Purpose
    - 1.2 Scope
    - 1.3 Definitions, Acronyms, and Abbreviations
    - 1.4 Refences
    - 1.5 Overview

2. Overall Product Description: describe general factors that affect product and its requirements, including subsections:
    a. 2.1 Product Perspective
    b. 2.2 Product Functions
    c. 2.3 User Characteristics
    d. 2.4 Constrains
    e. 2.5 Assumptions and Dependencies
    f. 2.6 Apportioning of Requirements
3. Use Case Diagram
4. Highlights of Functional Requirements
5. Non-Functional Requirements

High Level Architectural Design

This document should identify and explain the overall architecture of the system and subsystems, while providing potential alternatives. The overall architecture should show the division of the system into subsystems with high cohesion and low coupling.

It should include the following sections:

1. Introduction: Provide an overview of the entire document, including subsections:
    1.1 Purpose
    1.2 System Description
    1.3 Overview
2. Analysis Class Diagram
3. CRC Cards

Detailed Design Document

This document should provide detailed State Charts for Controller Classes, Sequence Diagrams, and a Detailed Class Diagram.

Acceptance Criteria for Document:

- All non-standard notations used are documented [53].
- Diagrams should be able to speak for itself without providing a text explanation [53].
- Architecture diagram shows all data flows, inputs, outputs, and interfaces as required by **IEC62304:2006** to reduce risk of cascading failures [54].
- Interfaces must conform to defined API specifications with no unresolved dependencies. This is required for hazard identification [55].
- All APIs and interfaces must be fully defined [56].

DO2 – Functional Design Outputs

Symptom Questionnaire Module

Purpose: Provide a description of questionnaire pathways for future verification and validation

This document should include and specify mandatory questions used, decision-tree logic, branching rules, risk scoring system, and requirements. It should ensure the risk categories match the standard dental caries risk framework [57].

Image Capture Protocol Document

Purpose: Provide the user with a standardized way of image collection to minimize error.

The document should include a guideline for how users should take photos of their mouth, indicating required anatomical views and image acceptance criteria.

Multi-label CNN Model Document

Purpose: Serve as a reference to evaluate prototypes and act as a benchmark for future verification and validation. It should be cross referenced to the main SRS document to align functions and requirements, ensuring the screening accuracy matches or exceeds market standards.

The document should contain quantitative model requirements such as model type, model architecture, preprocessing steps, size, sensitivity, specificity, accuracy, ROC AUC etc.

Specificity $\geq$ 75%

ROC-AUC $\geq$ 0.80

Model size $\leq$ 50 MB (mobile deployment constraint)

Inference time $\leq$ 10 seconds per image

Input resolution: 224×224 RGB images

Triage Logic Specification Document

Purpose: Ensure triage logic is safe and consistent, meeting mandatory health and safety requirements.

The document should indicate how the integration of questionnaire scores and CNN analysis of image outputs combine to produce different triage categories.

DO3. User Interface (UI/UX) Design Outputs

UI/UX Wireframes & Style Design and Assessment

Purpose: Demonstrate how users should navigate the system, ensuring consistency in UI and style.

A full document containing annotated wireframes, interaction flows, visual style guidelines, and meeting accessibility specifications (WCAG 2.1) [58]. It should also state design elements used and justify design decisions (i.e. colour palette choices). **Interview results** of prototypes must be included and provide adjustments based on feedback.

Usability Performance Assessment

Purpose: Outline usability targets such as learnability and memorability, provide acceptance criteria for evaluating performance.

This document should cross-reference the main SRS, outline usability goals, how to meet them, measurement methods and test protocols. **Usability Survey** should be included with survey results.

DO4 – Data Schema & Encryption

Purpose: Document how data and encryption are handled in the system.

The document should describe the structure of data storage, what encryption method is used for data (AES-256) and transmission (AES-256), how they are implemented, and how it aligns with privacy policies.

DO5 — Privacy, Consent, and Access Control Documentation

Personal Health Information, Privacy and Data Collection Document

Purpose: Ensure data collection is aligned with legal requirements.

The document should indicate what the user data rights are, how data are collected/handled, and specification for access control.

Terms of Service

Purpose: Create a legal blinding contract for using a service.

This document should be written consent that is signed by the user before using the software; it should specify what data is being collected, how they are being used and handled, how to contact the developers if the user wishes to delete their data. The Terms of Service document should align with SRS non-functional requirements and all compliances listed (e.g. PIPEDA).

DO6 – Risk Management File

Purpose: Create traceable risk management and safety documentation.

The document should include preliminary risk assessments, risk analysis including hazard identification, FMEA tables, control measures and updates.

DO7 – Verification & Validation Plan Output

Purpose: Ensure every requirement and goal is achieved; everything is measurable, tractable, testable, and verifiable.

A written V&V protocol mapping every design input to **verification tests** and **validation tests**.

2.2 Final Decision Concept

The final concept selected for this capstone project is a hybrid oral-health assessment platform combining:

1. A multi-label CNN capable of detecting several oral conditions simultaneously (dental calculus, cavities, gingivitis, periodontitis, missing teeth)
2. A rule-based questionnaire designed to integrate patient-reported symptoms, experiences and risk factors that are not always visible in images.

This hybrid approach was chosen because image-only models cannot fully capture clinical context, such as bleeding frequency, pain, or dental history, while questionnaire-only systems cannot localize visual pathology. Together, the two branches produce a more reliable and informative preliminary screening tool, aligning with the project objective of non-diagnostic risk estimation.

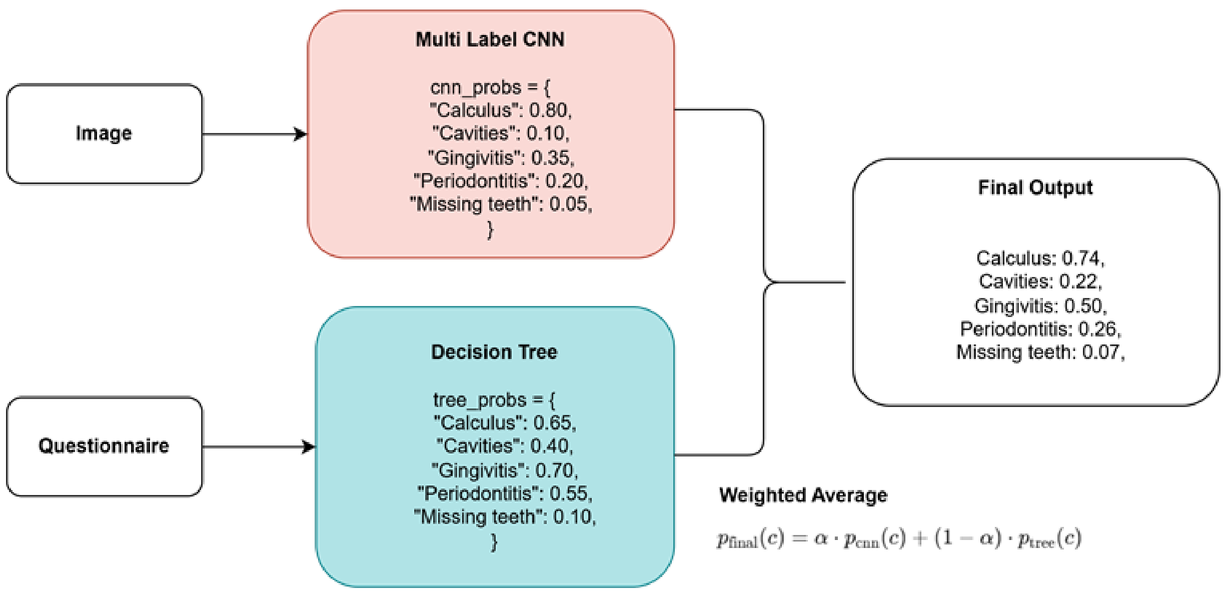


***Figure 1**. Overall structure of proposed final concept including both CNN and decision tree pathways.*

The selection of this concept is supported by the design outputs from Section 2.1 as the safest, most functional, and most compliant model:

- **DO1 (Software Architecture Document):** The system architecture supports a modular pipeline image upload, CNN inference, questionnaire scoring, and result integration.
- **DO2 (Functional Design Outputs):** Multi-condition prediction and symptom capture were identified as core functions. A hybrid model directly satisfies these requirements.
- **DO3 (UI/UX Design Outputs):** The proposed interface includes both an image upload flow and a short symptom questionnaire, it will also display the end results.

- **DO4 (Data Schema & Encryption):** The architecture requires storing only anonymized images and questionnaire responses, both of which fit the schema constraints.
- **DO5 (Privacy, Consent, Access Control):** A rule-based component allows meaningful risk output **without requiring cloud-based diagnosis**, reducing privacy risk. On-device inference is feasible (MobileNetV3: 7 MB; ResNet18: 43 MB).
- **DO6 (Risk Management File):** A hybrid model reduces risk by avoiding over-reliance on image-only decisions and reinforcing interpretations with clinically validated symptom logic.
- **DO7 (Verification & Validation Plan):** This concept allows for unit testing of both subsystems independently (CNN accuracy metrics and questionnaire logic validation) as well as final comparison of results vs actual diagnosis provided by our stakeholder thus meeting V&V requirements.

Analysis of the Final Concept Using Relevant Theory

A multi-label classifier predicts multiple conditions simultaneously. For an input image x, the model produces logits [59].

Where each condition is independent and uses sigmoid activation.

The CNN in the final concept will produce five independent probabilities, allowing the model to identify multiple co-occurring oral conditions, such as Calculus + Gingivitis or Calculus + Cavities + Missing Tooth etc. This is particularly important for dentistry where diseases often overlap.

The decision-tree questionnaire contributes risk-based reasoning that cannot be captured visually, assigning weighted scores to each condition based on symptoms and behavioral risk factors. These scores are normalised using a softmax transformation to produce interpretable, probability-like outputs [60].

The decision tree utilizes Bayesian risk analysis, where each symptom is independent evidence toward a certain condition [61].

In the full concept, both the CNN outputs and questionnaire probabilities would be fused through a weighted average. The weights are adjustable, which makes the platform more clinically relevant since everything can be adjusted as needed.

The selection of a CNN over traditional image processing or simpler machine learning models is based on its superior ability to automatically learn hierarchical visual features such as textures, edges, and lesion patterns directly from raw image data. Unlike rule-based or classical methods, CNNs do not require manual feature engineering and have demonstrated higher accuracy in medical image classification tasks. Lightweight architectures such as MobileNetV3 and ResNet-18 were specifically chosen due to their balance between computational efficiency and performance, making them suitable for on-device inference.

Similarly, a decision-tree–based questionnaire was selected over more complex probabilistic models (i.e. Bayesian networks) due to its interpretability, low computational cost, and ease of validation against clinical guidelines. Decision trees allow transparent reasoning pathways, which is critical for user trust and regulatory compliance in healthcare applications.

The final output will be presented to users in an interpretable and user-friendly format. Each condition will be displayed as a color-coded risk indicator (green = low, yellow = moderate, red = high) accompanied by a percentage risk score and a short explanation (e.g., "Moderate risk of gingivitis based on gum redness and reported bleeding"). The system will also provide actionable recommendations, such as improving oral hygiene practices or seeking professional care. This presentation aligns with UI/UX design inputs that emphasize clarity, accessibility, and non-alarming communication.

Assumptions, Limitations, and Constraints

The concept relies on several important assumptions. It assumes that users will input clear intra/extra-oral images and answer the questionnaire honestly. It also assumes that the CNN has been trained on a diverse and representative dataset that captures real-world variability.

However, there are still some limitations. Although the system is explicitly non-diagnostic and should not replace professional evaluation, the current dataset is still too small to fully realize the multi-label model. The rule-based questionnaire simplifies complex pathology, which may limit diagnostic precision in borderline cases.

Several constraints also exist. Ethical and regulatory limitations require careful handling of identifiable images, which justifies the emphasis on local processing and minimizing stored data. Technical constraints include limited availability of labelled datasets and the need for lightweight models suitable for mobile deployment. Computational constraints further justify selecting compact backbones such as MobileNetV3 and ResNet-18, which allow on-device inference while maintaining acceptable performance.

2.3 Final Decision Risk and Compliance Management Plan

FMEA – Key Failure Modes

The **Failure Modes and Effects Analysis (FMEA)** will be the primary tool for preliminary risk analysis. All Key Failure Modes are documented in *Table A2: Key Failure Mode Names and Risk Priority Number.* Please refer to *Table A3: Final FMEA Table for Selfie Dental Assistant Application* in **Appendix A** for the details of each Failure Mode (Cause, primary function, etc.). The subsystems in Table 3 correspond to Figure 1. High-level Software Architecture and Figure 2. Internal Software Architecture. Below includes a list of key failure modes, what sections they also belong to, and their respective **RPN** (RPN = S×O×D) calculated by multiplying **Severity (S)**, **Occurrence (O)**, and **Detectability (D)** on a standard 1–10 scale.

2.3.1 Health and Safety

Although the proposed solution is identified as a "software application", it qualifies as **SaMD** because it is intended to be used for one or more medical purposes outlined in the device definition of the Act. It also performs the intended function without being part of a hardware medical device: it processes user-specific health data (symptoms and photos) to generate risk-based triage recommendations.

Since the solution is a SaMD all the key failure modes identified are risks that would affect the end-users' health and safety (**FM1 – 11**). **FM1** to **FM4** ties to the core function of the designed solution, directly affecting the result of the analysis. Although **FM5,6** look like they are only a technical concern, but any delay or unresponsive behaviour will result a delay care. This is extremely dangerous if the user has undiagnosed severe oral conditions, hence, they are also considered to be risks categorized under Health and Safety. Mitigation and compliance strategies for **FM8 – FM 11** are addressed in 2.3.3 Standards and Codes since they are risks related to misalignment with regulatory standards and code.

This section will only address the failure modes with the 3 highest RPN: **FM1** (RPN =252), **FM3** (RPN =175), **FM8** (RPN) = 162. Please refer to Table 3 for details of other failure modes.

FM1 — False Reassurance / False Negative Triage: RPN = 252

FM1 has the highest RPN out of all failure modes. Any incomplete questionnaire or inaccurate response from the user, poor triage scoring system, mis-calibrated threshold may lead to this failure. False negatives affect the users' health most heavily because it is extremely dangerous when a serious condition is classified as low risk. This may result in delayed care, progression of disease, increase in pain, potential rise of infection/complications, or the worst-case scenario ---- death.

Prevention: To prevent this, red-flag override rules (flag for high triage scores/severe symptoms and override other results) are implemented. Additional prevention include key questions are mandatory to answer, apply conservative high-sensitivity threshold, include unit tests on triage logic and most importantly provide a suggestion to visit a dentist no matter what the results are (i.e. "There are no signs of calculus, please remember to have your regular dental checks" vs. "Please attend to a dental clinic immediately.")

Detection: This risk can be detected by monitoring the triage distributions and having a clinical validation against labelled cases.

FM3—Improper Use / Incorrect Image Capture: RPN = 175

FM3 is most likely caused by the user: rushing, ignoring overlay guides, user taking a photo with a broken/flawed camera, or confusion. These may lead to inaccurate ML features and result in potential false alarms.

Prevention: Provide the user with detailed instructions, image taking protocol, and overlay guidelines. In addition, enforce blur/exposure checks before sending the image to the CNN model. If possible, filter out poor quality images and ask the user to retake images. Include warnings for skipping questionnaires and poor image quality.

Detection: Monitoring image-quality scores and analyze repeated retakes instead of analyzing one photo only.

FM8 — Data Breach / Unauthorized Access to Health Data: RPN = 162
Weak encryption, insecure storage, misconfigured cloud and inadequate access control are the causes of **FM8**. This could put the users at risk when their Personal Health Information (PHI) is being accessed by unauthorised parties.

Prevention: This can be prevented by implementing strong AES-256 encryption at rest and in transit and routine audits.

Detection: Implement detective measures such as security monitoring, log reviews, and breach-detection alerts.

2.3.2 Ethic Concerns

FM7 — Algorithmic Bias / Unequal Performance Across Subgroups: RPN = 144

The use of algorithms, ML training, and AIs may induce algorithmic bias into the system. Biases in training data, algorithm design, proxy data, and evaluation are the root causes of algorithmic bias. Algorithmic bias could arise when there is limited, non-representative training data or flaws in dental labelling. It might lead to inequitable care, having higher false negatives or false positives for certain user groups (**FM1** and **FM2**).

Prevention: This could be prevented by having diverse training data, tracking subgroup metrics, and having a clinician review and validating the high-risk cases.
Detection: Perform periodic bias audits on validated data and monitor the subgroup confusion matrix.

Further strategies for reducing algorithmic bias include adding fairness evaluation metrics to acceptance criteria before release and tune threshold values separately for each group.

2.3.3 Standards and Codes Compliance Strategies

Medical Device & Software Standards

SaMD Classification – Health Canada [62]

> **Strategy**: Document software architecture, risk management, validation results aligned with the SaMD guidance.

ISO 14971:2019 – Risk Management for Medical Devices [63]

> ISO14971:2019 is an international standard that defines a systematic life-cycle approach to risk management and includes a comprehensive process (identify hazards, estimate and evaluate risks, implement controls and monitor their effectiveness) for medical devices including SaMD.
>
> **Strategy**: Create a Risk Management File summarizing the FMEA, risk-control measures, and verification (unit testing of safety functions, usability studies, etc.). Ensure risks are reassessed after significant software updates or model retraining.

IEC 62304 – Medical Device Software Life Cycle [64]

Strategy: For the prototype, implement a scaled-down IEC 62304-style process: version control, documented requirements, unit/integration testing, defect tracking, and controlled deployment environments. Maintain algorithm releases and keep log changes.

Usability (IEC 62366 principles) [65]

Strategy: Apply usability concepts in prototype by defining primary users and critical tasks. Detect errors related to usability through formative tests.

PIPEDA [49]

Strategy: Design privacy policy and on-device notices aligned with PIPEDA's principles. Should indicate what information is collected and how to contact the developer for access or correction requests in Terms of Service. Strengthen encryption and introduce audit logs when accessing PHI to prevent data breach and unauthorized access.

Ontario's PHIPA [51]

Strategy: Treat all user input (dental images and symptom data) as PHI. Ensure these data are encrypted, need authorization before accessing and logging all access events. Follow PHIPA's guidance to create a basic breach-response plan.

Canadian Software Security Code of Practice & Cybersecurity Guidance [52]

Strategy: Adopt key principles in code of practice for designed prototypes such as secure-by-default settings, secure dependency management, code review, minimal privileges, vulnerability patching, and clear security documentation.

CASL [50]

Strategy: Obtain explicit user consent for app installation via the app store and avoid hidden functions or deceptive behaviors (no unauthorized data exfiltration, no spyware-like features). Provide a clear description of what the app does and how it interacts with the device

2.4 Final Decision: Proof-of-Concept and Verification

As stated in Section 2.2, the final decision concept is automating identification of oral health conditions from images using CNNs, complemented by a rule-based questionnaire system. Since the full project involves building a multi-label model capable of detecting multiple oral diseases simultaneously, the POC focuses on validating the feasibility of this approach through a simplified binary classification task, distinguishing between Calculus and Healthy teeth using two different pretrained CNN architectures. A secondary POC component evaluates the decision-tree questionnaire intended to supplement image-based predictions with symptom-based reasoning. Together, these components verify the individual foundational elements of the final concept.

The dataset contained extraoral labeled images that were sourced from 2 different datasets [66] [67]. The dataset underwent a manual quality-control process in which images that were blurry, partially obstructed, improperly framed, or otherwise unsuitable for diagnostic evaluation were removed resulting in 307 images that were retained for training and validation. Our stakeholder also verified that the labels were correctly applied. The dataset was randomly split into an 80% training and 20% testing partition (245 training, 62 testing). All images were resized to 224×224 pixels and normalized using ImageNet statistics [68]. Standard data augmentation including random rotation (±15°), brightness jitter (0.9–1.1), and horizontal flipping to increase dataset diversity, improve robustness and reduce overfitting [69]. The same preprocessing pipeline was used for both models. The models were evaluated using accuracy, confusion matrices, ROC curves, and Grad-CAM visualization.

The first model utilized MobileNetV3-Small architecture, chosen for its efficiency and suitability for deployment on resource-limited devices such as laptops or mobile applications [70]. The model was initialized with pretrained ImageNet weights (MobileNet_V3_Small_Weights.IMAGENET1K_V1) to leverage generalized low-level feature extraction, while all convolutional backbone layers were frozen to prevent overfitting and to allow the model to focus training capacity on the classification head. The default classifier was replaced with a compact multilayer perceptron consisting of a 1024→256 linear layer, a ReLU activation, a 40% dropout layer to improve regularization, and a final 256→2 linear layer producing logits for the "Calculus" and "Healthy" classes. The model was trained using the Adam optimizer (learning rate = 1e-4, weight decay = 1e-4) with cross-entropy loss, and training history was recorded each epoch, including training/test loss and accuracy. During training, the model checkpoint with the highest validation accuracy was stored and later applied to the test set.

The second model utilized ResNet-18 architecture, which is slightly deeper but still light enough for mobile development [71]. The network was also initialized with ImageNet-pretrained weights (ResNet18_Weights.IMAGENET1K_V1), and all parameters in the convolutional backbone were frozen so that only the final classification layer was trained. The original fully connected layer was replaced with a new linear layer mapping the 512-dimensional global average pooled feature vector to 2 output logits, corresponding to the two classes. Training used the Adam optimizer on the final fully connected layer only, with a learning rate of 1×10^{-3}, weight decay of 1×10^{-4}, and cross-entropy loss as the objective.

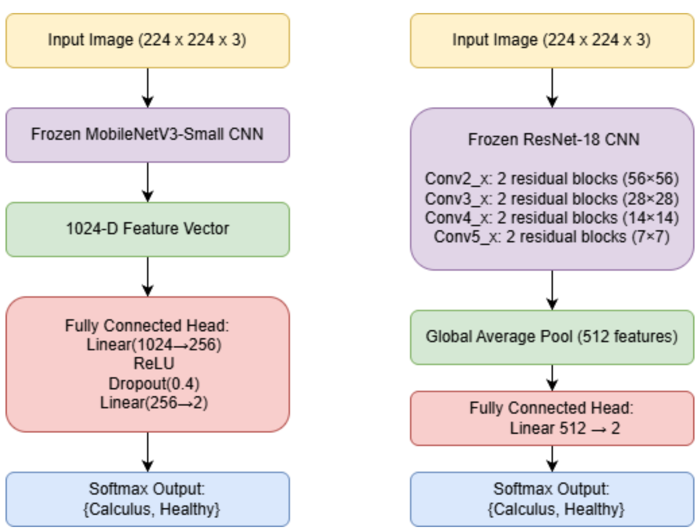


Figure 2. *ML layer architecture of the MobileNetV3-Small (left) and the ResNet-18 (right) with customized FC head.*

The MobileNetV3-Small prototype achieved an accuracy of 0.7419 and an ROC AUC of 0.7715. Its confusion matrix demonstrated moderate performance, showing reasonable ability to identify calculus, but some misclassifications still exist. In contrast, the ResNet-18 prototype performed better, achieving an accuracy of 0.7903 and an ROC AUC of 0.8208 with better precision and recall. This improvement suggests that the deeper residual architecture is more effective at extracting relevant features from dental photographs. These results validate that CNN-based calculus detection is technically feasible, and that model performance is likely to improve with larger and more diverse datasets.

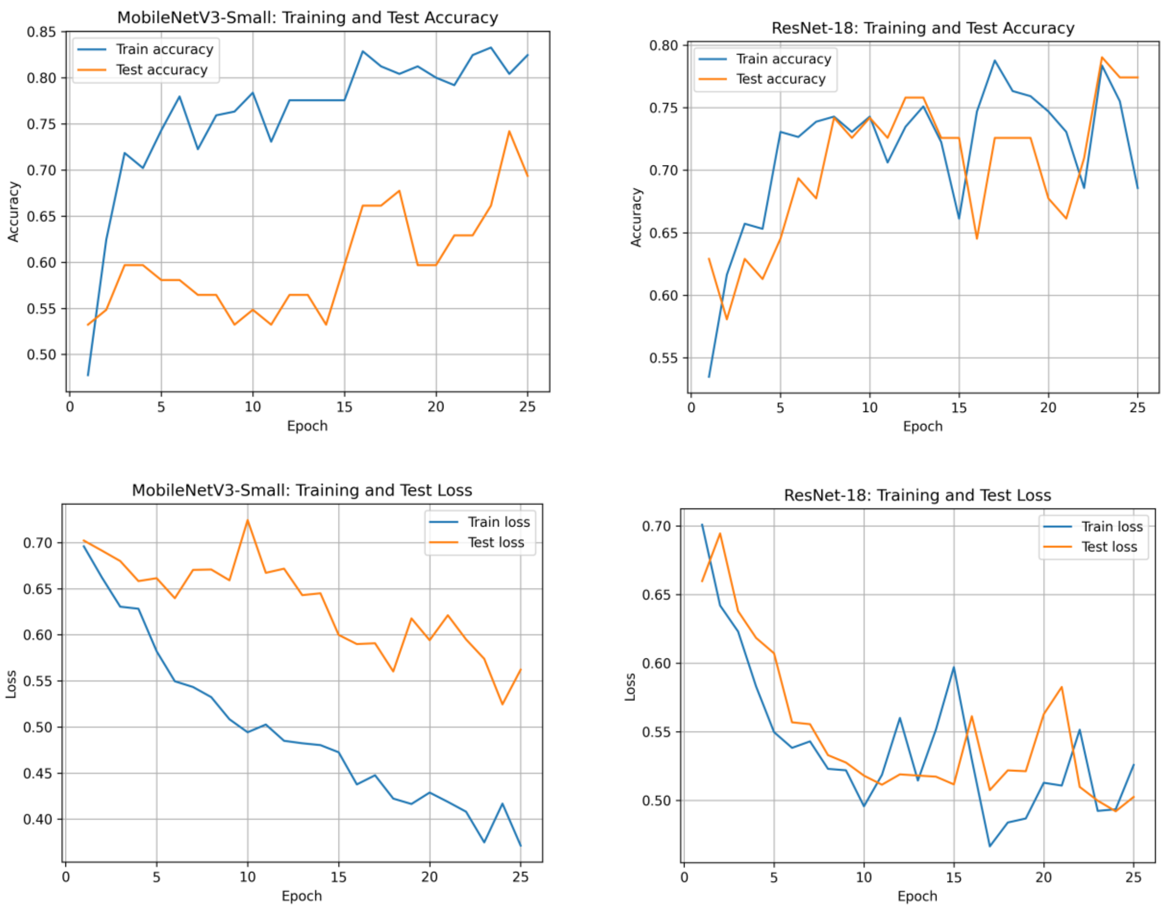


***Figure 3**. The training curves in Figure 3 demonstrate convergence, as both models show decreasing loss and increasing accuracy over epochs, indicating effective learning without severe overfitting. However, there is a slight divergence between training and testing*

Actual Values \ Predicted Values	Positive	Negative
Positive	31	8
Negative	8	15

Actual Values \ Predicted Values	Positive	Negative
Positive	22	8
Negative	5	27

Figure 4. *Model Confusion Matrices. The confusion matrices in Figure 4 highlight that minimizing false negatives is critical, as misclassifying diseased cases as healthy poses greater clinical risk. ResNet-18 shows improved sensitivity, making it more suitable for screening applications where high recall is prioritized*

The statistical verification results demonstrate that system performance is primarily limited by dataset size and variability rather than model capability. The observed AUC and sensitivity values indicate that the CNN is effective at distinguishing between classes; however, confidence intervals suggest variability due to limited test samples.

Furthermore, robustness testing shows that performance degradation under augmented conditions remains within acceptable limits ($\Delta AUC \leq 0.10$), indicating that the model generalizes reasonably well to variations in real-world input conditions. However, increased false positives in lower-quality images highlight the importance of enforcing strict image capture protocols. These findings directly inform future development by emphasizing the need for larger datasets, improved preprocessing pipelines, and enhanced quality control mechanisms to stabilize model predictions.

To further evaluate the reliability and generalization capability of the proof-of-concept models, an additional test was conducted using three external images that were not part of the training or testing dataset. These images represented three categories: Healthy, Light Dental Calculus, and Heavy Dental Calculus. This experiment simulates a real-world scenario in which the model is asked to classify unseen images with variable lighting, positioning, and calculus severity. Both the MobileNetV3-Small and ResNet-18 models were used to classify the images, and their predictions were subsequently visualized using Gradient-Weighted Class Activation Mapping (Grad-CAM) to assess whether the models were looking at the relevant regions during inference.

Figure 5. *The three test images used, Healthy, Light Dental Calculus, Heavy Dental Calculus (from left to right)*

Both models correctly classified all three external images. The Healthy image was confidently predicted as healthy by both networks, showing low activation intensity over the tooth surfaces in the Grad-CAM heatmaps, which suggests the models did not detect structural irregularities or calculus-like textures. The Light Calculus image was correctly labeled as Calculus by both models, although the probability margins were smaller, which is expected because mild calculus has more subtle visual features that are difficult to see. Grad-CAM visualizations for this case demonstrated moderate activation along the gum line. The Heavy Calculus image produced the strongest activation responses, and both networks classified it correctly with high confidence. Grad-CAM maps showed dense, focused activation over the tooth surface where large calculus deposits were located, confirming that the networks were responding to the correct pathological features rather than unrelated regions and background artifacts.

Figure 6. *Grad-CAM heatmaps generated by MobileNetV3-Small model. Respective class probabilities are C: 0.1604 H: 0.8396; C: 0.7974 H: 0.2026; C: 0.9957 H: 0.0043.*

Figure 7*. Grad-CAM heatmaps generated by ResNet-18 model. Respective class probabilities are C: 0.1793 H: 0.8207; C: 0.8297 H: 0.1703; C: 0.9618 H: 0.0382.*

In addition to the image-based models, a rule-based questionnaire was developed as part of the POC. This system collects user-reported symptoms and experiences, such as bleeding gums, tooth mobility, and pain, and assigns weighted contributions to the five oral conditions: Calculus, Cavities, Gingivitis, Periodontitis, and Missing Teeth. The questionnaire was developed through the help of our stakeholder as well as relevant research and can be seen in **Appendix B**. The system converts these scores into normalized probabilities and provides a risk distribution across conditions. Although the questionnaire does not yet integrate directly with the CNN outputs, the POC demonstrates that a rule-based inference system can effectively incorporate clinical logic for features not visible in images. This supports the viability of the hybrid diagnostic approach outlined in Section 2.2.

When comparing the POC results with the modelling and analysis presented earlier, the findings are consistent with expectations. Section 2.2 predicted that CNNs should be capable of learning visual indicators of oral disease, and this is supported by the POC's accuracy and ROC scores. The model also proposed that a hybrid system could compensate for limitations of purely image-based detection, which is validated by the successful implementation of the decision tree. However, the POC also highlights several gaps relative to the full multi-label concept. The current models address only a binary classification task due to dataset limitations, and no framework has yet been implemented to combine CNN probabilities with questionnaire-based estimates. Nevertheless, the POC provides strong preliminary evidence that both components work individually and can be integrated once appropriate datasets become available.

The proof-of-concept assumes that users provide clear, usable intra/extra-oral images and respond honestly to symptom-based questions. Its limitations are mainly the relatively small and narrowly sourced dataset and the lack of clinical validation for the questionnaire. Additional constraints include the time available for dataset collection and training, as well as privacy and ethical restrictions around medical imaging. Despite these limitations, the POC successfully verifies the technical feasibility of the final decision concept and demonstrates that a combined multi-label CNN and rule-based questionnaire system is a promising approach for automated preliminary dental assessment.

Future work and iterations

Future iterations of the system will focus on expanding from binary classification to full multi-label detection of multiple oral conditions. This will require the collection of a larger, more diverse, and clinically validated dataset. Model performance metrics obtained from the current proof-of-concept, including accuracy, ROC-AUC, and confusion matrices, will guide model selection and architecture refinement.

Additionally, future work will integrate CNN and questionnaire outputs into a unified fusion model, allowing real-time combined risk assessment. User feedback and usability testing data will inform improvements in interface design, workflow efficiency, and clarity of results presentation.

Further enhancements may include:

- Personalization based on user history
- Longitudinal tracking of oral health trends
- Improved bias mitigation through diverse datasets
- Clinical validation studies with dental professional

These iterations will progressively improve system accuracy, usability, and clinical relevance while maintaining compliance with safety and regulatory standard

Chapter 3

3.1 Design Prototype Plan

3.1A Backend Prototype Plan

Expanding upon the POC developed in Chapter 2: While the POC validated the feasibility of convolutional neural network-based disease detection and rule-based reasoning independently, the final prototype integrates both components into a unified, deployable mobile system with emphasis on modularity, safety, and privacy.

The prototype will be implemented as a four-layer modular architecture consisting of: (1) image acquisition and preprocessing, (2) CNN inference model, (3) questionnaire decision tree, and (4) triage fusion and user interface layer. Separating these components ensures high cohesion within modules and low coupling between subsystems, improving maintainability and testability.

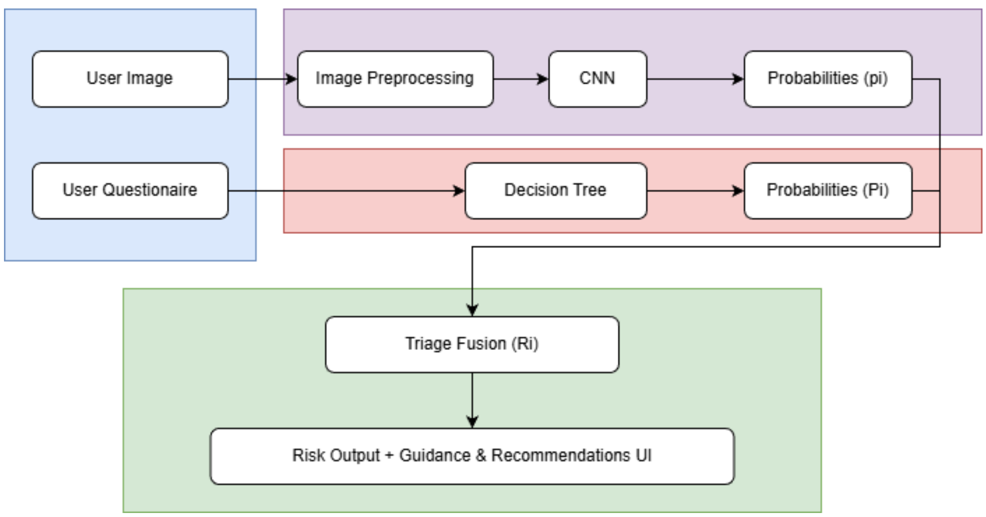


Figure 8. *Overall structure of proposed final concept as a four-layer modular architecture.*

Image Analysis Subsystem

The image-analysis subsystem will implement a multi-label convolutional neural network using a lightweight backbone architecture (ResNet-18; MobileNet-V3 can be used as verification) selected based on computational efficiency and suitability for on-device deployment [72]. For an input image x the model produces:

$$\hat{y} = f(x) \in \mathbb{R}^5$$

where each logit corresponds to one oral condition. Independent sigmoid activations generate per-class probabilities:

$$p_i = \sigma(\hat{y}_i)$$

Training will use binary cross-entropy loss to allow simultaneous detection of multiple co-occurring diseases (dental caries, gingivitis, periodontitis, calculus buildup, and tooth loss) [73]. Preprocessing steps include resizing to 224×224 pixels, intensity normalization, and data augmentation techniques (rotation, brightness scaling, blur simulation) to improve generalization (data augmentation is done for the training set only). The trained model will be exported to a mobile-compatible format (TensorFlow Lite or PyTorch Mobile) to enable local inference without need for cloud transmission if necessary [74].

Questionnaire Subsystem

The questionnaire subsystem will be implemented as a deterministic rule-based decision tree. Symptoms like bleeding gums, pain, tooth mobility and dietary habits are assigned weighted scores derived from clinical guidance and external research. For each condition i, the cumulative score s_i is converted to a normalized probability using softmax:

$$P_i = \frac{e^{s_i}}{\sum_j e^{s_j}}$$

The decision tree utilizes mimics Bayesian risk analysis, where each symptom is independent evidence toward a certain condition [75]. This approach provides interpretable outputs while maintaining deterministic and explainable behavior which is particularly useful for health data sets [76].

Fusion and Triage

The probability outputs of both subsystems will be combined using weighted probabilistic fusion:

$$R_i = \alpha p_i + (1 - \alpha) P_i$$

where α controls the relative influence of each subsystem. This parameter can be tuned to balance sensitivity and specificity. The final risk vector R will be mapped to discrete triage categories (low, moderate, high risk) and paired with personalized educational recommendations through our UI.

3.1B Front-End Integration Design Prototype Plan

High-level System Architecture

The Selfie Dental Assistant uses **a mobile-first design** with an overall **client-server layered architecture** as shown in Figure 2 where both the client (Mobile App) and the sever (Cloud/Sever) are presented. The client includes the hybrid screening app using questionnaire subsystem, image analysis subsystem for image quality checks and the fusion triage for risk scoring as mentioned in the section above. The server side is responsible for storing user information, updating analytics and modelling and ensuring secure storage through data encryption.

The mobile first design benefits the project greatly because it allows for clear prioritization of core features, ensuring improved usability and performance, scalable design and a cleaner UX. The design will enhance the approach of being user-centered, creating a better user experience and improved accessibility [77] [78].

The front-end component will be specifically using a **client-side layered mobile application** to align with the mobile-first design as illustrated by Figure 3 [79] [80]. The mobile-first design and client-side layout naturally reinforce each other. The properties and logic of the mobile-first design such as fast feedback and touch interactions are pushed directly to the client-side architecture, where it handles UI logic locally. This allows for smooth transitions and animations without having constant server calls. Having such layered structure promotes separation of concerns which improves the system modularity, reusability, scalability, and testability [81].

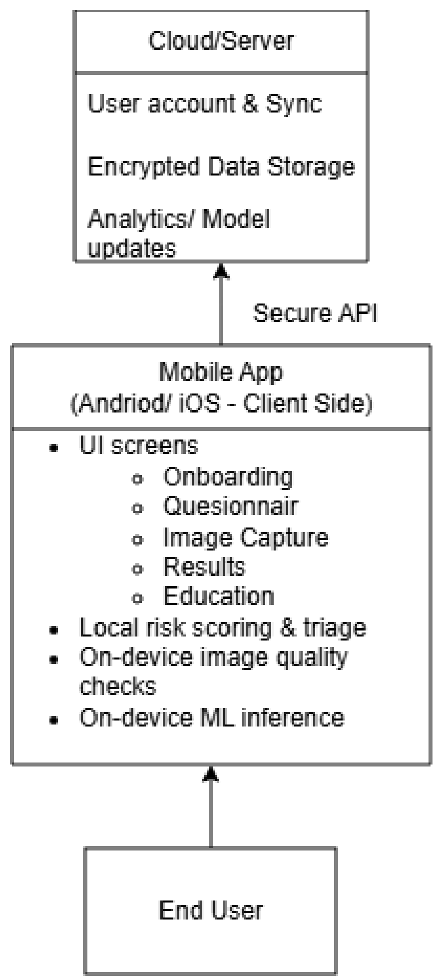


Figure 9. *System Architecture Client-Server View.*

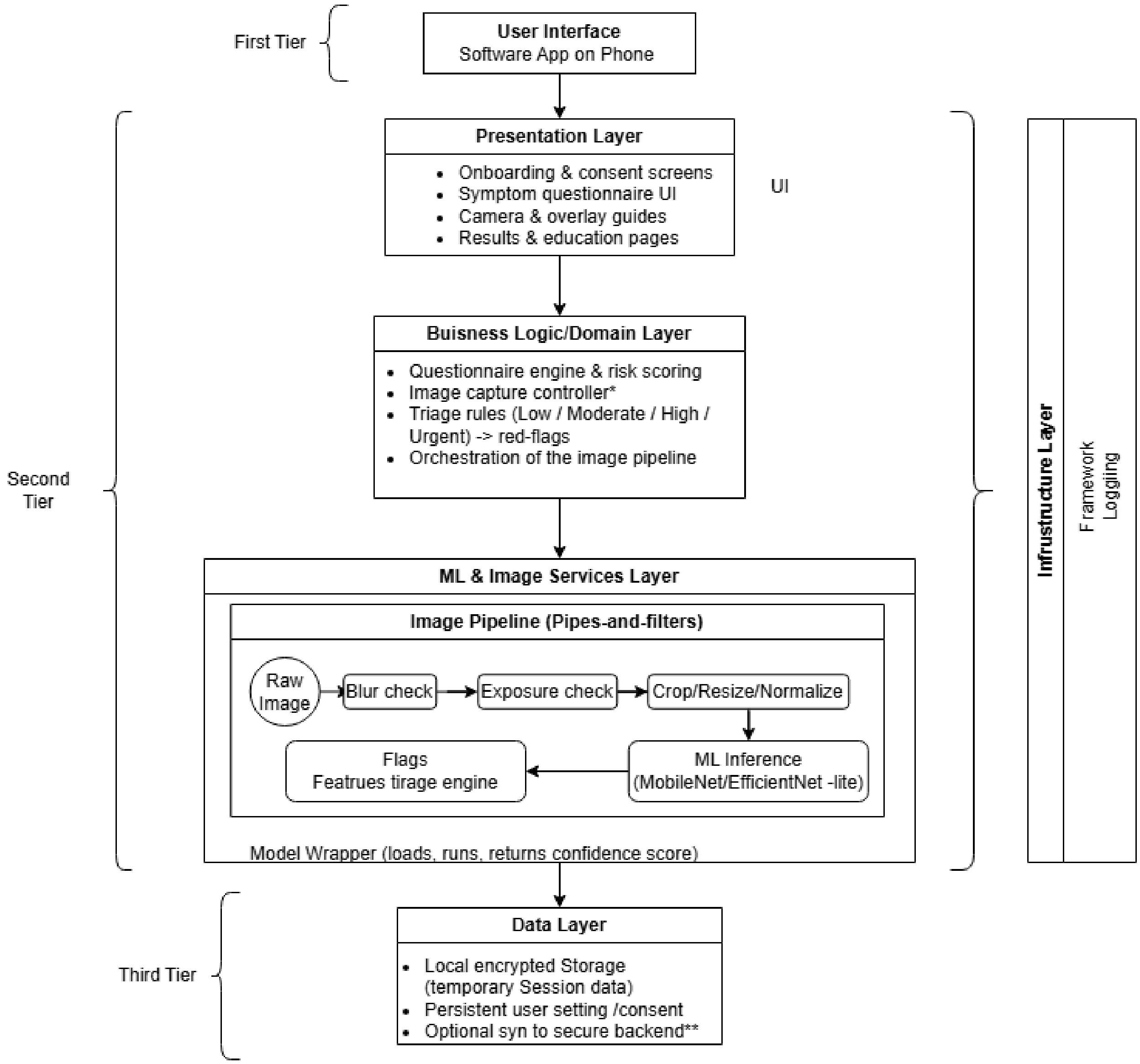


Figure 10. *System Architecture – Internal Client-Side Software Architecture.*

Frontend Prototype Plan [82]

The front-end prototype follows the agile development. The entire design and delivery process will include a total of 4 stages including the UI wireframe prototypes design, interactive prototype, API integration, and the final delivery. Between each of these stages, feedback from potential clients will be gathered through a combination of a 5-minute survey (**Appendix D**) and group interview. Feedback will be further analyzed to create or adjust requirements for the next stage and used in the future as a method of verification.

Stage 1 — UI Wireframe Prototypes Design

- Using personas, scenarios, use cases, business events (BE) and hierarchical task analysis (HTA) created in **DO1 – Software Architecture Document** to formalize deign ideas and create user flow mapping, sketches and story boards.
- Create low-fidelity mock-ups starting from paper prototypes and progress towards basic wireframing.
  - Paper Prototypes to test:
    - Conceptual Model
    - Functionality
    - Navigation and Task Flow
    - Terminology
    - Screen contents
  - Wireframing is used to define layout and establish functionality

Stage 2 — Interactive Prototype

Use Figma to implement the basic wireframes and create medium-fidelity interactive prototypes that implement navigation structure. It must include all core functionalities of the project and align with HTA. Design principles. Explore and consider different interface, interactions and design principles such as **Gestalt Principles of Perception** and **Norman Principles** [83] [84].

Stage 3 — Frontend Framework and API Integration

In stage 3, the focus shifts from purely visual design to the functional linking of the mobile interface with backend services via well-defined APIs. First use the pre-established interactive prototype in Stage 2 to simulate API behavior by including loading states and success/failed error flows. Next, use the interactive prototype as a guide to create a frontend framework using React Native.

This stage requires the implementation of multiple endpoints to support core application behavior, including secure image upload, questionnaire data submission, and risk result retrieval.
Image upload endpoints: Accept user-captured dental selfies, ensure proper validation and storage, and return status updates to the mobile client.
Questionnaire endpoints: Capture structured user responses and transmit them to backend services for further processing.
Retrieval endpoints: Deliver risk scores or diagnostic results computed by server-side logic or CNN models, ensuring that information is synchronized and accessible in real time. The front-end code will be connected to backend, replacing the simulated data on the prototype with live responses.

Stage 4 — Testing

Stage 4 includes a comprehensive testing regime designed to validate not only the functional and nonfunctional requirements listed in **DO1** but also its accessibility and user experience quality. Special attention will be given to edge-case handling to ensure that states and conditions that may lead to identified failure modes (**FM**s, as discussed in Chapter 2) are appropriately mitigated. Iterative testing cycles will inform refinements to both client-side logic and backend services, ensuring the final product is robust, intuitive, and inclusive.

Implementation of Risk Mitigation Strategies

- **Image quality variability:** A standardized image capture protocol and automated quality checks (contrast, blur detection, lighting) will reject unusable images before model inference and instruct the user step by step to take new images.
- **False negatives:** High-sensitivity thresholds and conservative triage rules will prioritize safety by recommending professional consultation when any uncertainty arises as to prevent possible oversight.
- **Privacy risks:** On-device inference and minimal data retention will reduce exposure of the user's personal health information.
- **Model misuse:** Any recommendations by the application will have explicit educational framing as well as disclaimers to prevent users from interpreting results as a clinical diagnosis.
- **Algorithmic bias:** Dataset diversification and augmentation are used to reduce demographic performance disparities. We actively sought out training data from underrepresented demographic groups to ensure the CNN model's visual inference remains reliable regardless of a user's age, ethnicity, or physical characteristics.
- **Subsystem failure:** Modular separation ensures independent testing and prevents cascading errors.

Assumptions, Limitations, and Constraints

Similar to the POC from Chapter 2, the final application still relies on several important assumptions. It assumes that users possess smartphones with camera capability and stable internet connectivity. Users will input clear intra/extra-oral images and answer the questionnaire honestly. It also assumes that CNN has been trained on a diverse and representative dataset that captures real-world variability.

Limitations include reduced performance under poor lighting, dependence on visual indicators that may not fully represent disease severity, and the non-diagnostic nature of the system. The questionnaire also simplifies complex clinical reasoning which may limit diagnostic precision in borderline cases and may not completely capture atypical presentations.

Several constraints also exist. Ethical and regulatory limitations require careful handling of identifiable images, which justifies the emphasis on local processing and minimizing stored data. Technical constraints highlight the need for lightweight models to maintain real-time inference justifying the selection of compact CNN backbones and local processing architecture. These constraints also directly impact system performance and design trade-offs. The requirement for on-device inference limits model complexity, resulting in the use of lightweight architectures (i.e. ResNet-18, MobileNetV3), which may reduce maximum achievable accuracy compared to larger models. Similarly, reliance on user-captured images introduces variability in lighting, angle, and resolution, which may decrease model robustness and increase false predictions. The absence of radiographic imaging further restricts the system to surface-level visual inspections, limiting sensitivity for non-visible or early-stage conditions. To mitigate these effects, data augmentation, conservative triage thresholds, and multimodal fusion with questionnaire data are implemented to maintain acceptable screening performance despite these constraints.

3.2 Design Verification Plan

There is a total of 7 design outputs listed in Chapter 2, they are:

- **DO1 – Software Architecture Document**
- **DO2 – Functional Design Outputs**
- **DO3 – User Interface (UI/UX) Design Outputs**
- **DO4 – Data Schema & Encryption Documentation**
- **DO5 – Privacy, Consent, and Access Control Documentation**
- **DO6 – Risk Management File**
- **DO7 – Verification & Validation Plan Output**

NOTE: *The following section of Chapter 3 covers the entirety of DO7 hence it would not be addressed in this section individually.*
Points of analysis and methods of analysis were judiciously selected based on their importance, relevance to system performance, user safety and values of **RPN** provided in Chapter 2.

DO1 DO2 Verification Plan

DO1 summarizes the scope of the project and lists all functional (FRs) and non-functional requirements (NFRs). This document's Section 4 FRs and Section 5 NFRs will overlap with some sections of **DO2-6** since each of the design outputs are a subsection of **DO1**.
The verification plan will verify the overall architecture design against the following industry standards for structural correctness:

- Google Android App Architecture Guidelines [85]
- Microsoft Application Architecture Framework [86]
- ISO/IEC 25002 Software Quality Model [87]
- ISO/IEC/IEEE 29148 – Software Requirements Engineering [88]
- Google Play Core App Quality Guidelines [89]

The architecture of the system will be verified through architecture traceability analysis, ensuring requirements are mapped one-to-one to components of the system. In addition, all diagrams (Noun-Verb Extraction, Class Diagrams, System Diagrams, use case Diagrams etc.) will be inspected and checked through peer review. Inputs, outputs, individual components, subsystems and their behaviours must be validated through functional test cases. Detailed verification plans of the core functionalities are included in the section below.
General Acceptance Criteria includes:

- ≥ **95% of system components** are modularized into clearly defined layers (UI, business logic, data) [85] [86] [87].
- **0 single-point-of-failure components** in critical user flows [85] [86] [87].
- **100% functional requirement traceability** [85] [86] [87] [88] [89].
- ≥ **95% functional test pass rate** under nominal conditions [88] [89].

Google and Microsoft Architecture Standards emphasize on separation of concerns and modularity to improve maintainability, testability, and fault isolation. These system attributes directly align with ISO/IEC 25002. All the standards above require every functional requirement to be verifiable and traceable. Google

Play Quality Guidelines requires all advertised functionality to have correct execution and guaranteed consistency.

Image Analysis Subsystem Verification

Similar to the POC, various methods of statistical analysis will be used to verify that the multi-label CNN accurately detects oral conditions and generalizes to unseen patients while minimizing false negatives. Since the classifier outputs independent probabilities $p_i \in [0,1]$ for each condition. Performance will be evaluated using statistical measures derived from the confusion matrix:

$$\textbf{Sensitivity} = \frac{TP}{TP + FN}; \qquad \textbf{Specificity} = \frac{TN}{TN + FP}$$

$$\textbf{Precision} = \frac{TP}{TP + FP}; \qquad \boldsymbol{F1} = \frac{2PR}{P + R}$$

ROC analysis will be used to compute AUC, which measures discrimination independent of threshold selection [90]. CI will also be calculated for all metrics.

Acceptance requires an AUC and Sensitivity of 0.85, Specificity of 0.80, and a CI width less than 0.10 [91]. Results can be corroborated through Cohen's kappa agreement analysis against stakeholder provided annotations and a secondary architecture comparison (MobileNet-V3) will be used to confirm performance stability.

Robustness, Generalization, and Explainability

During training mages are stratified and augmented to simulate varying camera hardware, lighting, motion blur, and demographic diversity. Performance degradation can be measured by:

$$\Delta AUC = AUC_{original} - AUC_{augmented}$$

With an acceptance criterion of $\Delta AUC \leq 0.10$, which is an industry standard [91].

Grad-CAM saliency maps will be generated for both correct and incorrect classifications. Activation regions will be examined and corroborated with stakeholders to ensure the model localizes teeth and gingival structures rather than background artifacts.

Image Capture Protocol Verification

To verify that the Image Capture Protocol via standardized instructions reduce input noise and improve model performance, we will conduct a comparative study of two groups: With Protocol vs. Without Protocol. Metrics include the AUC improvement, blur detection rates, and the percentage of rejected images. Hopefully a statistically significant improvement in model performance will be confirmed via a paired t-test (p < 0.05).

Questionnaire Decision-Tree Subsystem Verification

To verify the correctness and completeness of the decision-tree logic, we will employ unit testing of all branches using simulated patient profiles and boundary-condition testing. Outputs will then be compared directly with stakeholder assessments to generate a confusion matrix. The system should achieve 90% path coverage and a minimum 90% agreement with stakeholder-labeled outcomes. This will ensure that deterministic behavior of the subsystem aligns with established clinical standards.

Fusion Triage Logic Verification

To verify that the weighted fusion of visual and questionnaire data improves diagnostic accuracy. A sensitivity analysis can be conducted by varying the weighting coefficient ∝ from 0 to 1 to maximize the AUC. The fused system should yield a statistically significant improvement over either the CNN or questionnaire subsystems alone, which can be validated via the DeLong test for AUC differences [92].

DO3 User Interface (UI/UX) Design Outputs

To verify DO3, the user interface of this project will be inspected against the following industry standards:

- Google Material Design [93]
- Microsoft Fluent Design System [94]
- WCAG 2.1 Accessibility Guidelines [95]

Due to the number of NFRs included in this section, the verifications and analyses will have an emphasis on UI performance with the following verification method and acceptance criteria:

- Interaction Latency Test: ≥ **95% of UI interactions** respond within **≤100 ms**
  - Human perception is quantifiable as Google and Microsoft states delays above 100ms are noticeable [93] [94]
- Frame Rendering Analysis: ≥ **95% of frames rendered** at **≥60 fps**
  - Industrial standard for smooth mobile rendering
- WCAG 2.1 AA compliance
  - Baseline of accessibility for mobile platforms [95]

DO4 – Data Schema & Encryption Verification Plan

Data flow analysis, schema inspection, encryption configuration testing and review will be used while referring to Microsoft Security Development Lifecycle (SDL) [96]. To verify DO4, both structural and cryptographic validation methods will be applied. Data schema integrity will be verified through schema validation testing to ensure all required fields (image metadata, questionnaire responses, timestamps) are correctly formatted and stored. Encryption will be validated using penetration testing and packet inspection tools to confirm that all data transmissions are secured using TLS 1.2+ and that stored data is encrypted using AES-256. Some ambitious hopes:

- 100% of sensitive fields encrypted at rest and in transit
- 0 plaintext exposure during transmission (verified via packet sniffing tools)
- Successful decryption only with authorized keys
- Schema validation pass rate ⩾ 99%

DO5 – Privacy, Consent, and Access Control Documentation

To verify DO5, the consent workflow must be inspected, and role-based access must be tested to ensure there are 0 unauthorized access paths during role testing and 100% of the personal data access are gated behind explicit user consent. Verification of DO5 will include both functional and security testing of consent workflows and access control mechanisms. Role-based access control (RBAC) will be tested by simulating different user roles and attempting unauthorized access to restricted data. Consent verification will ensure that no personal data is collected, stored, or processed prior to explicit user approval.

- Google Play User Data Policy [97]
- PIPEDA Principles [49]
- 0 unauthorized access attempts succeed during penetration testing
- 100% of data access events logged and traceable
- 100% of sensitive operations gated behind explicit user consent
- Consent withdrawal results in complete data deletion within defined time (<24 hours)

DO6 – Risk Management File Output Verification Plan

FMEA identified in Chapter 2 needs to be verified through fault injection, stress and edge-case testing. All the identified risks and failure modes must be mapped to mitigation strategies. Stress and edge-case testing will be used to verify the mitigation effectiveness to meet $\geqslant$ 95% mitigation effectiveness under simulated fault conditions. The following documents must be aligned with to meet the industrial standard: Verification of DO6 will be conducted by validating that all identified failure modes (from FMEA) are effectively mitigated through targeted testing. Fault injection testing will simulate failure conditions such as incorrect inputs, system crashes, delayed processing, and incorrect model outputs [98].

- $\geqslant$95% of identified failure modes successfully mitigated under test conditions
- System maintains safe behavior (fail-safe outputs) under all simulated faults
- No critical failures result in unsafe user guidance
- ISO 14971 – Risk Management [63]
- Google App Stability Metrics [99]

Chapter 4

4.1 Implementation

Prototype Description (Frontend)

The Selfie AI Dental Assistant prototype is implemented as a mobile-based oral health screening application with a client-server architecture.

The frontend is developed in React Native with Expo using Type Script, and it is responsible for the following FRs created in DO1:

- Login/Log out
- New user Signup
- Educate users
- Photo Guidelines & Image Capturing
- Presenting & Answering Questionnaire
- Result & Recommendation Presentation
- Present Terms of Service
- Present Privacy & Security
- Local Storage of Past Results & Recommendations
- Save Time & Date of Questionnaire Response

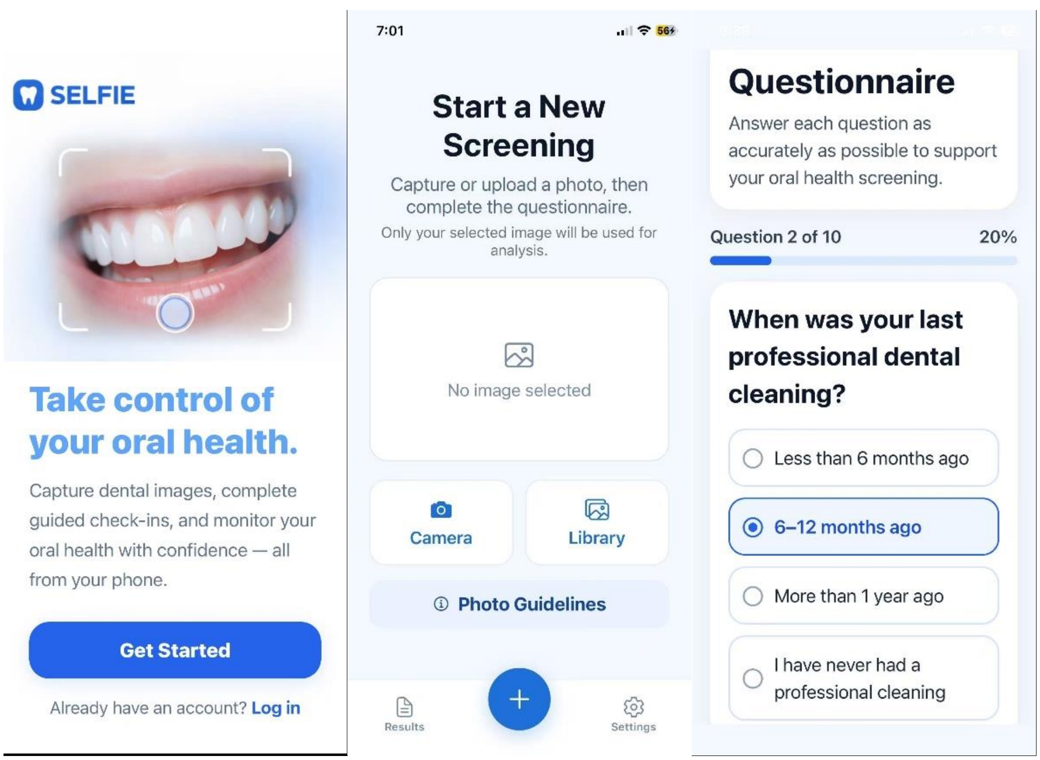


Figure 11. *SELFIE Application User Interface I. (a) Landing page, (b) Home page, (c) Questionnaire progress. The interface is designed to provide accessible screening, result interpretation, and user guidance.*

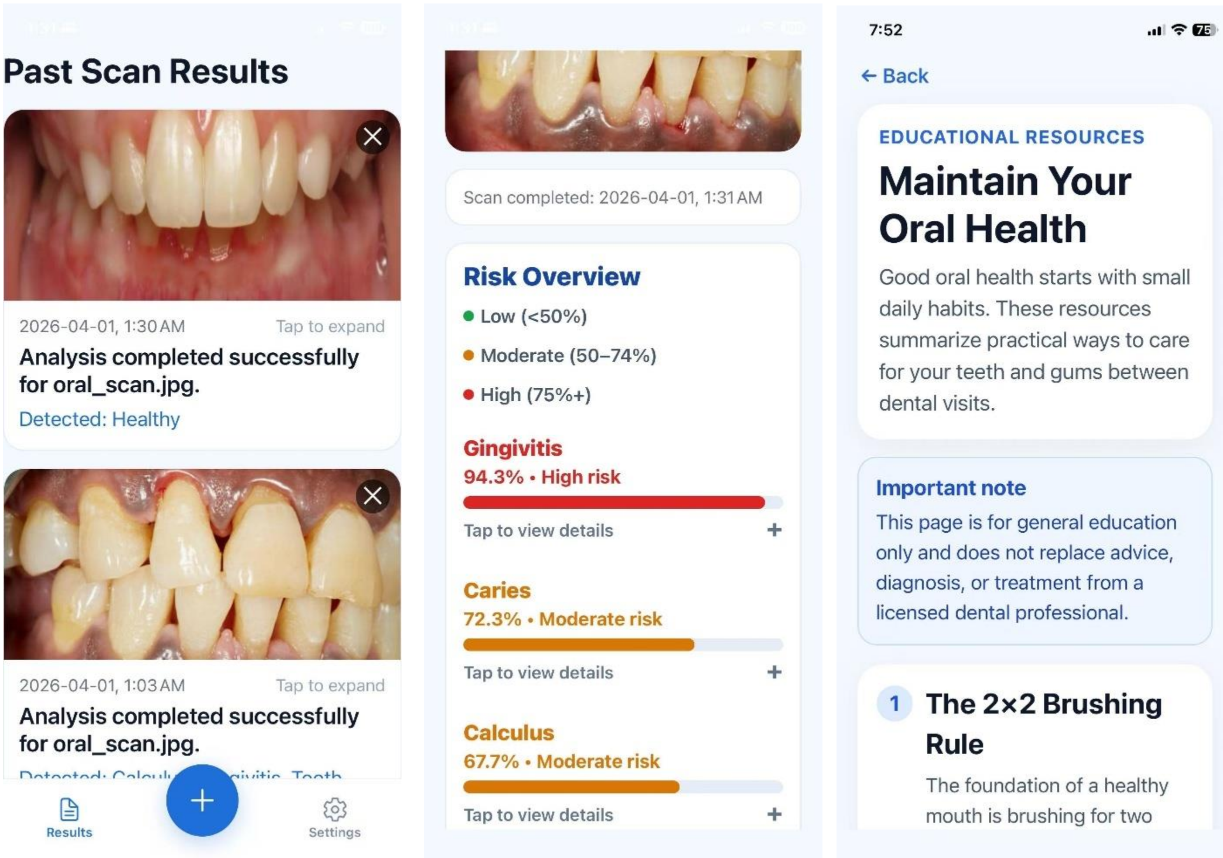


Figure 12. *SELFIE Application User Interface II. (a)Patient History Tracking, (b)Results display, (c) Educational resources. The interface is designed to provide accessible screening, result interpretation, and user guidance.*

Prototype Implementation (Frontend)

As described in Chapter 3, there are three stages of agile prototype development:

1. Wireframe Paper Prototype
2. Interactive Figma Prototype
3. Frontend Framework and API Integration
    - React Expo + Fast API

The survey feedback for each iteration is stored in **Appendix E**.

Wireframe Paper Prototype:

The initial design of the application was developed using a paper prototype to explore the overall workflow and user interaction sequence. It is highly simplified and abstract, created using pencil and paper to outline the key FRs of the proposed product. Each fold of the paper represents a key stage of the application shown in Figure 13 and 14 below.

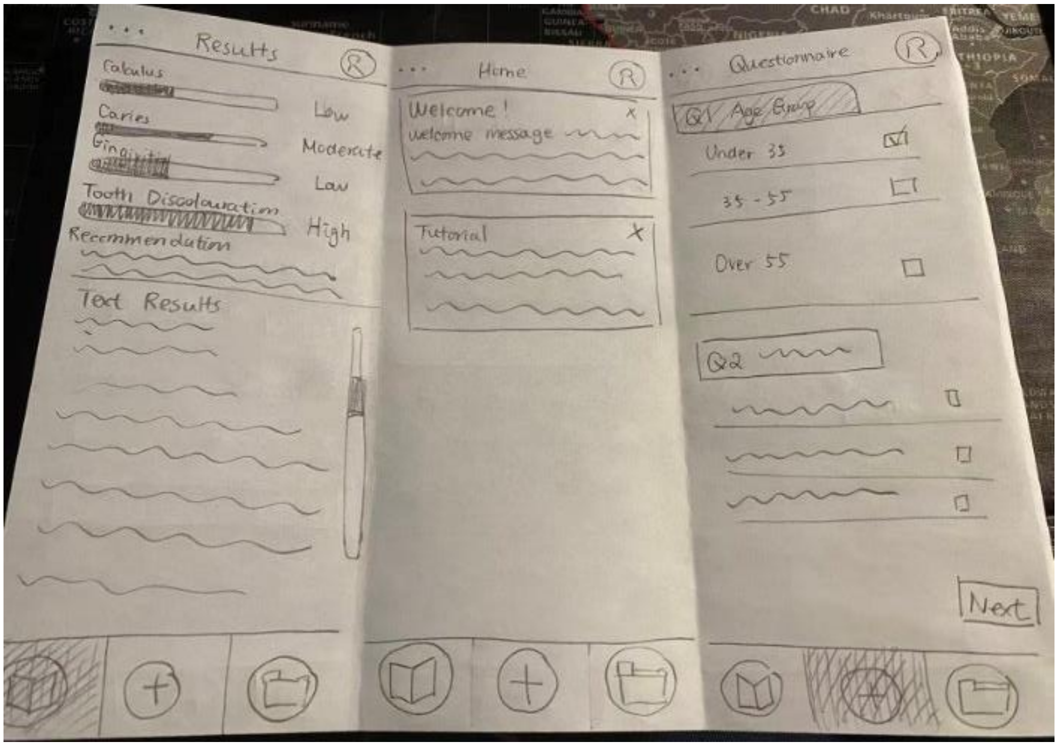


Figure 13. *Selfie Paper Prototype with Results Page (left), Home Page (middle) and Questionnaire Page (right).*

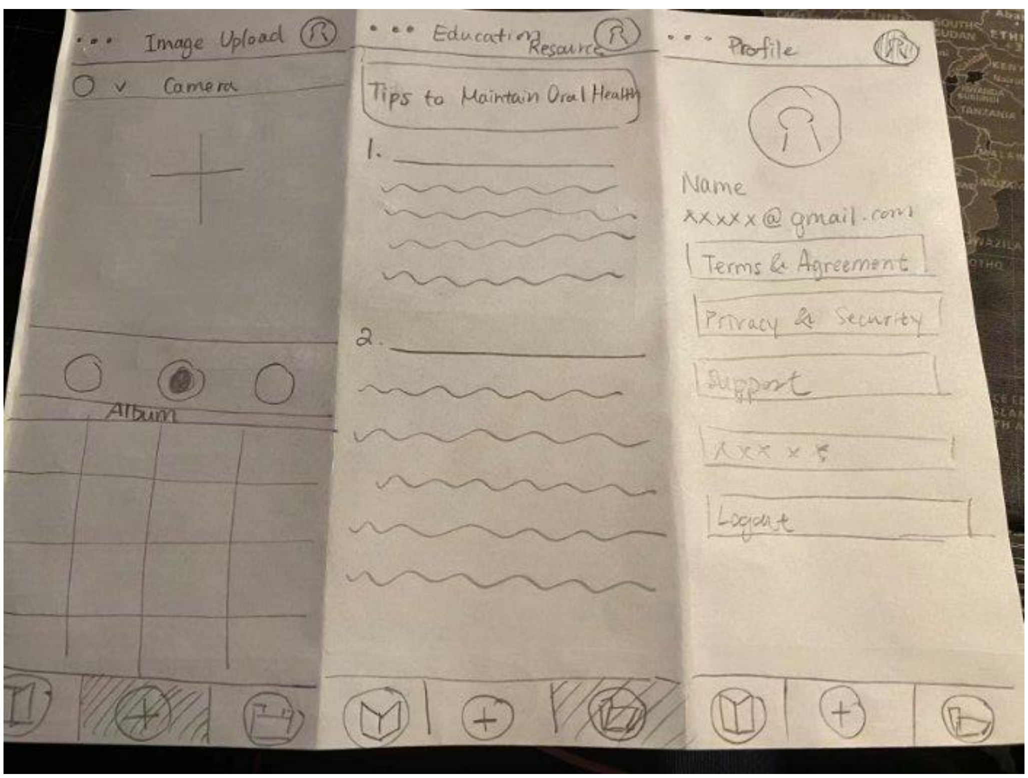


Figure 14. *Selfie Paper Prototype with Image Upload Page (left), Education Resources (middle) and Profile Page (right).*

The low-fidelity paper prototype was the foundation of the digital interface and was used to validate the logical flow of the application. It is a conceptual model that ensures the users could clearly understand the screen contents, function of each page, how to navigate through pages and the general task flow. Surveys provided in **Appendix D** and group interviews were conducted to gather feedback. Due to the abstraction and simplicity of the paper prototype, it caused confusion for the online surveys since no guidelines were provided online. However, it was very useful in a group interview setting where both group members could guide and answer questions in real time.

General feedback includes:

- Looks too overwhelming
- Confusing without guidance
- Complaints of paper prototype being poorly drawn with no proper guidance
- Complicated process of scanning
- Crowded information with too much text
- Good general app layout, however, need to work on details

Feedback collected at this stage helped refine screen transitions and UI components, reduce unnecessary steps, and improve clarity in instructions. The application was aligned to Google Material Design, Microsoft Fluent Design System and WCAG 2.1 Accessibility Guidelines when integrating the feedback [93] [94] [95].

Interactive Figma Prototype

The interactive Figma Prototype was created to implement and improve the basic wireframes. It was created to define the layout, structure, and interaction design of the application. It is kept as low fidelity for faster improvements and iterations. The focus was on reducing content, improving white space, enhance navigation structures and core functionality presentation. Navigation between screens was refined to minimize user effort and ensure a linear scan workflow. General layouts were changed to explore and consider different interface, interactions and design principles following the **Gestalt Principles of Perception** and **Norman Principles [83] [84].**

However, according to **Appendix D Survey Responses 2** and group interviews, the users preferred the paper prototype layout better. Many commented they would prefer a more professional layout because the current layout looks untrustworthy due to the old style and color choice.

Frontend Framework and API Integration

The integrated prototype combines both the mobile front end with the back-end machine learning pipeline explained in the next section. The integrated system was developed using React Native with Expo for the mobile application and Fast API for the backend server. This prototype represents the functional version of the Selfie Dental Assistant capable of real-time image submission and analysis. The integration of both frontend and back end will be explained in a separate section after the backend description and implementation.

There were multiple iterations at this stage, where UI components, navigation and tasks flow were adjusted frequently. Feedback was gathered on a regular basis to continuously refine the prototype and improve the user experience.

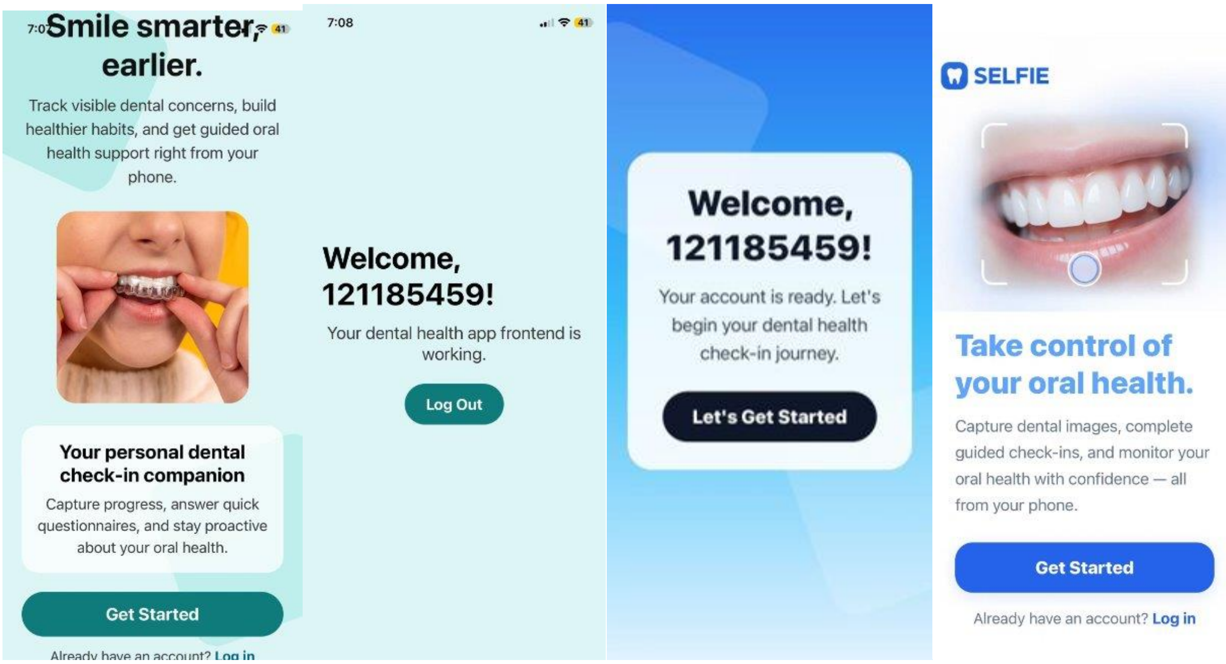


Figure 15. *Design iterations of Selfie Integrated Prototype. (a) (b) Early iterations of Selfie experimenting with different fonts, theme colors and layout (c) Final landing page design.*

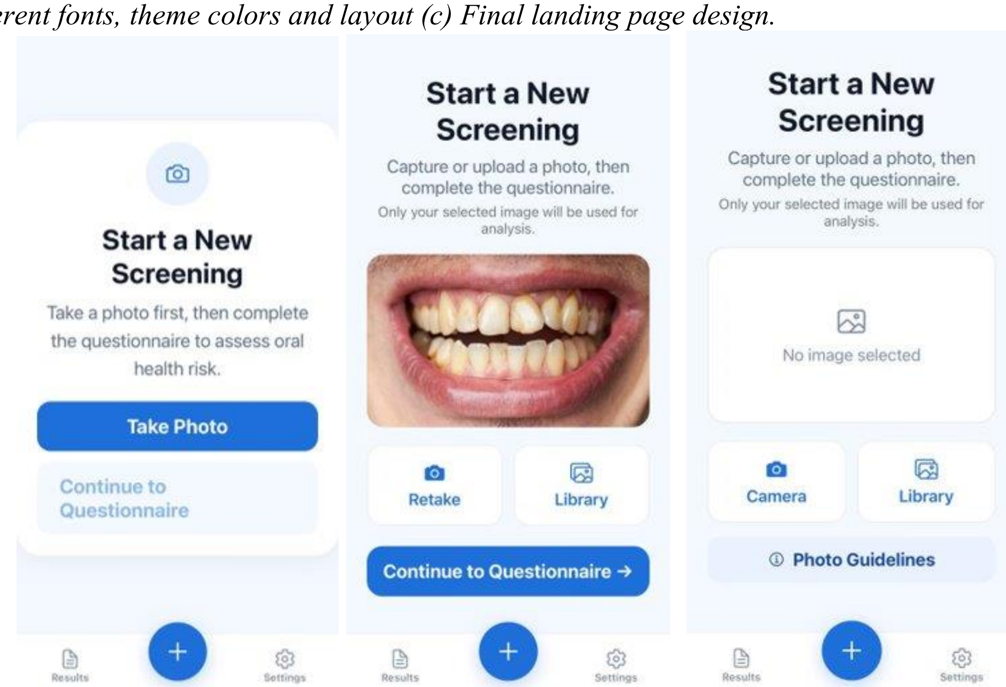


Figure 16. 3.X Design iterations of Selfie Integrated Prototype. (a) 3.0 New scan Screen (b) Implementing feedback from stakeholders to directly upload photo from album (c) Adding guidelines for compliance and mitigation.

Further feedback from stakeholders and users was continuously incorporated leading up to the Capstone Expo. The following items were prioritized and implemented to improve usability and clarity of the prototype:

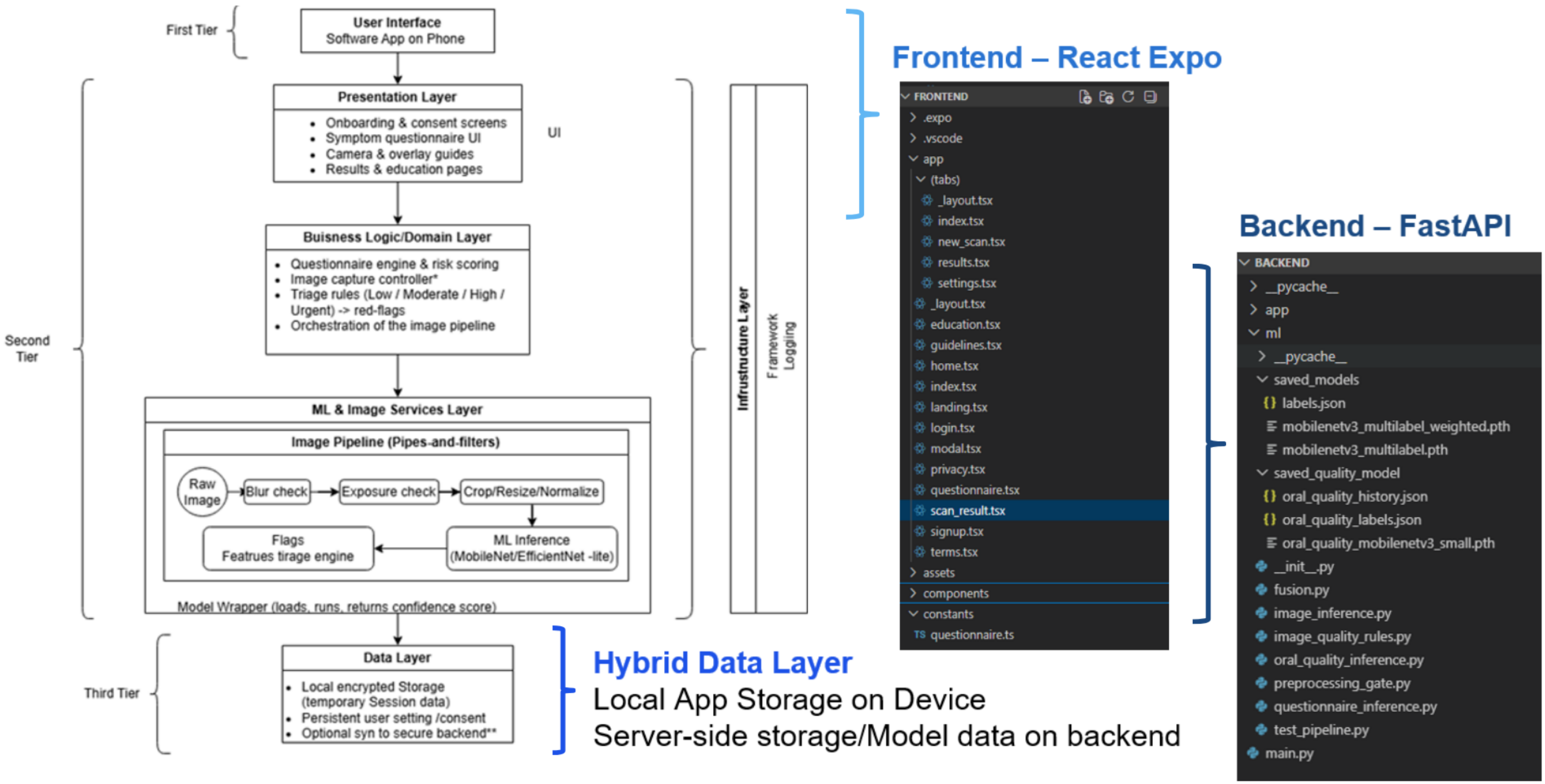
1. Added color-coded risk levels to improve readability of condition severity
2. Removed demo-only features to streamline the user interface
3. Removed image quality status from the results page to reduce visual clutter
4. Stored recommendations instead of questionnaire responses in scan history
5. Added timestamp including date and time for each scan
6. Introduced more visual elements to reduce dense text presentation
7. Improved layout to highlight key results more clearly
8. Allowed users to resubmit invalid images without repeating the questionnaire

***Figure 17**. Feedback integration of Feedback 1,3,4,5,6,7.*

Testing & Validation (Frontend)

Testing for the front end was very straightforward and direct with unit and functional testing. It is obvious whether a single function has succeeded or failed.

Since the architecture is already defined prior to the prototype design, ≥ **95% of system components** are modularized into clearly defined layers (UI, business logic, data) [85] [86] [87] meeting the requirement of DO1. 0 single-point-of-failure components in critical user flows, 100% functional requirement traceability and ≥ 95% functional test pass rate under nominal conditions were all achieved [85] [86] [87] [88] [89].



The project follows the same layered structure conceptually, but in implementation the ML & image services layer is deployed in the backend server rather than fully on-device.

Figure 18. *System Architecture Translated into Coding Structure.*

As described in chapter 3, the testing of the front end include:

1. Interaction Latency Test
   - ≥ **95% of UI interactions** respond within **≤100 ms**
2. Frame Rendering Analysis
   - ≥ **95% of frames rendered** at **≥60 fps**
3. WCAG 2.1 AA compliance

The results of these tests will be discussed in section 4.2

Prototype Description and Implementation (Backend)

This section focuses more on the backend implementation of the application, which is responsible for all computational processes, including image validation, model inference, questionnaire processing, and probability fusion.

The backend system has five components and was implemented in Python using the PyTorch deep learning framework and integrated with Fast API:

1. **Image Preprocessing and Quality Gate**
   This module assesses user-submitted images prior to analysis. It identifies and rejects images that have low quality.
2. **Oral Image Quality Classification Model**
   A lightweight CNN, based on MobileNetV3 architecture, is used to classify images as either valid oral images or invalid inputs. Serving as an additional validation layer beyond rule-based checks.
3. **Multi-label Disease Classification Model**
   A fine-tuned MobileNetV3 model is used to perform disease inference. It outputs independent probabilities for multiple dental conditions, including calculus, caries, gingivitis, and tooth discoloration.
4. **Questionnaire Scoring Module**
   This module implements a rule-based probabilistic scoring system that converts user-provided questionnaire responses into condition-specific risk probabilities.
5. **Fusion Engine**
   The fusion component integrates predictions from both the image-based model and the questionnaire module to produce final screening probabilities.

The architecture follows a sequential processing pipeline where each component operates independently and communicates via structured data outputs. This design enhances maintainability, scalability, and testability.

Preprocessing and Validation

Prior to any model inference, a rule-based image quality assessment module is implemented to ensure that only suitable inputs are processed by the machine learning models, saving time and resources. This preprocessing stage serves as a quality gate, improving the reliability and robustness of the overall system. The module evaluates multiple aspects of image quality using computer vision techniques implemented with the OpenCV library:

- **Resolution Check**

The spatial dimensions of the image are verified to ensure a minimum resolution of 180 × 180 pixels. Images below this threshold are rejected because insufficient resolution may degrade model performance.

- **Brightness Analysis**
  The image is converted to grayscale, and the mean pixel intensity is computed. Images are flagged as unsuitable if they fall below a predefined darkness threshold or exceed an upper brightness threshold, indicating underexposure or overexposure.
- **Blur Detection**
  Image sharpness is quantified using the variance of the Laplacian operator. A low variance indicates insufficient edge information and is interpreted to be blurry. Images below the defined blur threshold are rejected due to loss of critical visual features.
- **Contrast Evaluation:**
  Contrast is measured by the standard deviation of grayscale pixel intensities. Low contrast images are identified when the standard deviation falls below a predefined threshold, suggesting limited differentiation between structures.

Each check produces a structured output that can be easily traced in the front end. It will indicate whether the criterion has been satisfied, along with relevant quantitative metrics (i.e. brightness level, blur score). Based on these results, the system categorizes image quality into three levels:

- Pass: All checks are satisfied
- Warning: Minor issues are detected (suboptimal brightness or contrast)
- Fail: Critical issues are present (low resolution or excessive blur)

Images classified as fail are rejected immediately, while warning cases may proceed to the next inference model with caution. This hierarchical system ensures that critical quality issues are filtered out while allowing some flexibility for borderline inputs. This preprocessing module was introduced to directly address the challenge of input variability which was highlighted as a major concern in previous chapters.

Oral Image Quality Model

Inputs that pass the previous rule-based image quality checks are sent to a separate MobileNetV3 Small model for image classification. This model performs binary classification using a SoftMax activation function to distinguish between valid and invalid oral images. This model acts as a gating mechanism, ensuring that only images containing the relevant teeth and gum areas are passed to the disease classification stage.

Disease Classification Model

The disease classification model is based on MobileNetV3-Small architecture, which is selected for its efficiency in resource-constrained environments like phones. The classifier head was modified to include fully connected layers with dropout regularization to reduce overfitting. The model outputs multi-label predictions using a sigmoid activation function, enabling independent probability estimation for each dental condition. This design allows simultaneous detection of multiple conditions without enforcing mutual exclusivity.

Questionnaire Integration

The questionnaire module was implemented as a rule-based probabilistic scoring system based on clinical diagnostic factors. User responses are mapped to condition-specific risk scores and then transformed using a sigmoid function. This approach ensures that multiple conditions can be identified simultaneously and that the system remains both interpretable and easy to modify. The questionnaire has been redeveloped since Chapter 2 several times under direct guidance of our stakeholder to reflect current needs. Additional research was also done to verify every path is consistent with clinical practice.

Fusion Strategy

As outlined in Chapter 3, the fusion component combines image-based and questionnaire-based probabilities using a weighted approach to generate the final screening output. This strategy prioritizes image-based predictions while allowing questionnaire-derived information to provide complementary support. This is particularly important since a single image is not enough to provide conclusive results. Most importantly, the weighting parameters used are configurable and can be easily adjusted to better reflect the characteristics of specific dental conditions.

Testing and Verification (Backend)

The backend system was evaluated using a multi-stage testing approach. Unit testing was conducted to verify the functionality of individual modules, including preprocessing functions, model inference accuracy, and questionnaire scoring logic. This was followed by pipeline testing (test_pipeline.py in models folder), where the complete system was assessed using input combinations of valid images, poor-quality images, and varied questionnaire responses. Additionally, targeted edge case testing was also performed to address possible use case scenarios identified in earlier chapters, including blurry or low-light images, images lacking visible dental structures, and conflicting or extreme questionnaire inputs.

Examples (using same set of sample questionnaire responses):

Figure 19*. Image of a table, unsuitable image for classification model.*

Image Quality Rules: Success

- (Resolution: Passed; Brightness: Passed; Blur: Passed; Contrast: Passed)

Image Inference Model: Failure

- Predicted label: "invalid_oral" with 0.7015 probability

Triggers immediate failure with message: “Image does not appear to be a usable oral/teeth photo” “Image quality is insufficient. Please retake the photo.”

Figure 20. *Unsuitable image for classification model, too dark, and lack of teeth and gumline for the classification model to perform inference on.*

Image Quality Rules: Success

- (Resolution: Passed; Brightness: Warning; Blur: Passed; Contrast: Passed)
- Warning Message: “Image is too dark.”

Image Inference Model: Failure

- Predicted label: "invalid_oral" with 0.7483 probability

Triggers immediate failure with message: “Image does not appear to be a usable oral/teeth photo” “Image quality is insufficient. Please retake the photo.”

***Figure 21**. Suitable image for classification model, with high visual quality and showing enough teeth and gumline.*

- Image Quality Rules: Success
  - (Resolution: Passed; Brightness: Passed; Blur: Passed; Contrast: Passed)
- Image Inference Model: Success
  - Predicted label: "valid_oral" with 0.9706 probability

Image Probabilities:
- Calculus: 0.9614011645317078,
- Caries: 0.6847766041755676,
- Gingivitis: 0.9706622958183289
- Tooth Discoloration: 0.6869508624076843

Questionnaire Probabilities:
- Calculus: 0.3775406687981454,
- Caries: 0.9820137900379085,
- Gingivitis: 0.9820137900379085,
- Tooth Discoloration: 0.2689414213699951

Final probabilities:
- Calculus: 0.757049991024961,
- Caries: 0.7739477599342699,
- Gingivitis: 0.9752028935061607,
- Tooth Discoloration: 0.582448502148262

Implementation of Front/backend Integration

In the integrated workflow, the user first interacts with the front end by selecting or capturing an image of their teeth and completing the oral health questionnaire. Once the required inputs are collected, the front end packages the image together with the questionnaire answers into a multipart form request and sends it to the `/analyze` API endpoint hosted on the back end.

On the server side, the FastAPI back end receives the uploaded image and questionnaire data, goes through each component of the backend to obtain the results. The server then returns a structured JSON response containing the detected conditions, associated probabilities, summary text, and any warnings or failures related to image quality.

After receiving the response, the front end interprets the returned JSON data and displays the analysis results in a user-friendly format demonstrated in previous sections. This includes condition-specific probabilities, detected oral health concerns, and recommendations. The front end also stores scan history locally using AsyncStorage so that previous results can be reviewed later without requiring repeated submissions.

Testing of Front/backend Integration

Testing mainly focused on verifying that the backend could successfully receive requests and data from the frontend. Additional tests were done on the implementation of TLS 1.2 + using render to create HTTPS endpoints. Both aspects will be discussed in detail in Section 4.2. .AES-GCM will be added in later implementations for data at rest since it is irrelevant at the current stage with local database.

4.2 Results and Discussion

DO2 – Functional Design Test

Frontend/Backend Integration Testing

Functional testing of front end and backend communication. As shown in Figure 22-24, communication between the client and server is successful.

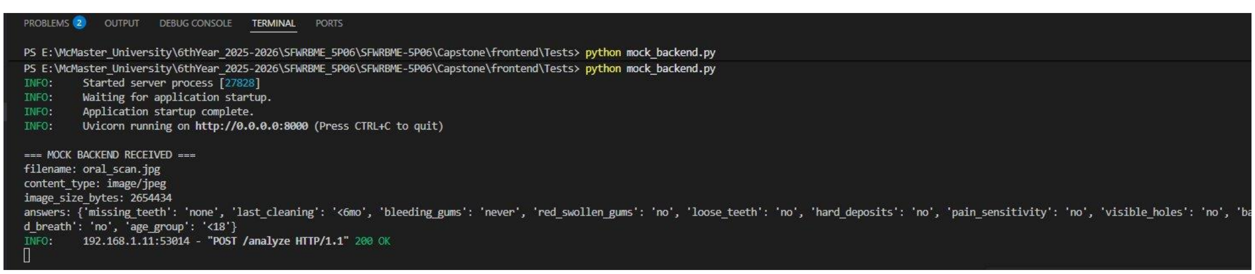


Figure 22. *Check if Data is received with mock_backend.py file, a fake backend that simulates responses without running the actual backend.*

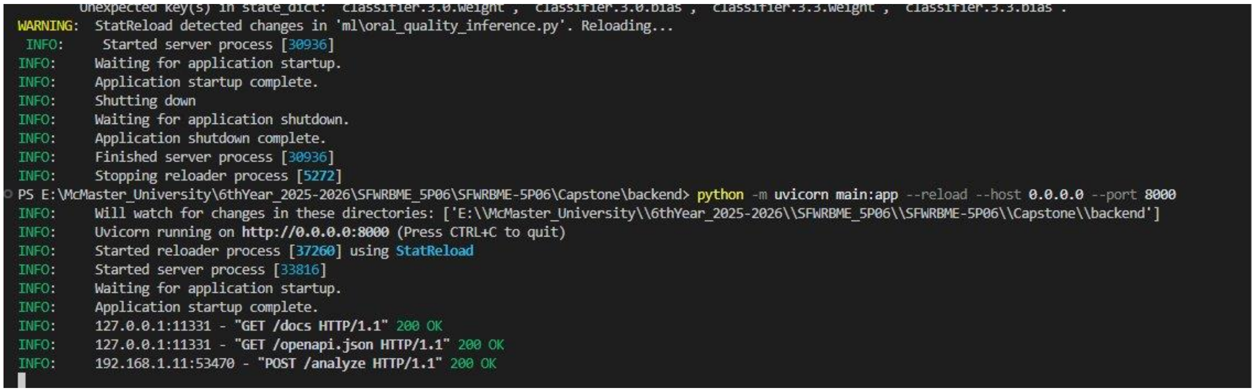


Figure 23*. Backend Created with HTTP using local IP. http://127.0.0.1:8000/docs showing that the data is received and analyzed by the ML*

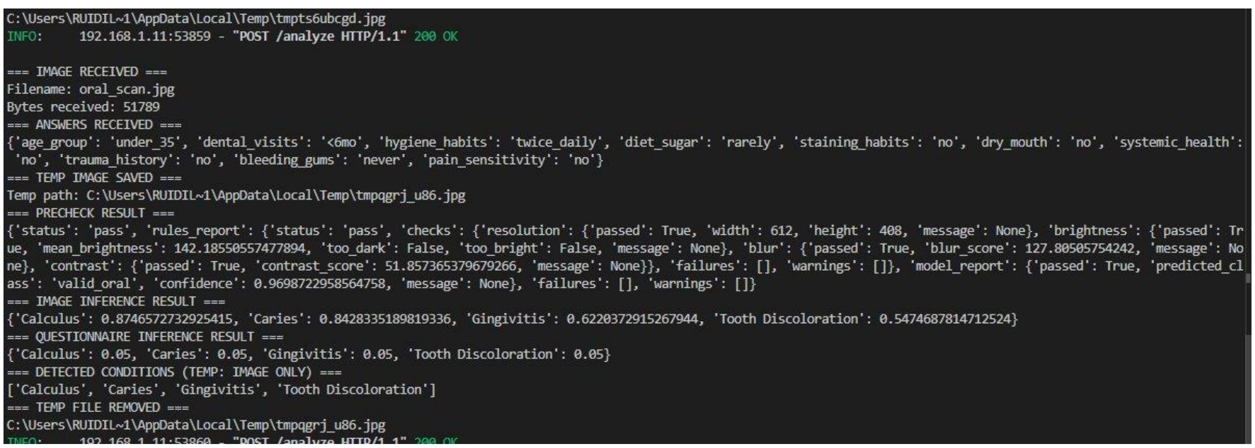


Figure 24. *Backend Created with HTTPS using local IP with http://127.0.0.1:8000/docs showing that the data is received and analyzed by the ML (old survey 1)*

Image Preprocessing and Quality Assessment

Theoretical Basis

The preprocessing module is based on classical image quality assessment techniques commonly used in computer vision.

- Blur detection using Laplacian variance is a well-established method for estimating image sharpness, where lower variance indicates reduced edge information and potential loss of features which is critical for image classification tasks [100].
- Brightness and contrast evaluation rely on statistical properties of grayscale intensity distributions, which directly affect feature extraction in CNNs. Contrast directly impacts the ability of CNNs to distinguish objects since poor contrast reduces the distinction between object features and background, limiting feature extraction capability [101].

The importance of high-quality input data is particularly critical in medical artificial intelligence applications. Poor image quality can significantly degrade model performance and lead to unreliable or misleading predictions. Many recent studies have demonstrated that variability in image quality, particularly noise and blur, can impact the accuracy and generalizability of deep learning models in medical imaging contexts [102]

Results and Discussion

The implemented preprocessing module serves as a critical quality control stage within the system by effectively identifying and filtering out images with critical quality deficiencies. Without this preprocessing stage, the model would still generate predictions for severely degraded or irrelevant images. However, such predictions are not dependable at all since the model would attempt to infer patterns from insufficient or misleading visual information. This could result in outputs that appear valid but are not clinically meaningful and even incorrect, which harms the user.

Example (without preprocessing step):

***Figure 20**. Unsuitable image for classification model, too dark, and lack of teeth and gumline for the classification model to perform inference on.*

Image Probabilities:

- Calculus: 0.725625816001836,
- Caries: 0.9241551739098323,
- Gingivitis: 0.9786548663102180,
- Tooth Discoloration: 0.9388574208460736

Overall, the preprocessing stage plays a crucial role in enhancing system robustness by mitigating the impact of low-quality inputs. This aligns with findings in the literature, which emphasize that ensuring high-quality input data is a fundamental requirement for achieving reliable performance.

<u>Disease Classification Model</u>

Theoretical Basis:

The disease classification model was implemented using a pretrained MobileNetV3 Small architecture, initialized with ImageNet weights. We chose MobileNetV3 architecture because it employs depthwise separable convolutions to reduce computational complexity while maintaining strong performance [103]. To further reduce computational cost and support efficient deployment, the convolutional feature extraction layers were frozen during training, and only the classifier head was optimized. The original classifier was replaced with a custom fully connected head consisting of a linear layer, ReLU activation, dropout regularization with a rate of 0.4, and a final linear output layer sized to the number of disease classes (4 in this case). This transfer learning approach allows the model to leverage pretrained visual feature representations while adapting the final decision layers to the dental image classification task [104].

The dataset was implemented using a custom multi-label wrapper. Images in the "healthy" class are assigned an all-zero target vector, whereas disease images are assigned a positive label only for the corresponding disease class, with remaining classes masked using a value of -1 (i.e. Caries is [-1, 1, -1, -1]). This masking strategy was paired with a custom masked binary cross-entropy loss, allowing the model to optimize only over valid target entries. As a result, the network was trained as a multi-label classifier without forcing all disease labels to be explicitly negative for every non-healthy sample. This approach saved so much time since partially labeled data from a larger number of datasets could be incorporated into the training set.

A sigmoid activation function is used for the multi-label classification, enabling independent probability estimation for each condition. Unlike SoftMax, which enforces mutual exclusivity, this approach allows multiple conditions to be detected simultaneously, which is particularly important in our applications where comorbidities are common [105]

To improve generalization, the training pipeline incorporated data augmentation, including resizing to 224 × 224 pixels, random rotation, brightness jitter, and random horizontal flipping, followed by normalization using standard ImageNet mean and standard deviation values. Training was conducted for 25 epochs using the Adam optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} these values were chosen after trying different combinations and produced the most stable results. Model selection was based on the best validation accuracy observed during training.

Results and Discussion

The implemented model produced stable classification performance across the training period and demonstrated strong discriminative ability across all disease classes. Because the CNN outputs raw logits during training and sigmoid-transformed probabilities during evaluation, each condition is assessed independently at inference time using a threshold of 0.5 (which can be customized to suit specific disease thresholds in the future if needed).

The performance of the disease classification model was evaluated using training and validation accuracy, training and validation loss, and class-wise Receiver Operating Characteristic (ROC) analysis. During training, both training and validation accuracy increased steadily over the 25 training epochs, indicating effective learning and convergence. The training accuracy reached approximately 0.96, while validation accuracy stabilized at approximately 0.93, suggesting good generalization performance.

Similarly, both training and validation loss decreased over time, with the validation loss remaining relatively close to the training loss throughout most of the training process. Although a small separation between the curves emerged in the later epochs, the gap was limited, indicating that overfitting was mild rather than severe (this is to be expected since our dataset was relatively small comprising of only 1436 images). The minimal overfitting suggests that the use of dropout, data augmentation, and transfer learning was indeed helpful in model regularization.

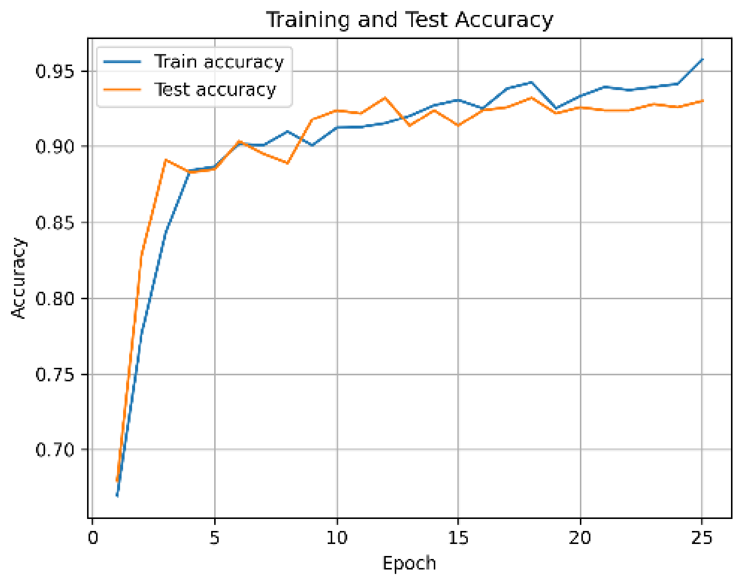


Figure 25. *Training and validation accuracy of the pretrained MobileNetV3 Small multi-label dental classification model over 25 epochs.*

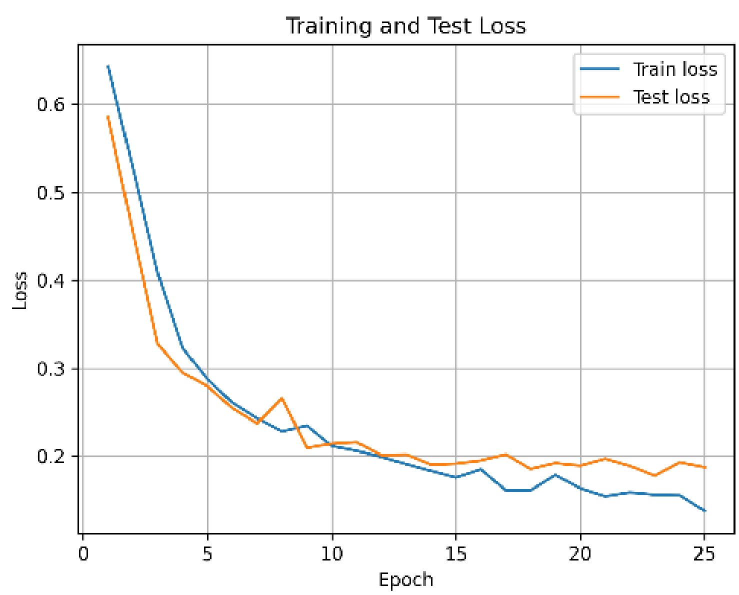


Figure 26. *Training and validation loss of the pretrained MobileNetV3 Small multi-label dental classification model over 25 epochs.*

Class-wise ROC analysis further demonstrated strong performance across all four dental conditions. The reported AUC values were 0.93 for calculus, 1.00 for caries, 0.98 for gingivitis, and 0.99 for tooth discoloration. These values indicate excellent discrimination between positive and negative samples for each condition, with particularly strong performance for caries and tooth discoloration since obvious visual characteristics are associated with these conditions. Overall, the quantitative results support the effectiveness of the selected model architecture and training strategy for multi-label dental screening.

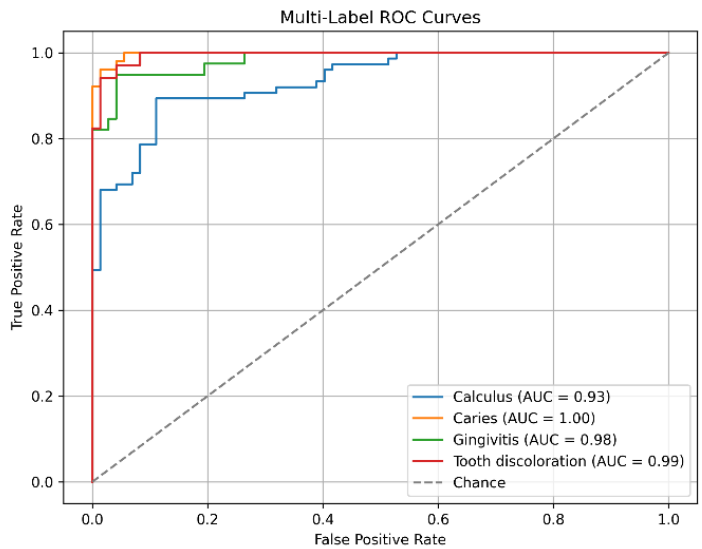


Figure 4.X: Class-wise ROC curves for the MobileNetV3 Small multi-label dental classification model, showing discrimination performance for calculus, caries, gingivitis, and tooth discoloration.

Questionnaire-Based Risk Scoring

Theoretical Basis

The questionnaire module was designed based on established clinical risk factors for common dental conditions and was validated through consultation with our stakeholder. This ensured that the selected questions align with diagnostic indicators used in practice in the real world.

In addition to stakeholder validation, the questionnaire design was informed by guidelines from recognized dental authorities, including the American Dental Association (ADA) and the American Academy of Periodontology (AAP). These sources provide evidence-based frameworks for assessing risk factors associated with dental diseases.

For dental caries, the questionnaire incorporates factors such as fluoride exposure, dietary habits, xerostomia (dry mouth), and pain symptoms. The ADA Caries Risk Assessment identifies frequent consumption of sugary foods and lack of fluoride as key contributors to high caries risk, while reduced salivary flow is known to impair natural remineralization processes [106] [107]. Advanced caries may also present as sensitivity or pain, reflecting deeper structural involvement of the tooth.

For gingivitis and periodontal disease, the questionnaire includes risk factors such as smoking, age, systemic health conditions, and bleeding gums. According to the AAP, tobacco use is a major dosedependent risk factor, while systemic conditions such as diabetes significantly increase susceptibility to periodontal inflammation [108]. Bleeding during brushing is also widely recognized as the primary clinical indicator of gingival disease.

For calculus formation, the questionnaire considers the frequency of professional dental cleanings. As calculus is mineralized plaque that cannot be removed through routine brushing, and extended intervals between dental visits are associated with increased accumulation.

For tooth discoloration, both extrinsic and intrinsic factors are captured. Dietary habits such as consumption of coffee, tea, or tobacco products contribute to surface staining, while dental trauma may lead to internal discoloration due to pulpal damage [109].

Results and Discussion

The questionnaire module provides clinically meaningful supplementary information that enhances the overall screening capability of the system. By incorporating symptom-based and behavioral risk factors, the system can account for conditions that may not be fully observable in an image.

For example, patient-reported symptoms such as pain during chewing or gum bleeding can increase the predicted likelihood of caries or gingivitis, even when visual indicators are subtle or absent. This reflects real clinical workflows, where diagnosis is based on both visual examination and patient history (patient-reported symptoms).

Fusion Strategy and System-Level Behavior

Theoretical Basis

The fusion component is based on a weighted probabilistic combination, a commonly used approach in multimodal systems. This method allows different data sources to contribute proportionally to the final prediction [110]. Such fusion strategies are widely used in medical AI systems to improve robustness and compensate for missing or uncertain data [111].

Results and Discussion

The fusion mechanism successfully balanced image-based and questionnaire-based predictions. Because the weighting scheme is fully customizable, flexibility enhances the clinical relevance of our application.

Backend Verification Results

The performance of the image-based classification subsystem was evaluated using standard statistical metrics derived from the confusion matrix, including sensitivity, specificity, precision, and F1-score. Additionally, Receiver Operating Characteristic (ROC) analysis was conducted to assess the model's discriminative ability.

The model demonstrated strong performance across all evaluated conditions, with Area Under the Curve (AUC) values exceeding 0.90 for all classes and reaching 1.00 for caries detection. These results satisfy the predefined acceptance criteria of AUC $\geqslant$ 0.85 from Chapter 3 and indicate good class separability. Overall, the results validate the effectiveness of the Image Analysis Model in identifying dental conditions from image data.

Although a formal controlled study was not conducted to verify the image capture protocol, the preprocessing module and image quality checks effectively served as a proxy for enforcing image capture standards. The system demonstrated consistent rejection of low-quality inputs, including blurry and underexposed images, reducing noise in downstream predictions.

The questionnaire decision-tree logic was validated through structured testing using simulated patient inputs. Predictions were compared against expected clinical diagnosis and stakeholder feedback. The system demonstrated consistent and interpretable behavior, correctly mapping symptom-based inputs to increased risk probabilities for relevant conditions. Both our stakeholder and background research confirms that the rule-based decision structure aligns with established clinical logic.

The fusion subsystem was evaluated by analyzing the combined outputs of the image analysis model and questionnaire module. The weighted fusion approach successfully integrated both sources of information, producing more context-aware predictions. In cases where image-based confidence was low or ambiguous, the questionnaire inputs were very meaningfully to the final prediction. This behavior supports the design objective of improving robustness through multimodal input integration. Although a formal statistical sensitivity analysis (i.e. DeLong test) was not conducted, stakeholder feedback indicate that the fused system provides more clinically relevant outputs than either subsystem alone.

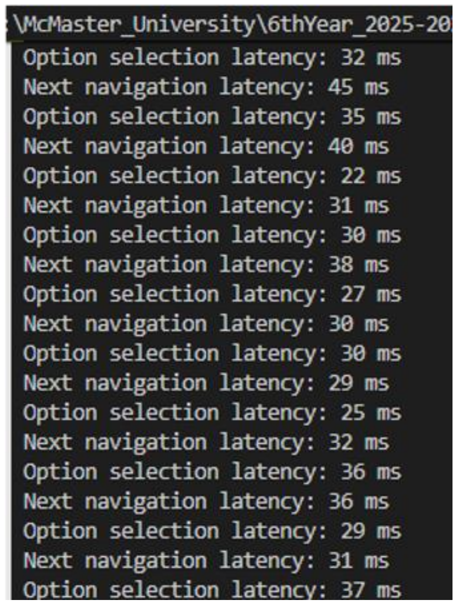
DO3 – User Interface (UI/UX) Design Outputs Frontend Results & Verification

The main components of DO3 are tested below:

Interaction Latency Test: ≥ **95% of UI interactions** respond within **≤100 ms**

Due to the time limitation, only the critical task flow has been tested which includes:

- Next & Option Navigation Latency (⩽100 ms)
- Backend Request Latency (⩽100 ms)
- Submit UI Latency (≥ 100ms, average around 4500 ms)



```
\McMaster_University\6thYear_2025-20
Option selection latency: 32 ms
Next navigation latency: 45 ms
Option selection latency: 35 ms
Next navigation latency: 40 ms
Option selection latency: 22 ms
Next navigation latency: 31 ms
Option selection latency: 30 ms
Next navigation latency: 38 ms
Option selection latency: 27 ms
Next navigation latency: 30 ms
Option selection latency: 30 ms
Next navigation latency: 29 ms
Option selection latency: 25 ms
Next navigation latency: 32 ms
Option selection latency: 36 ms
Next navigation latency: 36 ms
Option selection latency: 29 ms
Next navigation latency: 31 ms
Option selection latency: 37 ms
```

***Figure 27**. Next and Option Navigation Latency Test*

```
LOG  Option selection latency: 29 ms
LOG  Next navigation latency: 31 ms
LOG  Option selection latency: 37 ms
LOG  submitScan imageUri: file:///var/mobile/Containers/Data/Application/0F215607-3BD1-4D62-AA41-26578D1A4C41/Library/Caches/ExponentExperienceData/@anony
agePicker/727EE717-B83D-42EF-A3A5-DD345FC58110.jpg
LOG  submitScan answers: {"age_group": "35_to_55", "bleeding_gums": "sometimes", "dental_visits": "6-12mo", "diet_sugar": "sometimes", "dry_mouth": "yes",
: "mild", "staining_habits": "tobacco", "systemic_health": "yes", "trauma_history": "no"}
LOG  Sending request to: https://sfwrbme-5p06.onrender.com/analyze
LOG  Backend status: 200
LOG  Raw backend response: {"success":true,"summary":"Analysis completed successfully for oral_scan.jpg.","image_quality_passed":true,"image_quality_status
atus":"pass","rules_report":{"status":"pass","checks":{"resolution":{"passed":true,"width":612,"height":408,"message":null},"brightness":{"passed":true,"me
,"too_bright":false,"message":null},"blur":{"passed":true,"blur_score":98.93493397255953,"message":null},"contrast":{"passed":true,"contrast_score":38.0229
s":[]},"model_report":{"passed":true,"predicted_class":"valid_oral","confidence":0.9653937816619873,"message":null},"failures":[],"warnings":[]},"received_
6-12mo","hygiene_habits":"once_daily","diet_sugar":"sometimes","staining_habits":"tobacco","dry_mouth":"yes","systemic_health":"yes","trauma_history":"no",
ld"},"image_probabilities":{"Calculus":0.9900022745132446,"Caries":0.7880412936210632,"Gingivitis":0.9940217137336731,"Tooth Discoloration":0.5013675093656
775406687981454,"Caries":0.9820137900379085,"Gingivitis":0.9975273768433653,"Tooth Discoloration":0.6224593312018546},"probabilities":{"Tooth Discoloration
alculus":0.77564071251296,"Gingivitis":0.99542397897755},"detected_conditions":["Tooth Discoloration","Caries","Calculus","Gingivitis"]}
LOG  Backend request latency: 4525 ms
LOG  Total submit UI latency: 4639 ms
```

***Figure 28.** Backend Request latency and Submit UI Latency.*

The next steps would be to identify the bottle neck of the submit process, test the latency of each backend component. Other ways to improve, include ensuring backend is always warmed up and resize or compress image if possible.

<u>Frame Rendering Analysis</u>: ≥ **95% of frames rendered** at **≥60 fps**

As shown in Figure 21, the frames are rendered at 60 fps according to the native React Expo testing kit. This is aligned with the industrial standard for smooth mobile rendering.

Figure 29*. 60 UI Frame Rate tested with React Expo native testing kit.*

- Industrial standard for smooth mobile rendering

<u>WCAG 2.1 AA compliance</u>

After reviewing the design against WCAG 2.1 AA guidelines, most accessibility requirements were addressed within the current prototype. However, some accessibility components were not fully implemented due to time and resource constraints. Features such as color-blind modes and voice-based accessibility were not considered core functional requirements for the demo and were therefore deferred to future iterations. These enhancements will be prioritized in subsequent development phases to further improve inclusivity and overall accessibility.

DO4 – Data Schema & Encryption Verification

Testing of TLS

Render is used for TLS that could automatically provide security certificates. There are 3 methods to verify this:

- **Verification 1**: https://sfwrbme-5p06.onrender.com/docs and the security certificate
- **Verification 2**: Curl.exe a command-line tool that sends HTTP/HTTPS requests
- **Verification 3**: Logs on Render

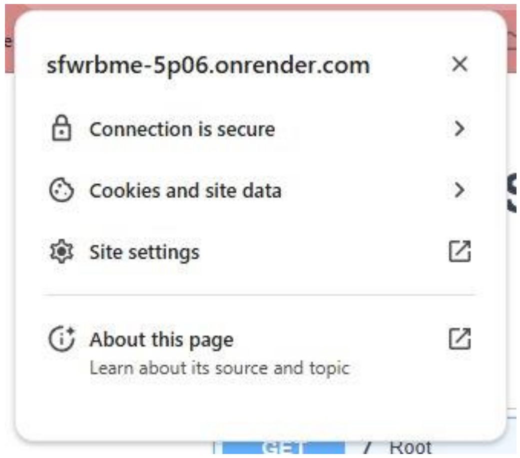
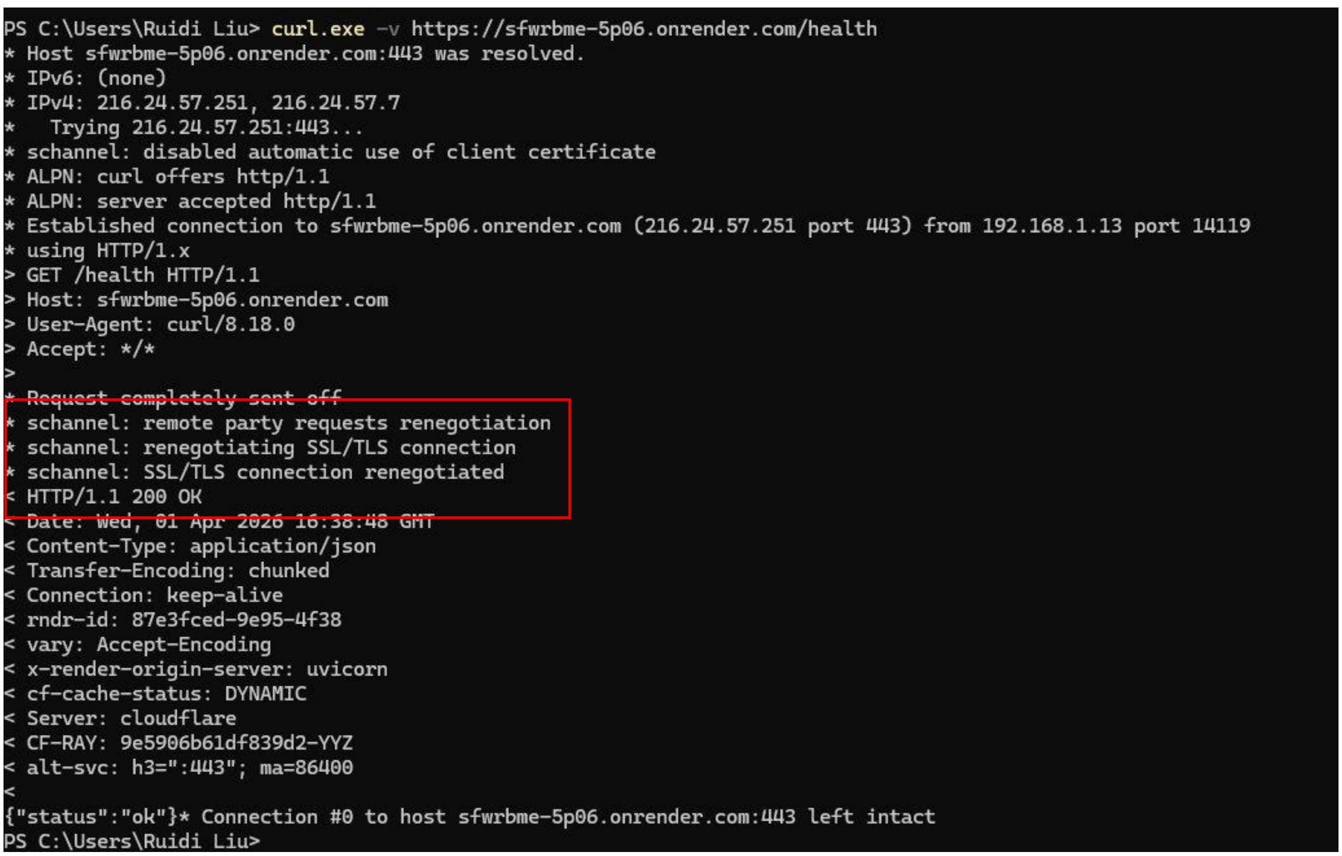


Figure 30. *Result of Verification 1: Security certificate provided by https://sfwrbme-5p06.onrender.com/.*



```
PS C:\Users\Ruidi Liu> curl.exe -v https://sfwrbme-5p06.onrender.com/health
* Host sfwrbme-5p06.onrender.com:443 was resolved.
* IPv6: (none)
* IPv4: 216.24.57.251, 216.24.57.7
*   Trying 216.24.57.251:443...
* schannel: disabled automatic use of client certificate
* ALPN: curl offers http/1.1
* ALPN: server accepted http/1.1
* Established connection to sfwrbme-5p06.onrender.com (216.24.57.251 port 443) from 192.168.1.13 port 14119
* using HTTP/1.x
> GET /health HTTP/1.1
> Host: sfwrbme-5p06.onrender.com
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
* schannel: remote party requests renegotiation
* schannel: renegotiating SSL/TLS connection
* schannel: SSL/TLS connection renegotiated
< HTTP/1.1 200 OK
< Date: Wed, 01 Apr 2026 16:38:48 GMT
< Content-Type: application/json
< Transfer-Encoding: chunked
< Connection: keep-alive
< rndr-id: 87e3fced-9e95-4f38
< vary: Accept-Encoding
< x-render-origin-server: uvicorn
< cf-cache-status: DYNAMIC
< Server: cloudflare
< CF-RAY: 9e5906b61df839d2-YYZ
< alt-svc: h3=":443"; ma=86400
<
{"status":"ok"}* Connection #0 to host sfwrbme-5p06.onrender.com:443 left intact
PS C:\Users\Ruidi Liu>
```

Figure 31. *Result of Verification Method 2 using curl.exe on command line.*

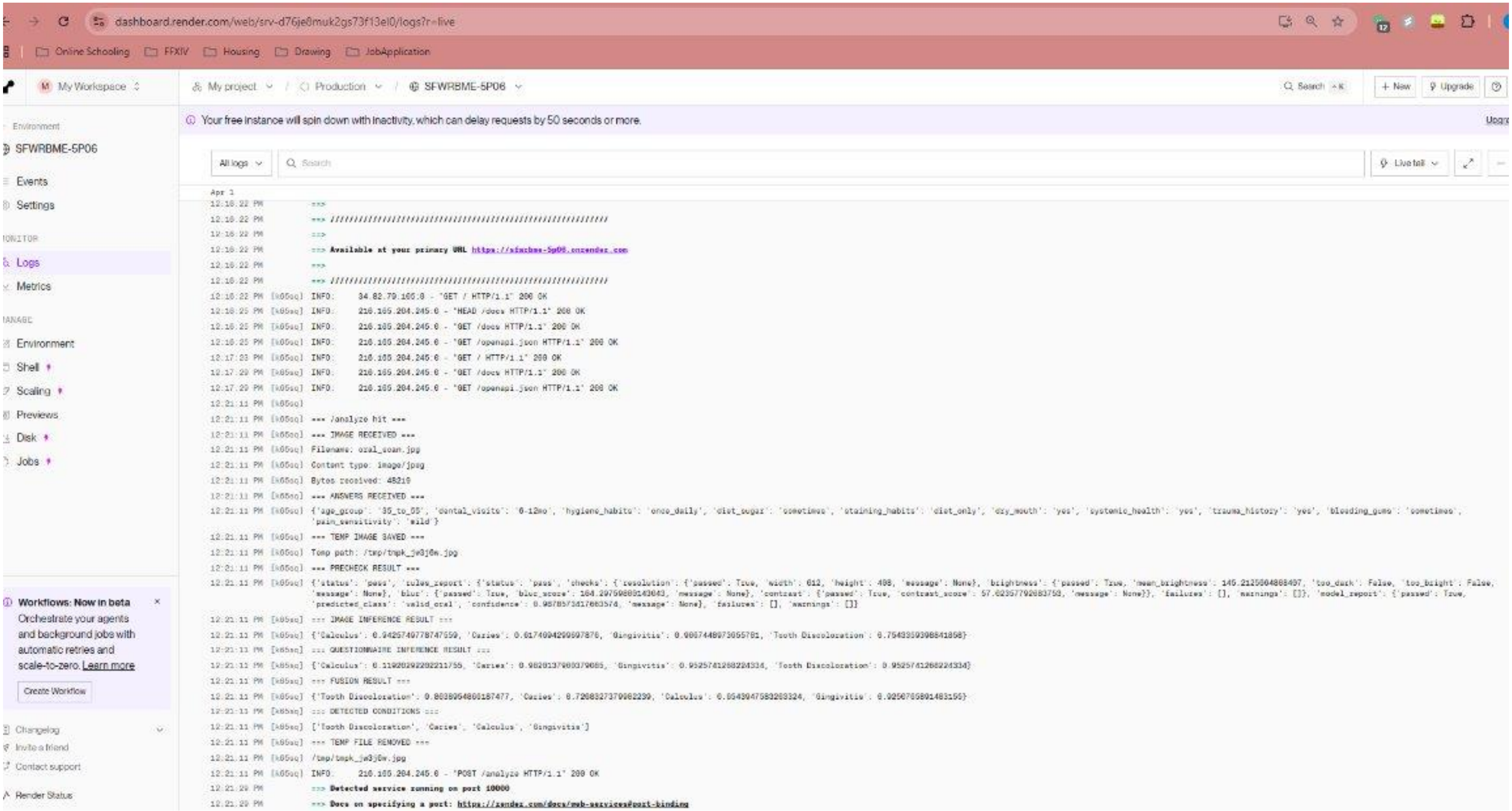


Figure 32. *Verification method 3checking logs on Render.*

Assumptions, Limitations and Constraints

Assumptions

Several assumptions were made in the design and evaluation of the SELFIE Application:

- **Image Content**
  It is assumed that user-submitted images contain sufficiently visible oral structures (i.e. teeth and gums) for meaningful analysis. While preprocessing filters out severely degraded inputs, it does not guarantee full clinical visibility of all regions. Especially since users are often limited to extraoral images that may not fully cover all the necessary tooth surfaces to provide a meaningful diagnosis.
- **User Compliance**
  The system assumes that users follow instructions for image capture (i.e. proper lighting, focus, and framing) and provide accurate questionnaire responses. Inaccurate or misleading inputs will reduce prediction reliability.
- **Dataset Representativeness:**
  The training dataset is assumed to be representative of real-world dental conditions. However, variations in demographics, imaging conditions, and disease presentation may not be fully captured since we did not collect our own data and only utilized the small number of relevant datasets able to be found online. We cannot guarantee if all subpopulations are covered.
- **Label Validity:**
  It is assumed that dataset labels are correct and reflect true clinical conditions. Any labeling inaccuracies may directly affect model performance.

Limitations

Despite achieving strong performance, the system has several inherent limitations:

- **Sensitivity to Image Quality:**
  Although preprocessing mitigates extreme cases, the model remains sensitive to variations in lighting, occlusion, and camera angle. Subtle degradations may still affect feature extraction and prediction accuracy.
- **Limited Clinical Validation:**
  The system has been evaluated using a curated dataset and simulated inputs rather than real-world clinical trials. As a result, its performance in clinical environments remains unverified.
- **Simplified Multi-label Representation:**
  The dataset structure assigns one primary disease per image, with other conditions masked. While the model supports multi-label outputs, it has not been trained on fully co-occurring disease cases, which may limit its ability to capture complex clinical scenarios.
- **Potential Algorithmic Bias:**
  If the dataset lacks diversity in terms of age, ethnicity, or imaging conditions, the model may exhibit biased performance across different populations.
- **Questionnaire Dependence:**
  The effectiveness of the questionnaire module relies on the user's honesty and understanding. Misinterpretation of questions or inaccurate self-reporting may introduce errors.

- **Threshold-Based Decision Making:**
  The use of fixed probability thresholds for classification may not be optimal for all conditions.

Constraints

The system design is subject to several practical constraints:

- **Data Availability Constraints:**
  The performance of the model is constrained by the size and quality of the available dataset. Limited labeled medical data restricts the ability to train more complex models or fully capture disease variability.
- **Deployment Constraints:**
  The system is designed as a screening tool rather than a diagnostic device, due to regulatory and ethical considerations. This restricts how predictions can be interpreted and communicated to users.
- **User Interaction Constraints:**
  The system depends on non-expert users for data input, which introduces variability that cannot be fully controlled, unlike clinical imaging environments.

4.3 Future Work

Development of an End-to-End Multimodal Learning Framework

A major next step for the backend system would be developing an end-to-end multimodal learning framework that learns from both the oral images and questionnaire responses, instead of combining them only at the final decision stage. The current prototype uses a late-fusion strategy in which image-based predictions and questionnaire-derived probabilities are weighted and merged after independent processing. While this approach is modular and interpretable, it does not allow the model to learn deeper interactions between visual and non-visual clinical features, which is quite important for oral health disease.

This extension would involve designing a unified neural architecture with two input branches: an image-processing branch based on a convolutional backbone and a structured-data branch for questionnaire features. These branches would then be fused at an intermediate or feature level, allowing the system to learn relationships between reported symptoms and image patterns during training [112]. Literature on multimodal medical AI has shown that combining image and non-image information can improve predictive performance, robustness, and clinical relevance by capturing complementary sources of information [113]. Multimodal fusion seems to be one of the major directions in medical AI because many clinical decisions depend on both imaging and patient context rather than either source alone [114].

For this project, such a system would be particularly valuable because most dental conditions are not always fully visible in a single photograph. For example, pain during chewing, xerostomia, or bleeding gums may provide important clinical evidence even when image-based features are subtle. In other cases, multiple photographs may be necessary to fully capture the user's oral health condition. An end-to-end multimodal model could learn these interactions directly rather than relying on manually specified fusion weights. This extension would involve redesigning the data pipeline, restructuring labels and inputs, retraining with paired multimodal samples, and systematic comparison of early, intermediate, and late-fusion strategies [115].

Evidence from literature suggests that moving from simple decision fusion toward learned multimodal representations is a meaningful technical progression for the SELFIE application.

Explainable AI and Increased Interpretability

Another major extension for the current backend system would be integrating explainable artificial intelligence (XAI) techniques to provide justifications for model predictions. Currently, the model outputs probabilities for each condition but it does not provide insight into why a particular prediction was made. This lack of interpretability can limit clinical trust and usability, particularly in healthcare settings where decision transparency is critical [116]. Similar feedback was also received during the UI/UX testing phase.

For our proof-of-concept, visualization methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) were explored to highlight salient regions of the input images to identify dental calculus. However, the current multi-label classification task made implementation of any similar techniques much more difficult. Unlike binary classification tasks, where Grad-CAM can be applied directly to a dominant class, multi-label models require generating separate activation maps for each condition [117]. And due to the lack of time, explainability features were not implemented in the final prototype.

Future work could focus on developing class-specific activation maps for each predicted condition and evaluating whether these visual explanations align with clinically relevant regions. This would allow for better assessment of the model to make sure it focuses on clinically relevant regions (i.e. tooth surfaces and gum lines) rather than irrelevant background features/noise. Additionally, visualization tools could be integrated into the user interface to provide interpretable feedback alongside the predictions for non-expert users. For example, disease locations could be highlighted on the input image to provide users with more information about their oral health.

For this project, integrating explainability would require modifications to the inference pipeline, implementation of the visualization module, and further evaluation of whether highlighted regions align with known clinical features. This extension would significantly enhance the clinical applicability of the system by improving transparency, supporting model validation, and increasing user trust.

Frontend UI Feedback Integration

There were additional suggestions and feedback collected from the surveys, group interviews and users from Capstone expo that have not yet been implemented. Two major pain points discussed by the users include:

1. iOS Photo Library Permission Behavior + Cropping Function

Currently the photo library of both iOS and android behave differently. iOS displays a system level prompt when users select an image from the photo library as shown in Figure X.

iOS requests access for the library and displays two buttons “Select More Photos” and “Keep Current Selection” during the user’s first scan. This occurs due to iOS limited photo library access permissions which is an OS-level behavior outside application control. Capstone Expo users stated that they would like to let the app request the access of camera and album before the scan instead of during the scan.

In contrast, Android allows users to select images directly from the photo library while also enforcing image cropping before submission. Based on feedback collected during the Capstone Expo, users preferred the Android-style cropping functionality. This preference was reinforced during live demonstrations, where poor overhead lighting significantly affected model performance. Allowing users to crop the image and focus specifically on the mouth region improved input quality and increased the model's accuracy when analyzing oral images captured under suboptimal lighting conditions.

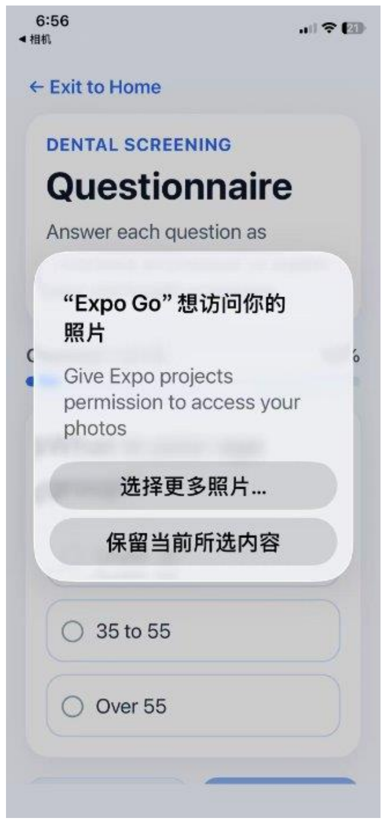


Figure 33. *iOS request access prompt.*

For the next iteration of the design, the implementation of image cropping prior to submission and creating native configuration options within Expo are considered key integrations. This enhancement will allow users to focus on the scan without disturbance, specifically on the mouth region, reducing background noise and improving the consistency of input images. Additionally, guided mouth-region framing will be introduced to assist users in capturing properly aligned images. Although we already have a preprocessing pipeline for lighting normalization, in the future the pipeline should be able to adjust the lighting metrics accordingly to the environment the user is in. Together, these improvements aim to enhance model robustness and improve performance under real-world conditions.

2. Survey being very repetitive

Both survey participants and Capstone Expo attendees noted that the questionnaire was repetitive. During the Expo, Dr. Fang suggested shortening the survey to include only key questions relevant to each scan. Additional information such as diet, age, and medical history should be collected once and stored within the user profile, rather than being repeatedly requested for every submission. This approach would streamline the user experience, reduce survey fatigue, and improve efficiency while still preserving important contextual information for analysis.

In future iterations, support for user profile creation should be implemented to allow users to store personal and health-related information for reuse across scans. A quick one-time survey will appear after the user's registration, asking personal factors that may affect the triage scoring and store it on the user's profile.

4.4 Reflections and Recommendations

Reflection on Design Process

Throughout the design and implementation of this project, several key successes, challenges, and lessons were identified. The initial project proposal went fine and the modular system design we chose enabled efficient development and integration of preprocessing, classification, and fusion components. However, a significant setback occurred during dataset preparation due to inconsistencies in labeling. The intended label order [caries, calculus, gingivitis, tooth discoloration] was not consistently followed across stakeholders, with one stakeholder inadvertently reversing the order of the first two classes. Unfortunately, this discrepancy was not detected prior to dataset aggregation, resulting in mislabeled data within the combined dataset. Given that supervised machine learning models are super sensitive to label accuracy, this issue posed a serious risk to model performance (very low accuracy for these two conditions), and a lot of time and effort were put into troubleshooting the problem. At the time of discovery, the dataset had already been expanded to approximately 500 images, making manual relabeling impractical within time constraints. To address this, a partial-label training approach was adopted, where uncertain labels were masked and excluded from loss computation, allowing the model to learn only from verified data. This approach also allowed us to increase the data set to include more external images that were labeled with only a single condition, improving overall data reliability. This experience highlighted the critical importance of data integrity, early validation of assumptions, and clear communication when working with multiple stakeholders. It also demonstrated the value of designing flexible systems capable of handling imperfect data.

The final Gantt chart reflects several adjustments compared to the initial project planning in Chapters 1–3. The most significant change was the development time for a single iteration. The iteration process was much longer, often spending a week and half designing, improve, and integrate feedback on a new iteration. In Chapter 1-3, these iterations are usually less than a week. It is likely due to additional debugging and testing communication between the mobile application and backend server. They require a lot more development time than anticipated, especially when users are actively providing feedback when working on a new iteration.

Furthermore, iterative user feedback gathered during testing and group interviews led to additional UI refinements, including visualization and user experience improvements, questionnaire simplification, and scan workflow adjustments. These changes were addressed by reallocating time from other prepared time slots such as accessibility development, comfort adjustments, presentation preparation and report writing.

As a result, some planned features such as color blind modes and cloud storage are moved to future iterations. Despite these deviations, core milestones including prototype integration, testing, and demo preparation were completed within the revised timeline, ensuring delivery of a fully functional high fidelity prototype.

Individual Reflection

Meet more frequently with instructors. Even when progress seems ok, early feedback can help identify and prevent critical issues later in the project. — Jasmine

I would like to know the Capstone demo setup, including table size and lighting, beforehand to reduce rushing and allow sufficient time for testing. — Ruidi

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Chapter 5 Conclusion

The Selfie AI Dental Assistant is a mobile-based oral health screening application that combines image analysis with questionnaire-based inputs to provide users with educational insights about their dental health. The primary objective of this project was to develop a non-diagnostic, educational tool capable of identifying potential oral conditions using smartphone cameras. The system uses a client-server architecture, where the React Native mobile front end communicates with a python-based backend via FastAPI. The backend is deployed on Render and is responsible for image validation, machine learning inference, and risk probability fusion.

The front end provides a cross-platform interface for capturing oral images, completing questionnaires and displaying results. Additional features include tracking patient history, personalized recommendations and educational resources. The interface was designed to be intuitive and accessible, with four condition-specific probability outputs and personal recommendations.

The back end receives uploaded images and questionnaire responses through a REST-based /analyze endpoint where images undergo preprocessing to verify sufficient brightness, contrast, and resolution. Then, validated images are processed through a CNN to generate probability estimates for dental calculus, caries, gingivitis, and tooth discoloration. These predictions are combined with questionnaire derived risk factors using a fusion weighted approach. The system returns the final results, with their corresponding recommendations, to the user interface.

System integration was successfully achieved through a modular pipeline separating frontend and backend responsibilities. This architecture enhances scalability, maintainability, and flexibility, allowing individual components to be modified or improved independently. The backend implementation demonstrated stable performance, with strong classification accuracy and high AUC values across all conditions, indicating effective discrimination between disease states.

User testing was conducted through surveys, group interviews, and live demonstrations at the Capstone Expo. Users enjoyed the intuitive task flow and particularly appreciated the visual presentation of results. However, testing also revealed key limitations. Model performance was highly sensitive to image quality, particularly under inconsistent lighting and improper framing. Users also suggested reducing the number of questionnaire items to focus on the most clinically relevant factors. Furthermore, the current late-fusion strategy limits the system's ability to capture complex interactions between image-based features and questionnaire inputs.

Based on these findings, several directions for future work are identified. A major backend extension involves developing an end-to-end multimodal learning framework that jointly learns from oral images and questionnaire responses. A unified neural architecture fusing image and structured inputs early on would potentially improving predictive performance and robustness. Another important future consideration is the integration of explainable artificial intelligence techniques. The current model outputs probability scores but does not provide insight into how predictions are generated. Implementing visualization techniques such as class-specific activation maps would allow the system to highlight clinically relevant regions of the input image to enhance interpretability, support model validation, and increase user trust in the system.

Frontend usability improvements were also identified through user feedback. Image selection workflows created confusion during testing, particularly due to differences in cropping functionality that seem to be an android only feature. Future iterations will integrate mouth-region guided framing and improved preprocessing metrics to normalize lighting conditions. These enhancements are expected to improve input consistency and overall model performance in real-world conditions.

Despite the successful implementation, several limitations remain. The dataset, consisting of approximately 14,000 images sourced from publicly available data, is constrained by variability and potential labeling inconsistencies. Additionally, privacy regulations in Canada limit access to large-scale clinical datasets, restricting further model training and validation. As suggested by Dr. Fang during the Capstone Expo, expanding the dataset through partnerships with clinics, hospitals, or professional organizations would significantly improve model generalizability and clinical relevance.

In conclusion, the Selfie AI Dental Assistant successfully demonstrates the feasibility of a hybrid mobile application that combines deep learning-based image analysis with questionnaire-driven risk assessment. The system provides an accessible, low-cost platform for preliminary oral health screening and user education. While its not intended for clinical diagnosis, the application represents a meaningful step toward improving awareness and accessibility in dental health. With continued development in multimodal learning, explainability, data expansion, and user experience design, the system has strong potential to evolve into a more robust, scalable, and clinically impactful tool.

Appendix A Tables

Table A1: Design Output Category

Key Failure Mode Index and Full Title	Risk Priority Number
FM1 — False Reassurance / False Negative Triage	**252**
FM2 — Over-Triage / False Positive Triage	100
FM3— Improper Use / Incorrect Image Capture	**175**
FM4 — Skipped or Inconsistent Questionnaire Responses	**140**
FM5 — App Crash / Freeze / Unresponsive Behaviour	72
FM6 — Server Downtime or Slow Backend Response	72
FM7 — Algorithmic Bias / Unequal Performance Across Subgroups	**144**
FM8 — Data Breach / Unauthorized Access to Health Data	**162**
FM9 — Misuse or Unclear Ownership of Personal Data	120
FM10 — Regulatory Non-Compliance	64
FM11 — Over-Reliance on the App / Misunderstanding Scope of Use	**140**

Table A2: Key Failure Mode Names and Risk Priority Number

Design Output	Categories
DO1 – Software Architecture Document	Technical, Societal
DO2 – Functional Design Outputs	Technical, Economic, Ergonomic, Societal
DO3 – User Interface (UI/UX) Design Outputs	Technical, Ergonomic, Societal
DO4 – Data Schema & Encryption Documentation	Technical, Environmental, Societal
DO5 – Privacy, Consent, and Access Control Documentation	Technical, Ergonomic, Societal
DO6 – Risk Management File	Technical, Societal
DO7 – Verification & Validation Plan Output	Technical, Societal

Table A3: Final FMEA Table for Selfie Dental Assistant Application

The FMEA Table is too large to be put in Word. An excel object is embedded below and the link for the shared excel file is linked to the Table name.

FMEA Spreadsheet Template

System / Product / Process	FMEA ID	Revision	Date
Selfie Dental Assistant Appication			Dec.1st 2025
Subsystem / Product / Process (if applicable)	Party Responsible	Approved By	Date Completed
	Team 9		
Team			
Team 9 Ruidi Liu & Jasmine Wang - Current Version 1.0			

Failure Mode and Effects Analysis (FMEA)

ID	System / Item / Process Step	Subsystems	Function	Potential Failure Mode	Effects of Failure	Severity	Causes	Occurrence
#	Name, ID number, etc.	What subsystem does this Failure Mode belong to?	Primary function	How could the system / item / process potentially fail?	Consequential impact on other systems, departments, etc.		All contributing factors	
1	**FM1**: False reassurance / false negative triage	Business Logic / Domain Layer	Perform accurate risk scoring and triage decision-making by combining questionnaire responses, image-analysis model, and 'red-flag' rules.	Serious condition classified as low risk or told user to monitor at home	Delayed care; progression of disease; increased pain; potential rise of infection and/or complications	9	Incomplete or inaccurate symptoms; poor fusion of questionnaire+image; miscalibrated thresholds; triage logic bug; skipping questionaire	4
2	**FM2**: Over-triage / false positive triage	ML & Image Services Layer	Process user images through the image pipeline (blur check → exposure check → crop/resize/normalize) and perform ML inference using MobileNet/EfficientNet-lite to identify potential dental issues.	Benign findings classified as high risk / urgent	Unnecessary anxiety and appointments; increased system burden	5	Lighting artefacts; reflective surfaces; overly conservative thresholds	5
3	**FM3**: Improper use / incorrect image capture	Presentation Layer	Support users in capturing high-quality intraoral photographs using guided overlays, instructions, and UI prompts that enforce correct image framing.	User takes poor-quality or incorrect images	Inaccurate ML features; potential false reassurance or false alarm	7	User confusion; low health literacy; ignoring overlay guides; rushing; Camera broken or has flaws	5
4	**FM4**: Skipped or inconsistent questionnaire responses	Presentation Business Logic Layers	Collect accurate and complete symptom and history data from the user and pass the structured responses to the risk-scoring engine.	Missing or contradictory symptom data used for risk scoring	Inaccurate risk assessment; potential under- or over-triage	7	Users rush; misunderstand questions; accidental taps; no logical checks	4
5	**FM5**: App crash / freeze / unresponsive behaviour	Infrastructure Layer	Provide a stable, responsive runtime environment for the mobile app, including UI rendering, ML model loading, memory management, and background processes.	App stops responding during assessment	Interrupted assessment; user frustration; possible abandonment and delayed care	6	OS/SDK incompatibility; memory limits; unhandled exceptions; heavy model on low-end device	4
6	**FM6**: Server downtime / slow backend response	Cloud / Server Layer	Enable secure data synchronization, storage, model updates, and user account services through a reliable backend accessible via encrypted APIs.	Backend unavailable or very slow	Images not uploaded; results delayed or absent; incomplete assessments	6	Network outages; server misconfiguration; overload; expired certificates	3
7	**FM7**: Algorithmic bias / unequal performance across subgroups	ML & Image Services Layer + Cloud	Maintain and update the machine-learning models, ensuring that inference pipelines remain accurate, fair, and up-to-date across user populations and devices.	Systematically worse performance for some demographic or device groups	Inequitable care; higher false negatives or positives for certain users	8	Non-representative training data; biased labels; untested subgroup performance	3
8	**FM8**: Data breach / unauthorized access to health data	Data Layer (Local) + Cloud / Server Layer	Store and protect personal health information (PHI) — including images, questionnaire responses, and consent settings — using encryption, access controls, and secure storage mechanisms.	PHI or images accessed by unauthorized party	Privacy violations; possible identity risk; regulatory penalties; loss of trust	9	Weak encryption; insecure storage; misconfigured cloud; inadequate access control	3
9	**FM9**: Misuse or unclear ownership of personal data	Data Layer + Regulatory & Compliance Process	Ensure proper collection, use, retention, and ownership of user data, including maintaining consent records and applying purpose-based data governance rules.	Data used for ML or shared without meaningful consent	Ethical violations; user distrust; legal / regulatory risk	8	Ambiguous consent text; broad default permissions; long retention	3
10	**FM10**: Regulatory non-compliance	System-wide / Regulatory & Compliance Process	Maintain compliance with regulatory standards and codes (CDA/CDHA, PIPEDA, PHIPA, CASL, SaMD, cybersecurity guidelines) through documentation, quality management, and controlled updates.	Misalignment with CDA/CDHA, SaMD, privacy, CASL, or cybersecurity guidance	Legal penalties; forced product withdrawal; reputational damage	8	Incomplete regulatory analysis; missing documentation; ad-hoc updates	2
11	**FM11**: Over-reliance on app / misunderstanding scope	Presentation Layer (results & education pages)	Communicate risk results, explanations, and educational guidance in clear, non-diagnostic language to support user understanding and promote appropriate follow-up care.	User treats app result as definitive diagnosis and delays seeing dentist	Delay in needed care; worsening of conditions	7	Weak or hidden disclaimers; users over-trust AI; limited health literacy	4
13								
14								
		**Regulatory & Compliance Process means it is outside of the software modules, they are organizational responsibilities						

RPN Columns contain a formula to auto-calculate; do not alter or delete.

ID	Current Controls: Prevention	Current Controls: Detection	Detection	RPN	Recommended Actions	Owner	Date Due	FMEA Results: Action Results	Date Completed	Severity	Occurrence	Detection	RPN
#	Prevention	Detection		Risk Priority Number	Steps required to reduce severity, occurrence, and detection	Organization, team, or individual responsible	Target date of completion	Actions taken	actual date of completion				Risk Priority Number
1	Red-flag override rules; mandatory key questions that doesn't allow for skipping; conservative high-sensitivity thresholds; unit tests on triage logic; 'Disease' has higher weight than healthy; Provide suggestion to see a doctor no matter what the results are	Clinical validation against labelled cases; monitoring of triage distributions after updates;	7	252	Develop and implement a 'drift' and model detection; Evaluate triage output trends; Flags when accuracy drops; Expand data set	Jasmine	Mid-Feburaray to March						0
2	Pre-processing (blur/exposure/normalisation); conservative wording of results; non-emergency framing	Review of confusion matrix in validation and pilot; monitor high-risk rate vs expectations;	4	100	Iterative testing & adjust treshold if needed; Expand data set	Jasmine	March						0
3	Step-by-step camera instructions; overlay guidelines; enforced blur/exposure checks; plain-language prompts	App-level quality scores; analytics on failed image uploads or repeated retakes	5	175	Perform Stakeholder Interview; Gather client feedback and update overall guidelines; Filter and check images before passing to model	Ruidi	Mid-Feburaray						0
4	Mandatory responses for high-impact items; confirmation for extreme values; simple consistency checks	Backend analytics on missing/changed responses; flags when internal checks trigger	5	140	Allow user to read through responses and confirm their answers; allow users to have a method to re-answer questions; Set all questions as mandatory;	Ruidi	Mid-Feubrary to March						0
5	Autosave; guarded model loading; try/except error handling; device/OS support matrix;data sync enable	Crash analytics and logs from pilot; user feedback reports	3	72	Optimize ML model using quantization and model-size reduction; expand device compatibility testing;badd automated crash reporting and analytics; ensure questionnaires auto-save every step	Both members	April						0
6	Graceful error messages; retry with backoff; symptom-only on-device fallback	Server uptime monitoring; API latency dashboards	4	72	Provide fully-offline questionair-based fallback if server is unavailable; Perforam load testing to identify bottle necks; Stagger model updates	Both members	March						0
7	Diverse training data; track metrics by subgroup; clinician oversight on high-risk cases	Periodic bias audits on validation and pilot data; monitoring of subgroup confusion matrices	6	144	Perform periodic bias audits; Expand dataset with intentionally diverse images (skin tone, age, camera types); Add fairness evaluation metrics to acceptance criteria before model release; Tune thresholds separately for groups if systematic patterns appear;Include clinician during pilot phase.	Jasmine	April						0
8	Encryption at rest and in transit; least-privilege access; secure OS storage; audits/logging; data minimisation	Security monitoring; log review; intrusion/breach detection alerts	6	162	Strengthen encryption (AES-256 for local storage; enforced TLS 1.2+); Introduce audit logs reviewing all access to PHI.	Ruidi	Late March						0
9	Explicit consent flows; separate opt-in for research/training; documented retention limits; transparent policy	Regular policy reviews; internal audits of data use vs consent records	5	120	Rewrite consent screen in plain language with explicit opt-in for data use in ML;add an in-app privacy dashboard or note showing what data is stored and why;implement data deletion on user request.	Both members	Feburary						0
10	Map requirements into design inputs; maintain risk management file; follow secure development guidance	Internal compliance review; external consultation if needed	4	64	Prepare SaMD-aligned risk management file (ISO 14971 template); Maintain design history and software lifecycle documentation);Review data-flows to maintain compliance with PIPEDA/PHIPA.	Both members	April						0
11	Prominent 'not a diagnosis' messaging; routine check-up recommendation even for low risk; clear escalation advice	User feedback; qualitative interviews in pilot; tracking recurrent high-risk follow-ups	5	140	Present "This is not a diagnosis" clearly and repeatedly in onboarding and results screen;add automatic follow-up >48 hours;include educational messages promoting regular dental exams even for "low risk." ; Provide clear instructions for when to seek urgent dental care.	Ruidi	Feburary						0
13				0									0
14				0									0

Table A4: Final Gantt Chart

The Gantt Chart Table is too large to be put in Word. An excel object is embedded below and the link for the shared excel file is linked to the Table name.

APPENDIX B Questionnaire & Weights

Questionnaire & Weights (Version 1)

```
# Q1: Missing teeth
if q1 == "none":
      scores["Missing teeth"] += 0.0
elif q1 == "1-2":
      scores["Missing teeth"] += 3.0
    scores["Periodontitis"] += 1.0
elif q1 == "3+":
      scores["Missing teeth"] += 4.0
    scores["Periodontitis"] += 1.5
# Q2: Last cleaning
if q2 == "<6mo":
      pass
elif q2 == "6-12mo":
      scores["Calculus"] += 1.0
      scores["Gingivitis"] += 0.5
      scores["Cavities"] += 0.5
elif q2 == ">1yr":
      scores["Calculus"] += 2.0
      scores["Gingivitis"] += 1.0
    scores["Periodontitis"] += 1.0
      scores["Cavities"] += 1.0
elif q2 == "never":
      scores["Calculus"] += 2.5
      scores["Gingivitis"] += 1.5
    scores["Periodontitis"] += 1.5  scores["Cavities"] += 1.5

# Q3: Bleeding gums
if q3 == "sometimes":
      scores["Gingivitis"] += 2.0
    scores["Periodontitis"] += 0.5
elif q3 == "often":
      scores["Gingivitis"] += 3.0
    scores["Periodontitis"] += 1.5

# Q4: Red/swollen gums
if q4 == "mild":
      scores["Gingivitis"] += 1.5
elif q4 == "severe":
      scores["Gingivitis"] += 2.5
    scores["Periodontitis"] += 1.0

# Q5: Loose teeth
if q5 == "mild":
    scores["Periodontitis"] += 3.0
```

```
elif q5 == "obvious":
    scores["Periodontitis"] += 4.0

# Q6: Hard deposits
if q6 == "yes":
      scores["Calculus"] += 4.0
      scores["Gingivitis"] += 1.0
    scores["Periodontitis"] += 1.0

# Q7: Pain / sensitivity
if q7 == "cold_sweet":
      scores["Cavities"] += 2.5
elif q7 == "chewing_or_constant":
      scores["Cavities"] += 3.5
    scores["Periodontitis"] += 1.0

# Q8: Visible holes/spots
if q8 == "yes":
      scores["Cavities"] += 4.0

# Q9: Persistent bad breath
if q9 == "sometimes":
      scores["Gingivitis"] += 1.0
    scores["Periodontitis"] += 1.0
      scores["Calculus"] += 0.5
elif q9 == "often":
      scores["Gingivitis"] += 2.0
    scores["Periodontitis"] += 2.0
      scores["Calculus"] += 1.0

# Q10: Age group
if q10 == "36-55":
    scores["Periodontitis"] += 0.5
elif q10 == ">55":
scores["Periodontitis"] += 1.0
```

Questionnaire & Weights (Version 2)

```
# 1. Age (Periodontal disease risk heavily increases with age)
if ans["age_group"] == "35_to_55":
    scores["Gingivitis"] += 0.5
elif ans["age_group"] == "over_55":
    scores["Gingivitis"] += 1.5

# 2. Dental Visits (Lack of scaling builds Calculus, increases
Gingivitis/Caries risk)
if ans["dental_visits"] == "6-12mo":
```

```
    scores["Calculus"] += 1.0;
    scores["Gingivitis"] += 0.5;
    scores["Caries"] += 0.5
elif ans["dental_visits"] == ">1yr":
    scores["Calculus"] += 2.5;
    scores["Gingivitis"] += 1.5;
    scores["Caries"] += 1.0
elif ans["dental_visits"] == "never":
    scores["Calculus"] += 4.0;
    scores["Gingivitis"] += 2.5;
    scores["Caries"] += 1.5

# 3. Hygiene & Fluoride (Lack of fluoride is a primary Caries risk)
if ans["hygiene_habits"] == "once_daily":
    scores["Calculus"] += 1.0;
    scores["Gingivitis"] += 1.0;
    scores["Caries"] += 1.0
elif ans["hygiene_habits"] == "rarely":
    scores["Calculus"] += 2.5;
    scores["Gingivitis"] += 2.5;
    scores["Caries"] += 3.0

# 4. Diet/Sugar (Frequent sugar is a primary Caries risk)
if ans["diet_sugar"] == "sometimes":
    scores["Caries"] += 2.0
elif ans["diet_sugar"] == "frequently":
    scores["Caries"] += 4.0

# 5. Staining & Tobacco (Tobacco heavily influences both Discoloration and
Periodontal issues)
if ans["staining_habits"] == "diet_only":
    scores["Tooth Discoloration"] += 3.0
elif ans["staining_habits"] == "tobacco":
    scores["Tooth Discoloration"] += 4.5;
    scores["Gingivitis"] += 3.0;
    scores["Calculus"] += 1.5

# 6. Dry Mouth (Saliva washes away acid; lack of saliva is a primary Caries
risk)
if ans["dry_mouth"] == "yes": scores["Caries"] += 2.5

# 7. Systemic Health (Diabetes/Genetics strictly influence
Periodontal/Gingivitis risk)
if ans["systemic_health"] == "yes": scores["Gingivitis"] += 2.5

# 8. Trauma (Dead nerves cause intrinsic tooth darkening)
if ans["trauma_history"] == "yes": scores["Tooth Discoloration"] += 4.0

# 9. Bleeding Gums (Hallmark symptom of Gingivitis)
if ans["bleeding_gums"] == "sometimes":
```

```python
    scores["Gingivitis"] += 2.5
elif ans["bleeding_gums"] == "often":
    scores["Gingivitis"] += 4.5

# 10. Pain/Sensitivity (Symptomatic indicator of advanced Caries)
if ans["pain_sensitivity"] == "mild":
    scores["Caries"] += 2.0
elif ans["pain_sensitivity"] == "severe":
    scores["Caries"] += 4.5
```

Appendix C: Equations

Image Analysis output (logits): results map one image to five independent disease scores before probability conversion.

$$\hat{y} = f(x) \in \mathbb{R}^5$$

Variables

- $\boldsymbol{x}$ input oral image after preprocessing (RGB tensor of size $224 \times 224 \times 3$)
- $\boldsymbol{f}(\cdot)$ convolutional neural network mapping learned during training
- $\hat{\boldsymbol{y}}$ raw model output (logits)
- $\hat{\boldsymbol{y}}_{\boldsymbol{i}}$ logit corresponding to condition i
- $\mathbb{R}^{\mathbf{5}}$ 5-dimensional real-valued output space
- $\boldsymbol{i} \in \{\mathbf{1}, \dots, \mathbf{5}\}$ disease index
    - **1:** calculus
    - **2:** cavities
    - **3:** gingivitis
    - **4:** periodontitis
    - **5:** missing teeth

Sigmoid probability conversion: Converts logits to probabilities in [0, 1]. Each disease is treated independently.

$$p_i = \sigma(\hat{y}_i) = \frac{1}{1 + e^{-\hat{y}_i}}$$

Variables

- $\boldsymbol{p}_{\boldsymbol{i}}$ predicted probability that condition $i$is present
- $\boldsymbol{\sigma}(\cdot)$ sigmoid activation function
- $\hat{\boldsymbol{y}}_{\boldsymbol{i}}$ logit for condition i
- $\boldsymbol{e}$ Euler's constant

Binary cross-entropy loss: Penalizes incorrect model predictions independently for each disease.

$$L = -\sum_{i=1}^{5} [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Variables

- $\boldsymbol{L}$ total training loss
- $\boldsymbol{y}_{\boldsymbol{i}} \in \{\mathbf{0}, \mathbf{1}\}$ ground-truth label for condition i
- $\boldsymbol{p}_{\boldsymbol{i}}$ predicted probability
- $\boldsymbol{i}$ condition index

Questionnaire Subsystem Softmax Normalization: Transforms raw scores into a probability distribution to make outputs comparable and interpretable

$$P_i = \frac{e^{s_i}}{\sum_{j=1}^{5} e^{s_j}}$$

Variables

- $\boldsymbol{s_i}$ cumulative symptom score for condition i
- $\boldsymbol{P_i}$ normalized questionnaire probability for condition i
- $\boldsymbol{e}$ Euler's constant
- $\boldsymbol{j}$ summation index over all conditions

How s_i is computed for the Decision-Tree: Each symptom contributes weighted evidence toward each disease, making a qualitative answer quantitative. Done in collaboration with stakeholder.

$$s_i = \sum_{k=1}^{m} w_{ik}\, q_k$$

Variables

- $\boldsymbol{q_k}$ binary or ordinal answer to symptom question k
- $\boldsymbol{w_{ik}}$ clinical weight linking symptom $k$to disease i
- $\boldsymbol{m}$ number of questionnaire items

Fusion / Triage Subsystem Weighted Probabilistic Fusion: implements linear fusion of results from the two subsystems.

$$R_i = \alpha p_i + (1 - \alpha) P_i$$

Variables

- $\boldsymbol{R_i}$ final fused risk score for condition i
- $\boldsymbol{p_i}$ CNN subsystem probability
- $\boldsymbol{P_i}$ questionnaire subsystem probability
- $\boldsymbol{\alpha} \in [\mathbf{0}, \mathbf{1}]$ weighting factor controlling contribution of image vs questionnaire
    - $\alpha = 1$ (image only)
    - $\alpha = 0$ (questionnaire only)
    - $0 < \alpha < 1$ (hybrid system using probabilities from both)

Appendix D: Survey Content

Selfie Dental Assistant App – UI/UX Prototype Survey

Thank you for participating in this usability study. This survey evaluates the user experience, usability, and perceived trust of a dental health mobile application prototype. Your responses are anonymous and will be used for academic research purposes only.

Section 1: Background Information

1. Have you visited a dentist in the past 12 months?

☐ Yes

☐ No

2. How often do you typically visit a dentist?

☐ Every 6 months

☐ Once a year

☐ Only when there is a problem

☐ Rarely / never

3. Have you ever used a health-related mobile app?

☐ Yes

☐ No

Section 2: First Impressions

4. What is your first impression of the app's interface?

☐ Very positive

☐ Somewhat positive

☐ Neutral

☐ Somewhat negative

☐ Very negative

5. Which words best describe the interface? (Select up to 3)

☐ Clean

☐ Professional

☐ Trustworthy

☐ Confusing

☐ Overwhelming

☐ Friendly

☐ Clinical

6. How visually appealing is the app?

(1 = Not appealing, 5 = Very appealing)

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

Section 3: Usability & Navigation

7. How easy was it to navigate through the app?

(1 = Very difficult, 5 = Very easy)

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

8. Were you able to find what you were looking for without assistance?

☐ Yes

☐ Somewhat

☐ No

9. Which part of the app felt most confusing, if any?

☐ Home screen

☐ Image upload

☐ Questionnaire

☐ Results page

☐ Navigation/menu

☐ Nothing felt confusing

10. If you got stuck at any point, where did it happen?

Section 4: Dental-Specific Features

11. How comfortable would you feel uploading a photo of your teeth to this app?

☐ Very comfortable

☐ Somewhat comfortable

☐ Neutral

☐ Somewhat uncomfortable

☐ Very uncomfortable

12. What concerns, if any, do you have about using this app? (Select all that apply)

☐ Data privacy

☐ Accuracy of results

☐ Medical reliability

☐ Ease of use

☐ Cost

Other ____________

13. How clear were the results or feedback provided by the app?

(1 = Very unclear, 5 = Very clear)

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

14. Did the app help you understand your dental health better?

☐ Yes

☐ Somewhat

☐ No

Section 5: Trust & Credibility

15. How much do you trust the information provided by this app?

(1 = Do not trust at all, 5 = Trust completely)

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

16. What would increase your trust in this app? (Select all that apply)

☐ Dentist endorsements

☐ Clear medical disclaimers

☐ Data privacy explanation

☐ Clinical accuracy explanation ☐ Professional visual design

☐ None of the above

Section 6: Overall Experience

17. How likely are you to use this app in real life?

(1 = Very unlikely, 5 = Very likely)

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

18. How likely are you to recommend this app to others?

(0 = Not at all likely, 10 = Extremely likely)

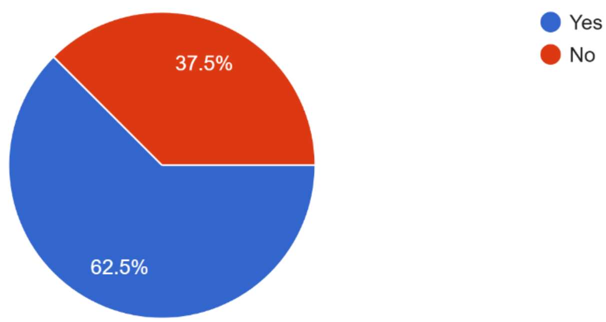
0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10 ☐

Appendix E: Survey Responses

Survey Responses for Paper prototype

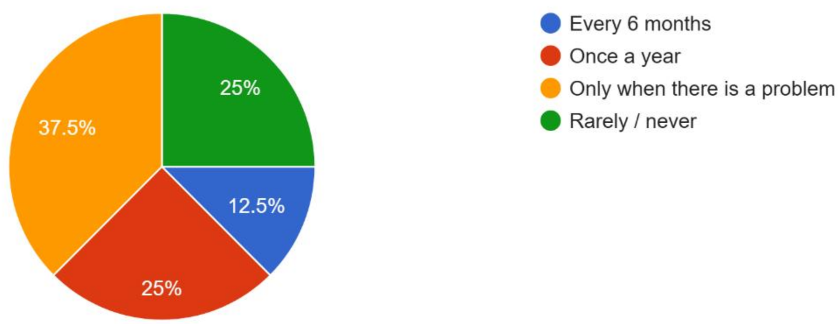
1. Have you visited a dentist in the past 12 months?

8 responses



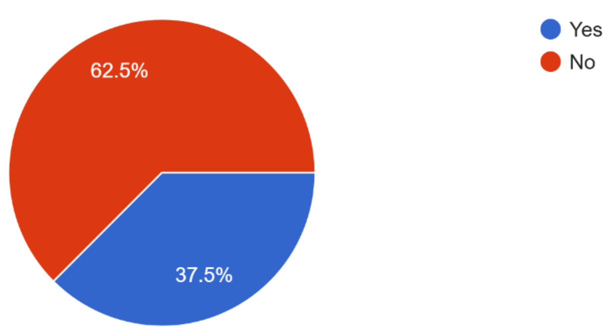
2. How often do you typically visit a dentist?

8 responses



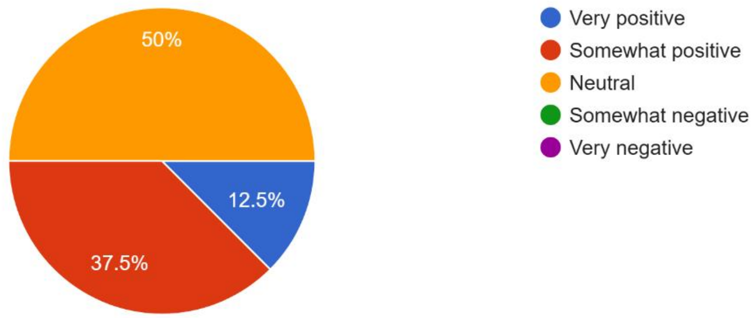
3. Have you ever used a health-related mobile app?

8 responses



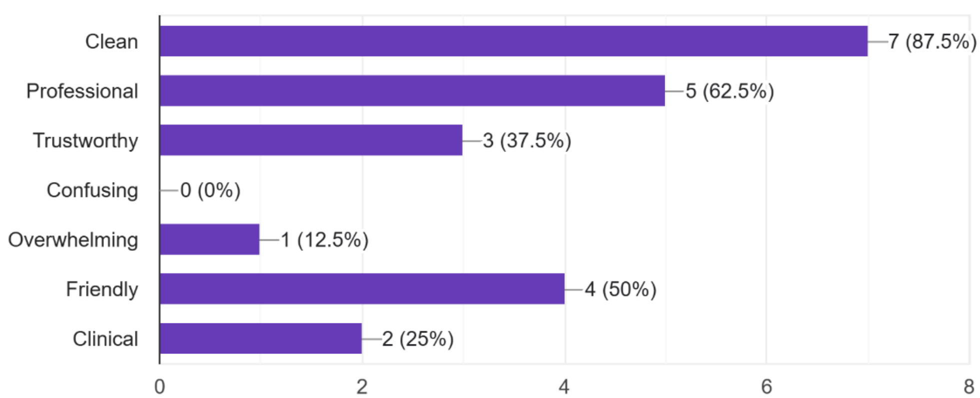
4. What is your first impression of the app's interface?

8 responses



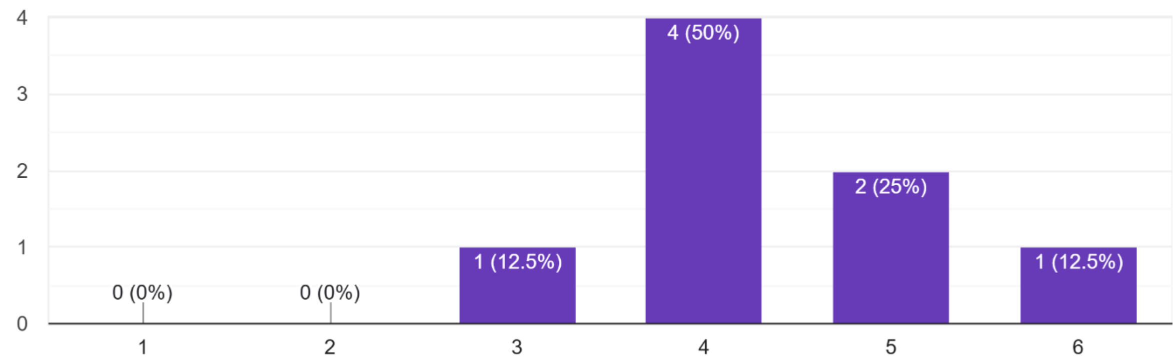
5. Which words best describe the interface? (Select up to 3)

8 responses



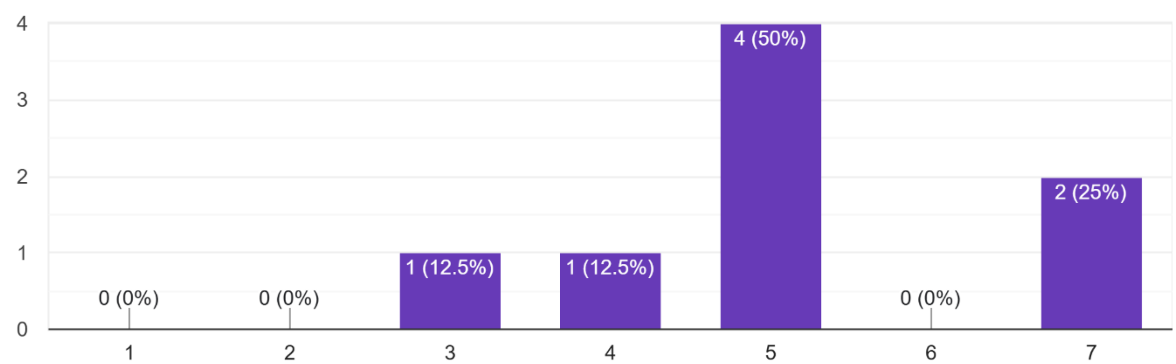
6. How visually appealing is the app? (1 = Not appealing, 5 = Very appealing)

8 responses



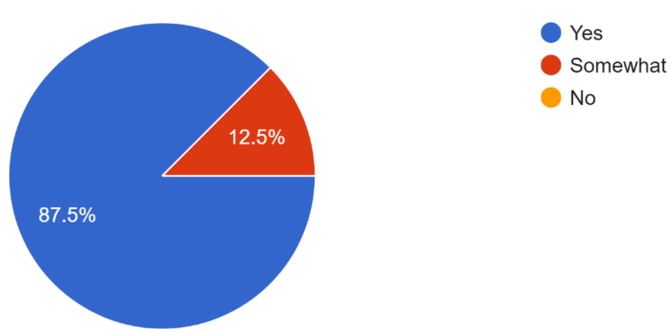
7. How easy was it to navigate through the app? (1 = Very difficult, 5 = Very easy)

8 responses



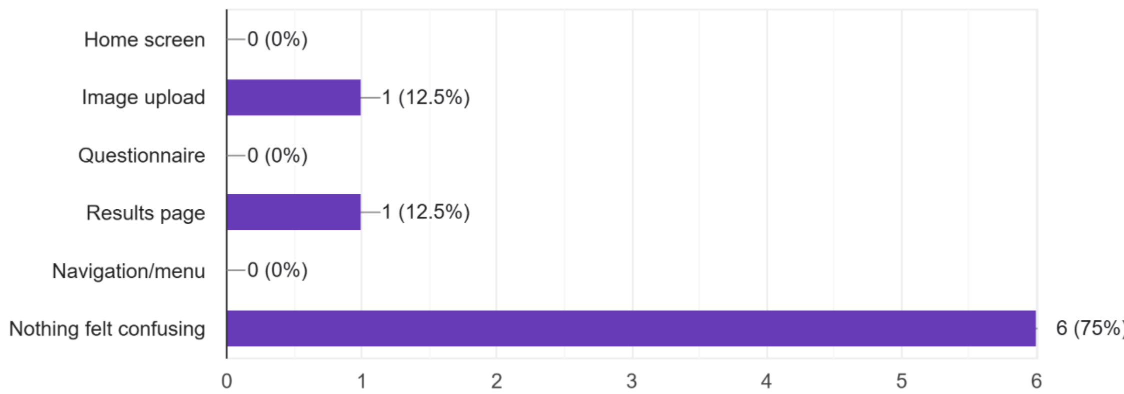
8. Were you able to find what you were looking for without assistance?

8 responses



9. Which part of the app felt most confusing, if any?

8 responses



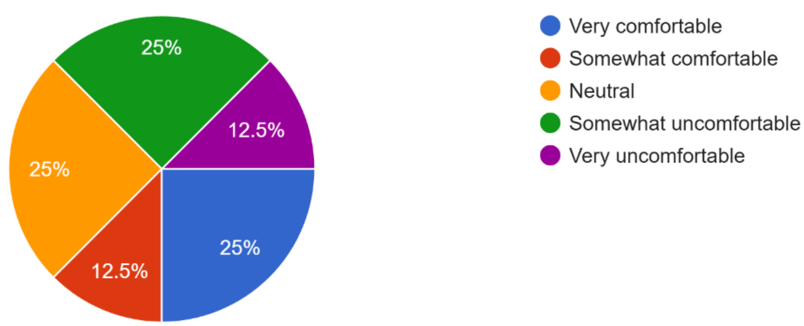
10. If you got stuck at any point, where did it happen?

1 response

I did not

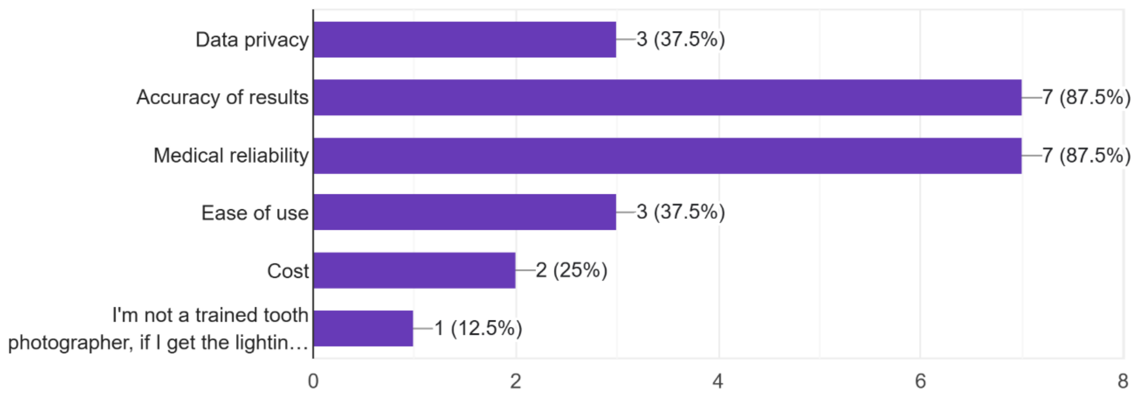
11. How comfortable would you feel uploading a photo of your teeth to this app?

8 responses



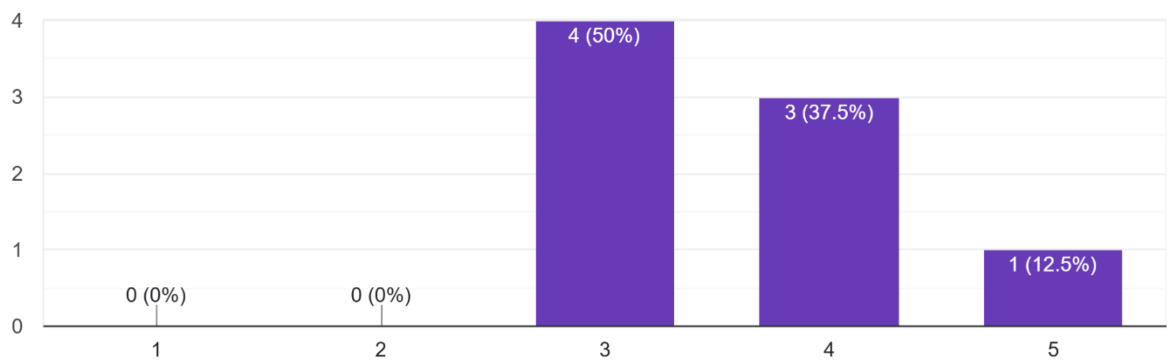
12. What concerns, if any, do you have about using this app? (Select all that apply)

8 responses



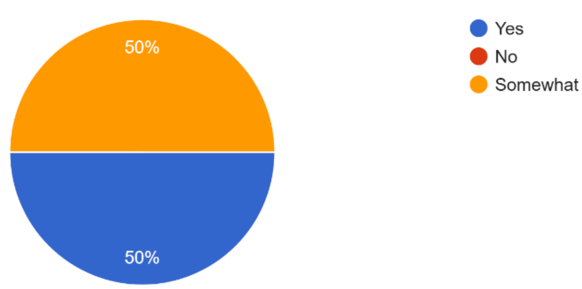
13. How clear were the results or feedback provided by the app? (1 = Very unclear, 5 = Very clear)

8 responses



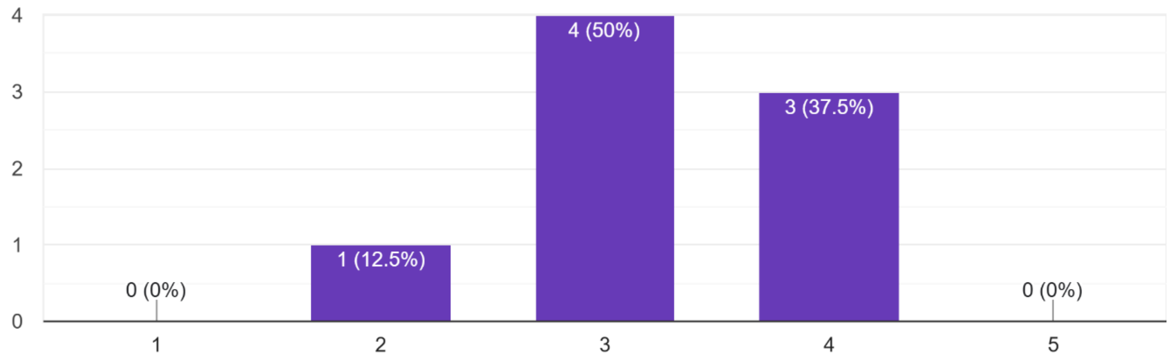
14. Did the app help you understand your dental health better?

8 responses



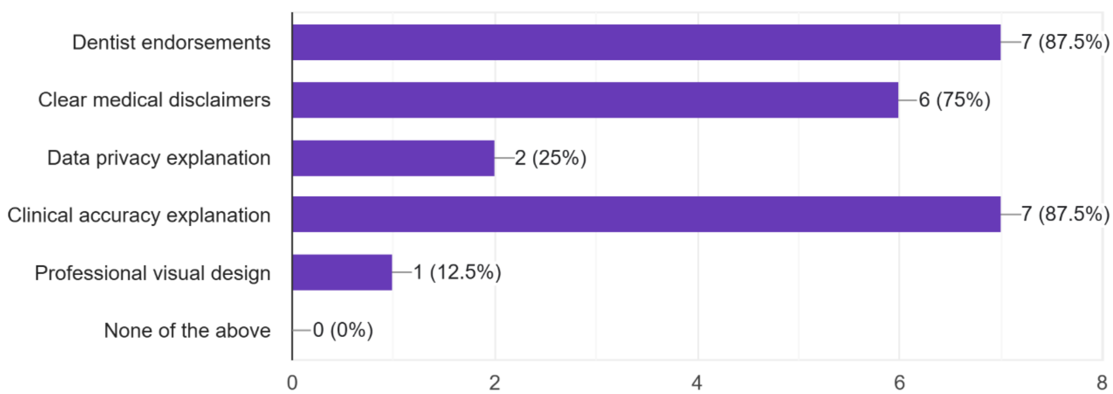
15. How much do you trust the information provided by this app? (1 = Do not trust at all, 5 = Trust completely)

8 responses



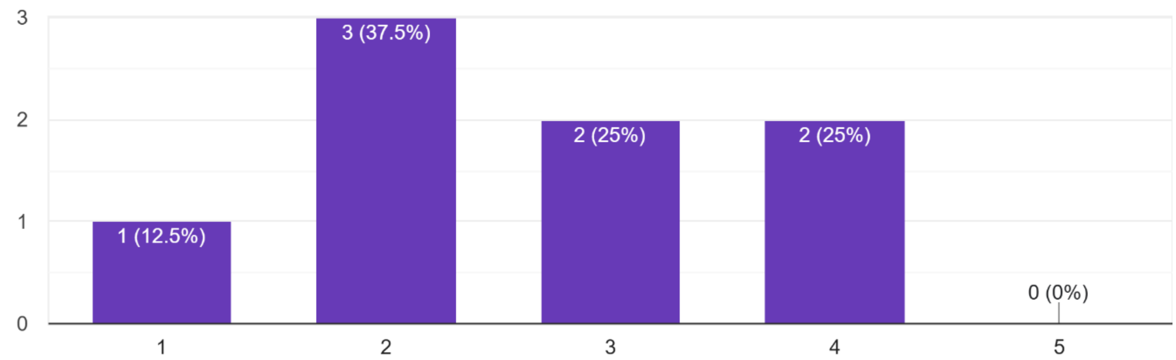
16. What would increase your trust in this app? (Select all that apply)

8 responses



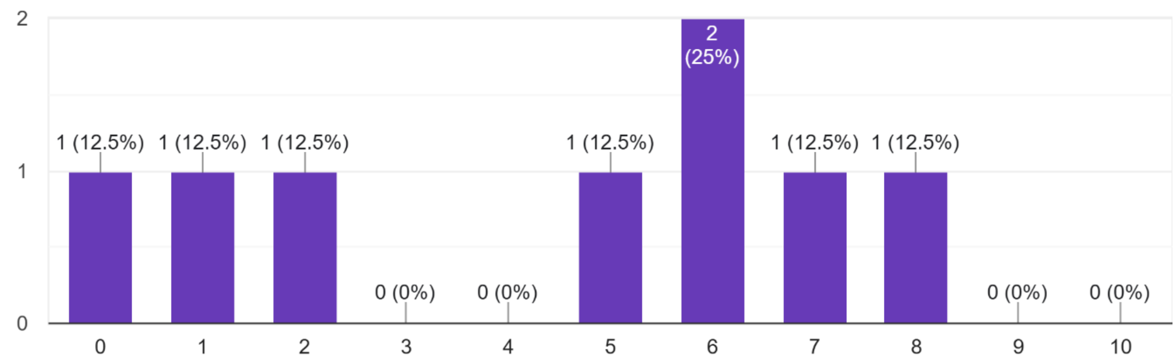
17. How likely are you to use this app in real life? (1 = Very unlikely, 5 = Very likely)

8 responses



18. How likely are you to recommend this app to others? (0 = Not at all likely, 10 = Extremely likely)

8 responses



Any other suggestions

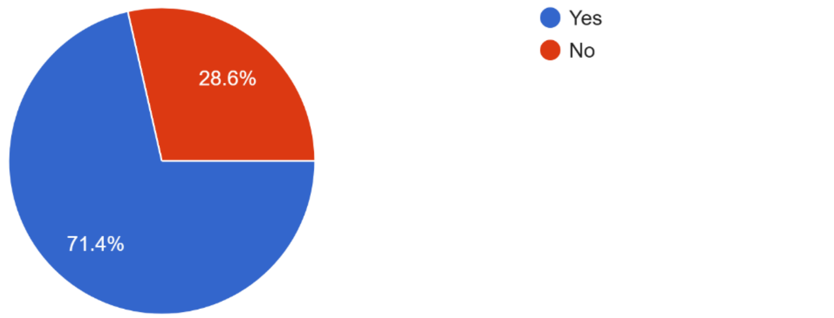
2 responses

I will more willing to read the result if it combine charts and data together.

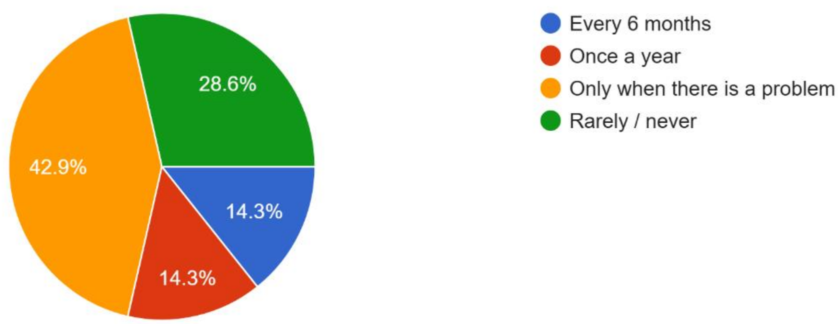
The image upload was an interesting panel, I'd rather than have to allow my entire gallery into the app just select a few images and allow them in like Instagram does

Survey Responses for Figma prototype

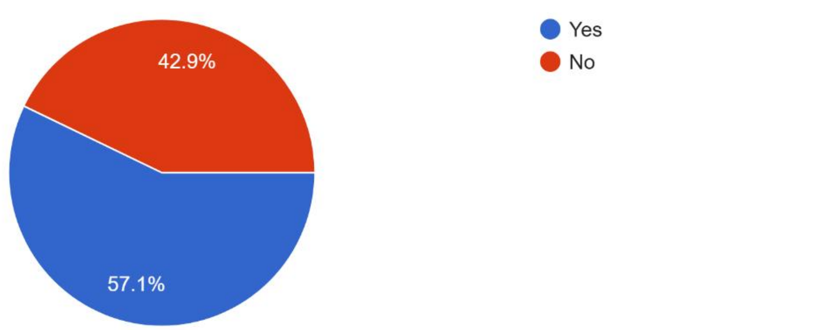
1. Have you visited a dentist in the past 12 months?
7 responses



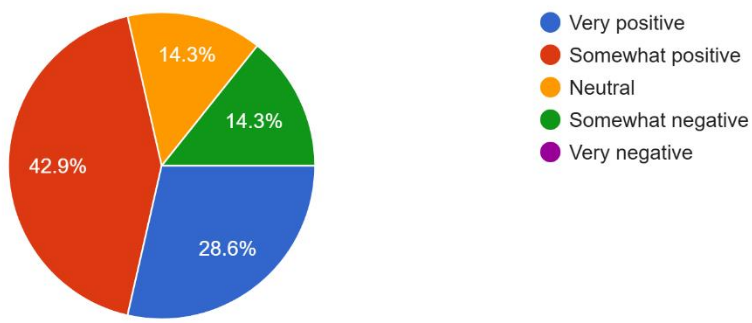
2. How often do you typically visit a dentist?
7 responses



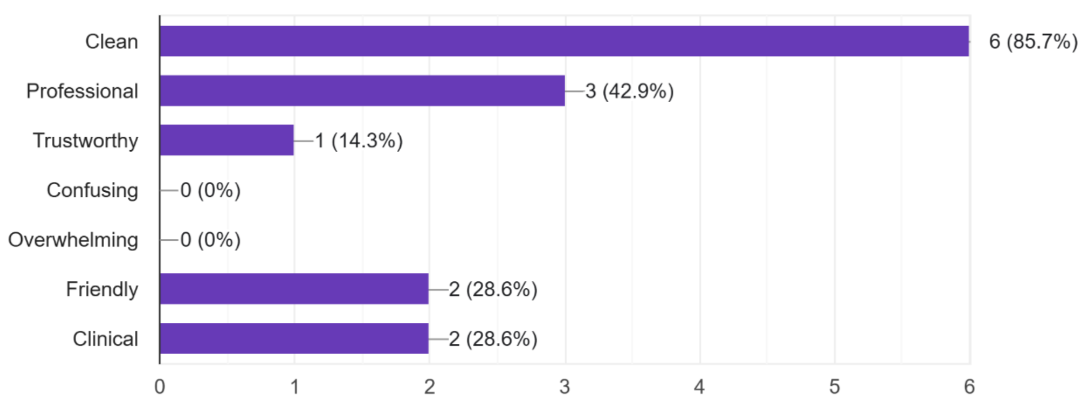
3. Have you ever used a health-related mobile app?
7 responses



4. What is your first impression of the app's interface?
7 responses

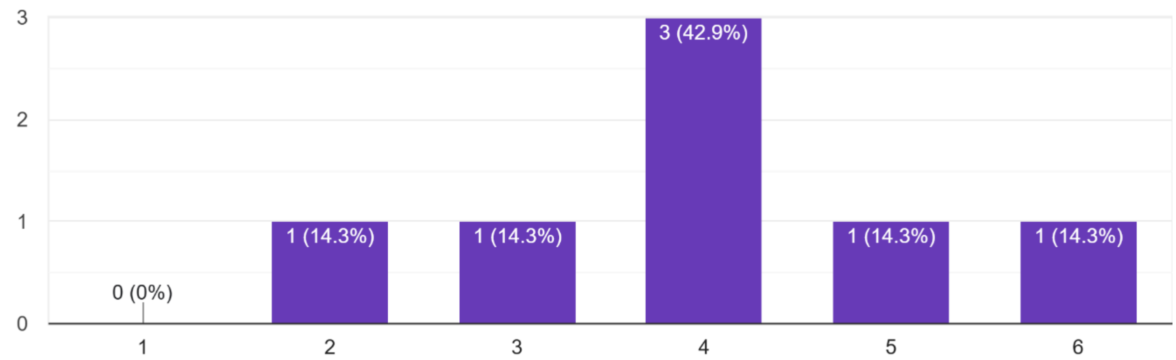


5. Which words best describe the interface? (Select up to 3)
7 responses



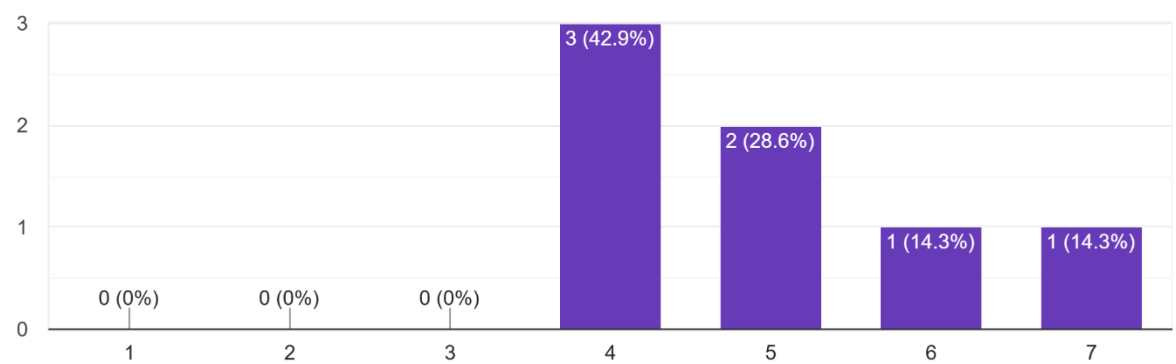
6. How visually appealing is the app? (1 = Not appealing, 5 = Very appealing)

7 responses



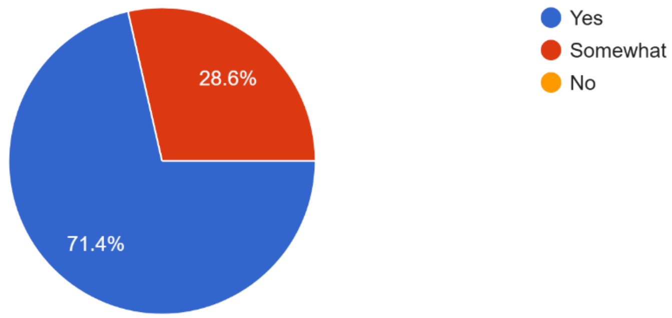
7. How easy was it to navigate through the app? (1 = Very difficult, 5 = Very easy)

7 responses



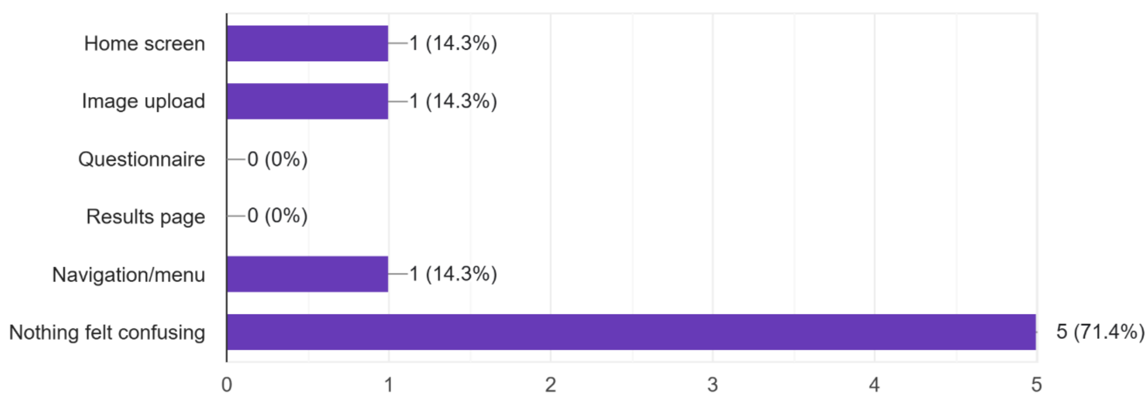
8. Were you able to find what you were looking for without assistance?

7 responses



9. Which part of the app felt most confusing, if any?

7 responses



10. If you got stuck at any point, where did it happen?

2 responses

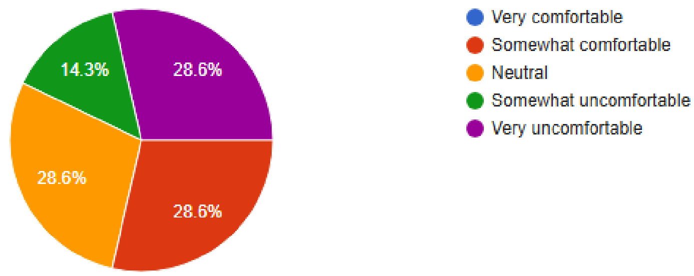
Just kidding, it just wasn't an option. The Resources tab freaked me out, it looks like one of those phishing scams where something looks just odd enough to catch me off guard. Maybe it's the amount of white space surrounding the text, or the scroll wheel just sitting there. It's like I opened a Kindle but I'm using a teeth app.

I did not

11. How comfortable would you feel uploading a photo of your teeth to this app?

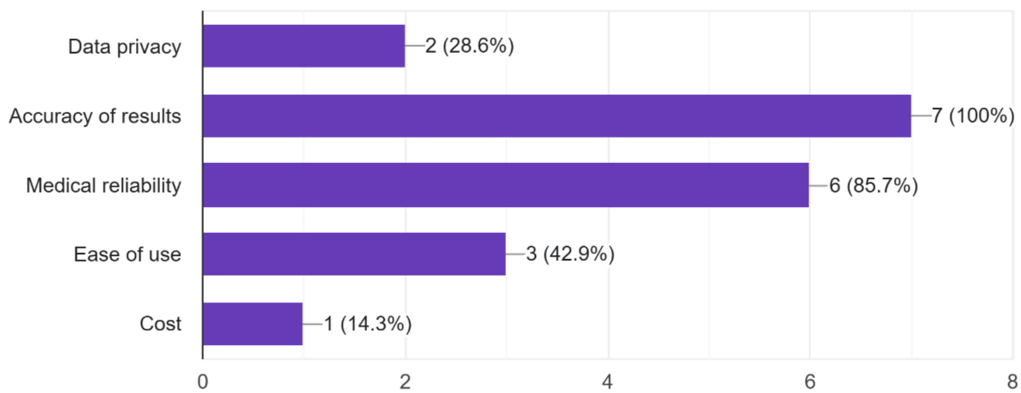
Copy chart

7 responses



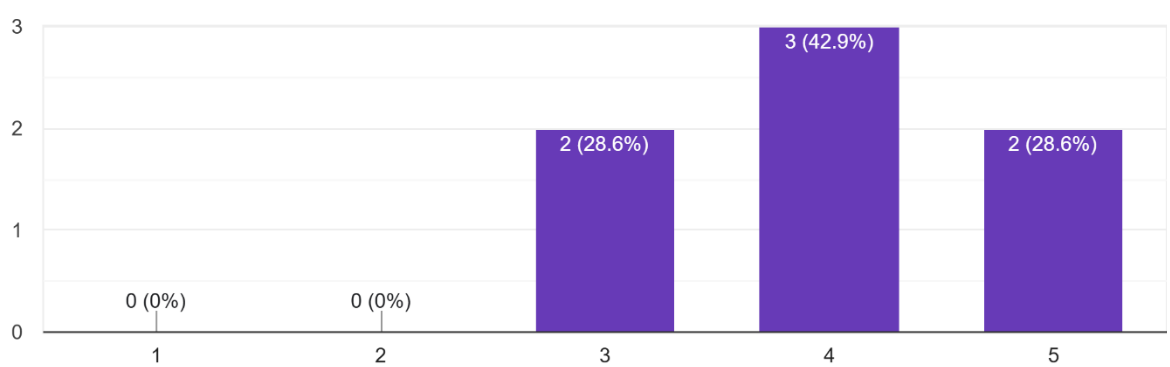
12. What concerns, if any, do you have about using this app? (Select all that apply)

7 responses

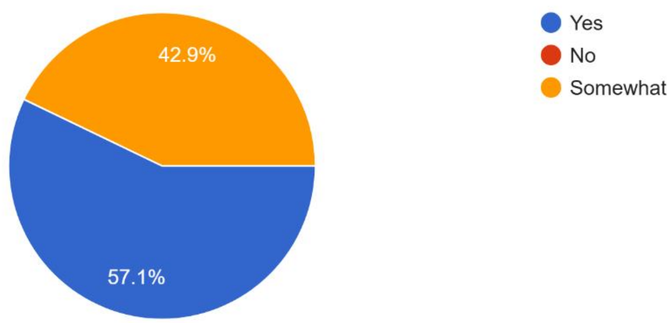


13. How clear were the results or feedback provided by the app? (1 = Very unclear, 5 = Very clear)

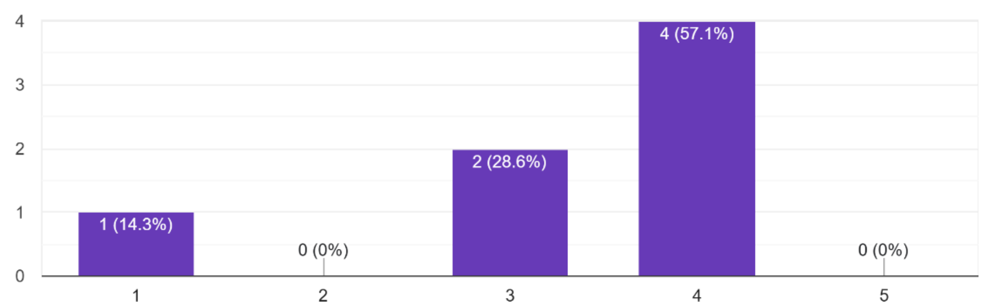
7 responses



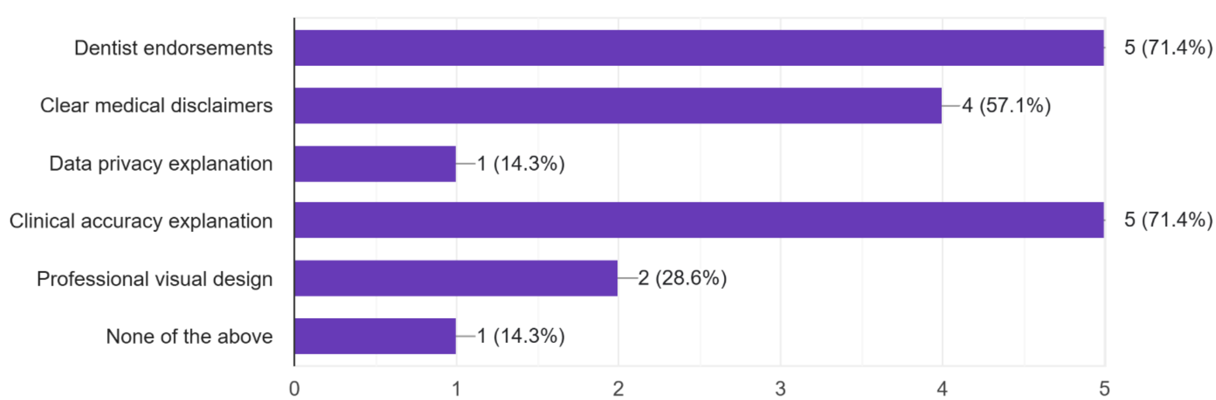
14. Did the app help you understand your dental health better?
7 responses



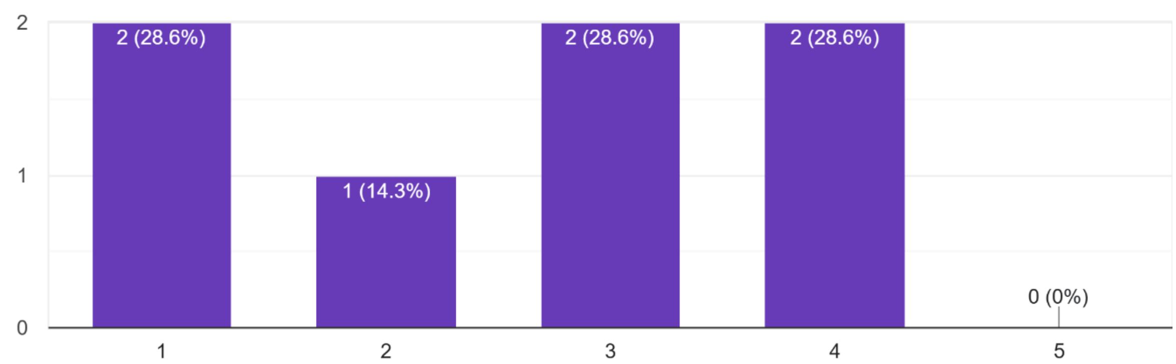
15. How much do you trust the information provided by this app? (1 = Do not trust at all, 5 = Trust completely)
7 responses



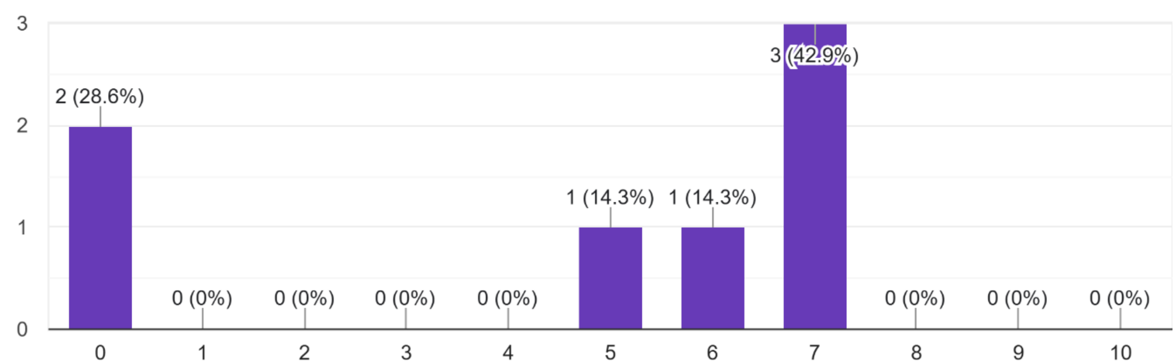
16. What would increase your trust in this app? (Select all that apply)
7 responses



17. How likely are you to use this app in real life? (1 = Very unlikely, 5 = Very likely)
7 responses



18. How likely are you to recommend this app to others? (0 = Not at all likely, 10 = Extremely likely)
7 responses



Any suggestions

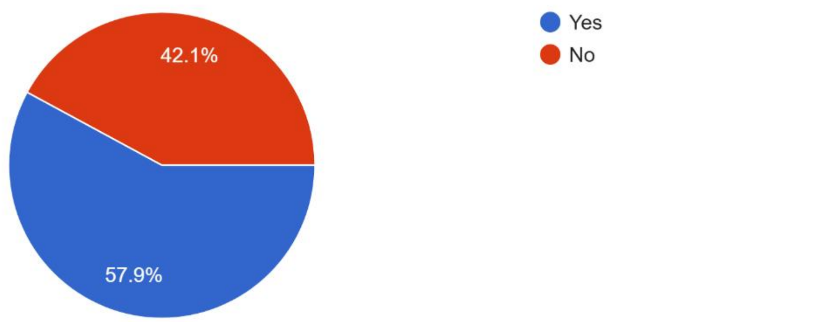
2 responses

AI is a pretty new technology and has a pretty bad reputation for many people, so I think this might struggle being perceived as trustworthy regardless of how accurate it is. I think a way to improve trust in people wary of the concept is an explanation of how the AI functions. A walkthrough simplified explanation in the tutorial section described in the paper prototype perhaps. The app seems good on a user interface level.

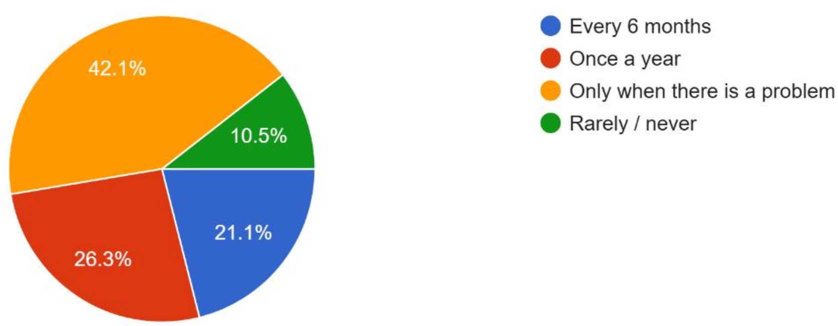
A visible outline or perhaps keeping the Home screen options nearby would make it feel nicer on the Resources page. Maybe an arrow on the top right with 'Back' and maybe on the History side on the top left showing 'Back' indicating the 3 pages nicely enough

Survey Responses for Integration Prototype

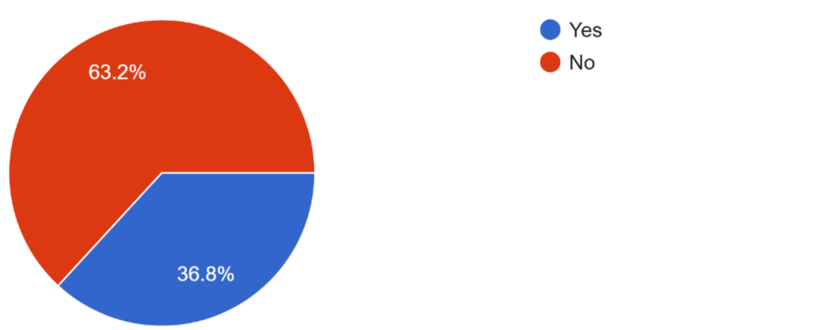
1. Have you visited a dentist in the past 12 months?
19 responses



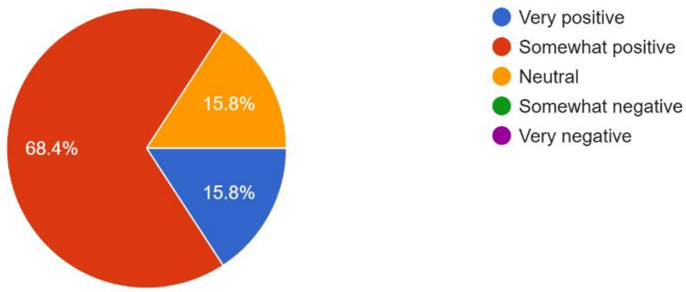
2. How often do you typically visit a dentist?
19 responses



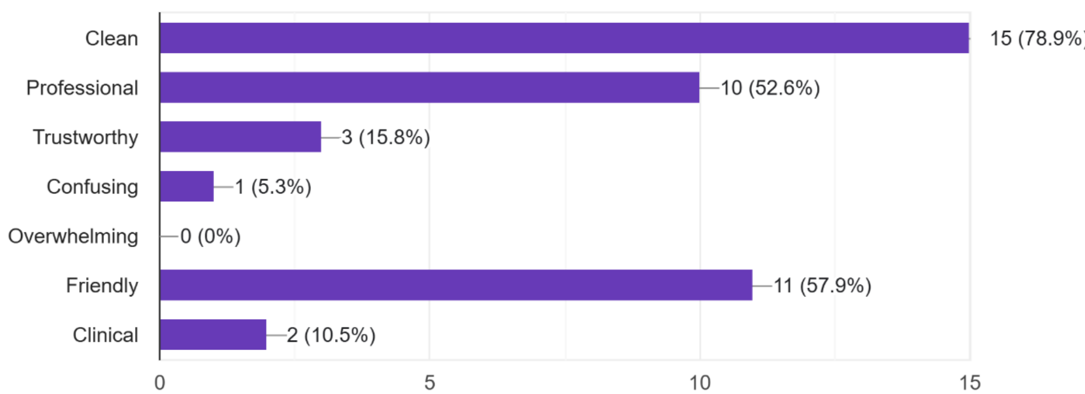
3. Have you ever used a health-related mobile app?
19 responses



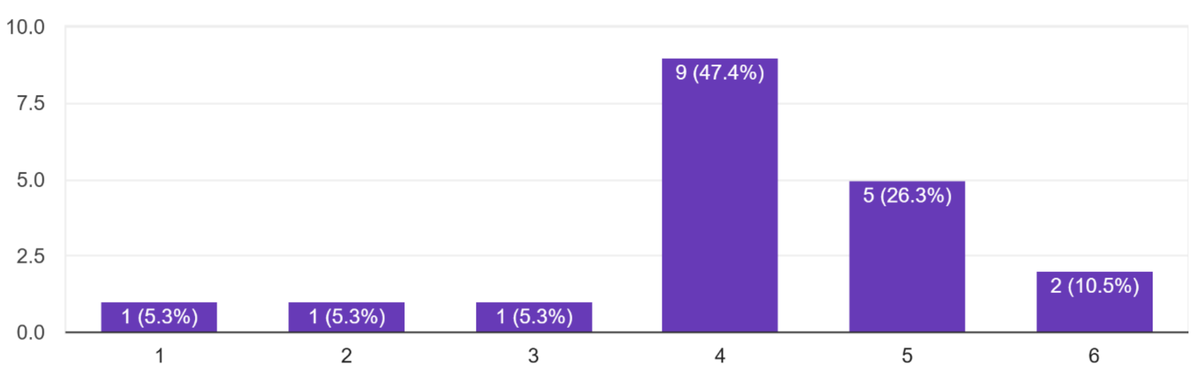
4. What is your first impression of the app's interface?
19 responses



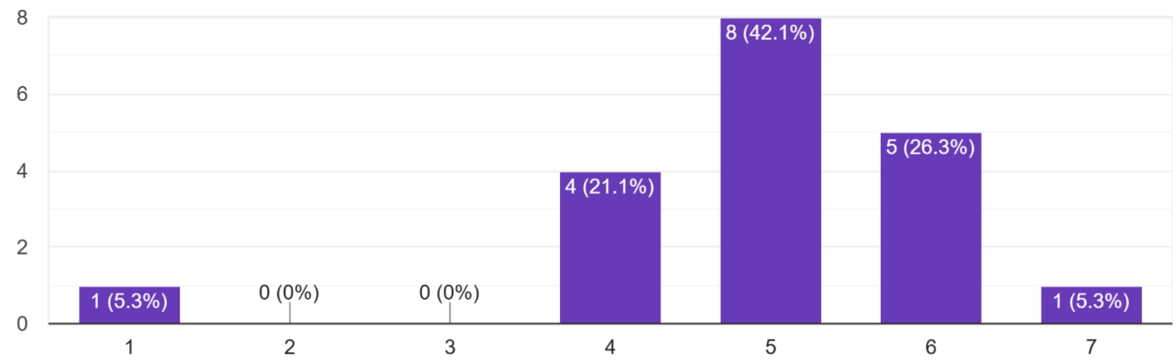
5. Which words best describe the interface? (Select up to 3)
19 responses



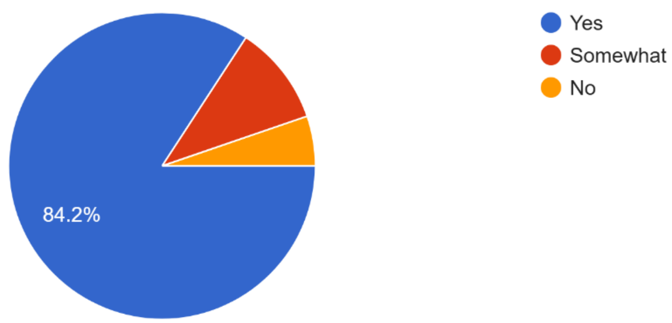
6. How visually appealing is the app? (1 = Not appealing, 6 = Very appealing)
19 responses



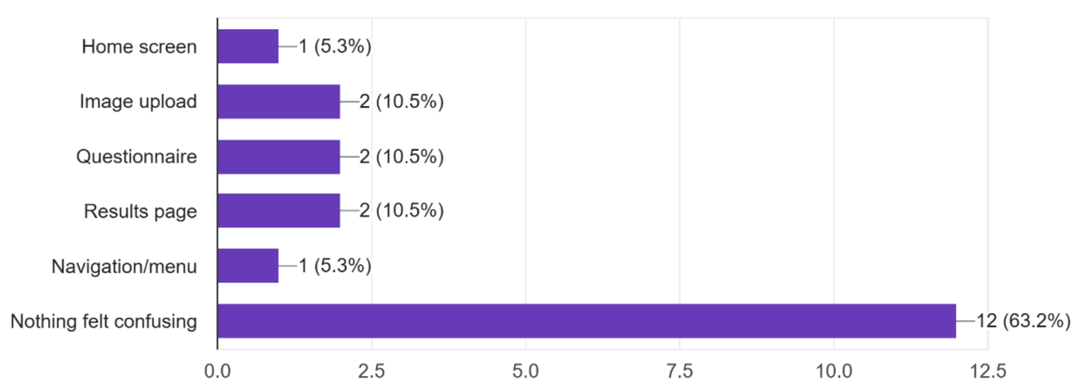
7. How easy was it to navigate through the app? (1 = Very difficult, 7 = Very easy)
19 responses



8. Were you able to find what you were looking for without assistance?
19 responses



9. Which part of the app felt most confusing, if any?
19 responses



10. If you got stuck at any point, where did it happen?

6 responses

N/A

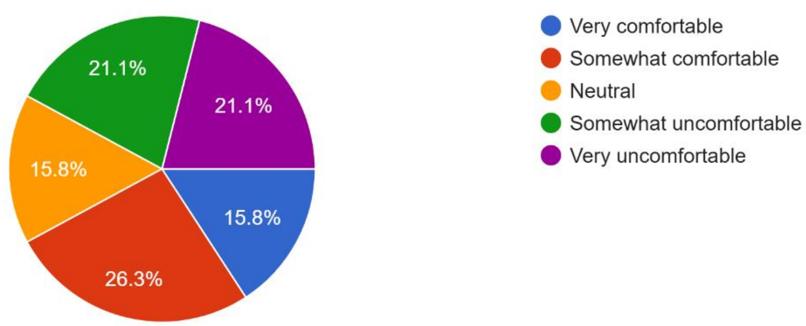
Not stuck but I'm an Android user, I think an easier Delete button for past scans/images uploaded would be nice. Never used an iPhone so seeing the slide to delete was a first.

Why was a questionnaire required every time a scan was being done? it seems a bit off putting and like more work than necessary

User Interface

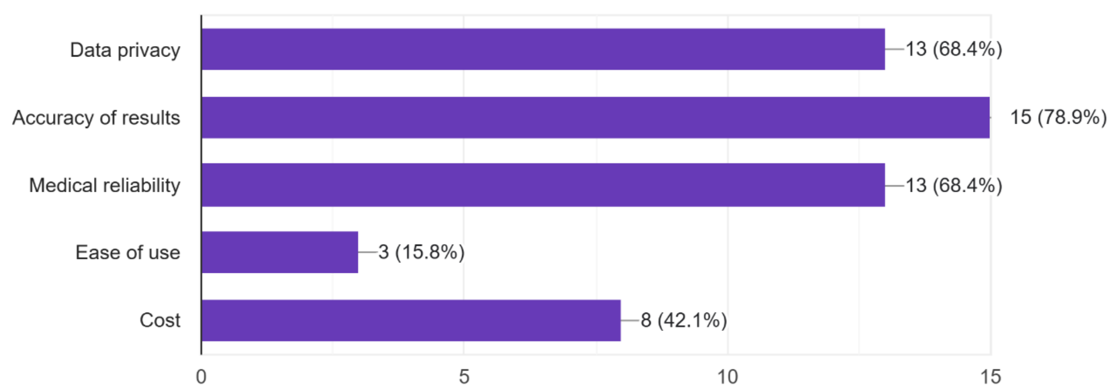
11. How comfortable would you feel uploading a photo of your teeth to this app?

19 responses

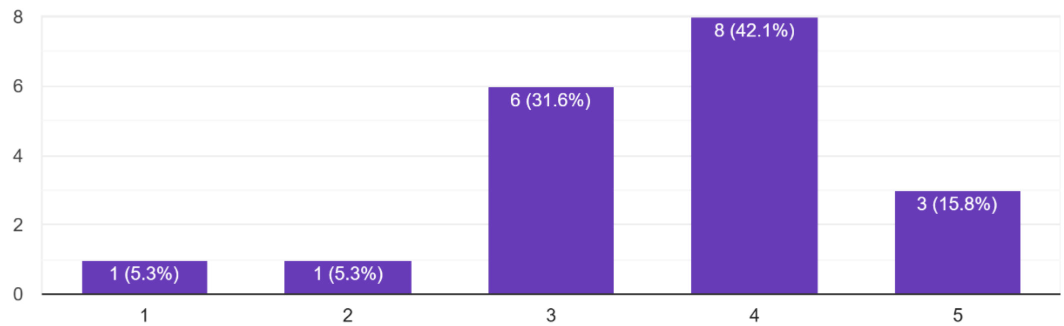


12. What concerns, if any, do you have about using this app? (Select all that apply)

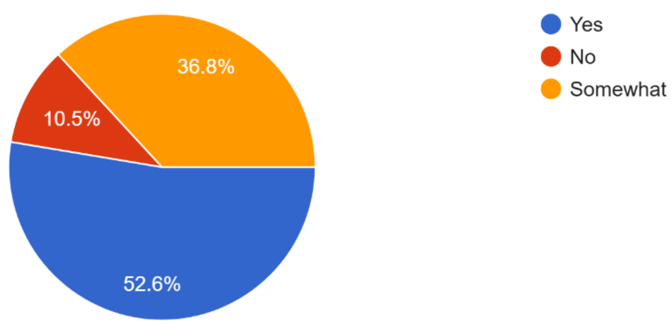
19 responses



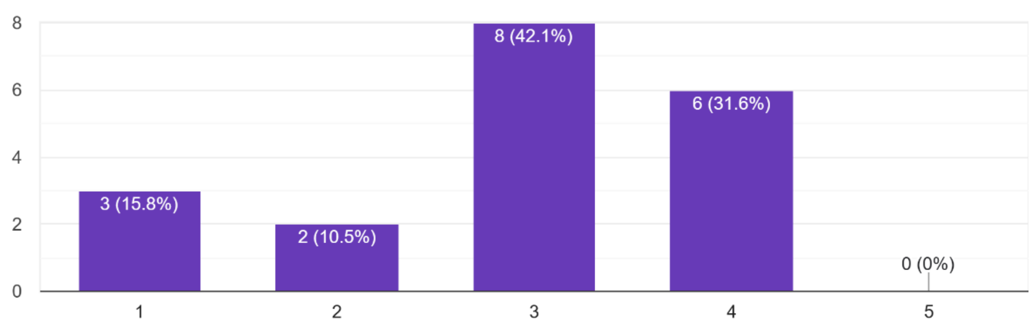
13. How clear were the results or feedback provided by the app? (1 = Very unclear, 5 = Very clear)
19 responses



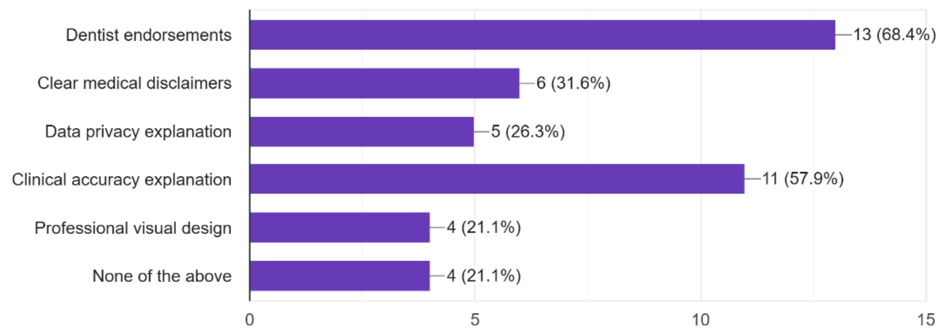
14. Did the app help you understand your dental health better?
19 responses



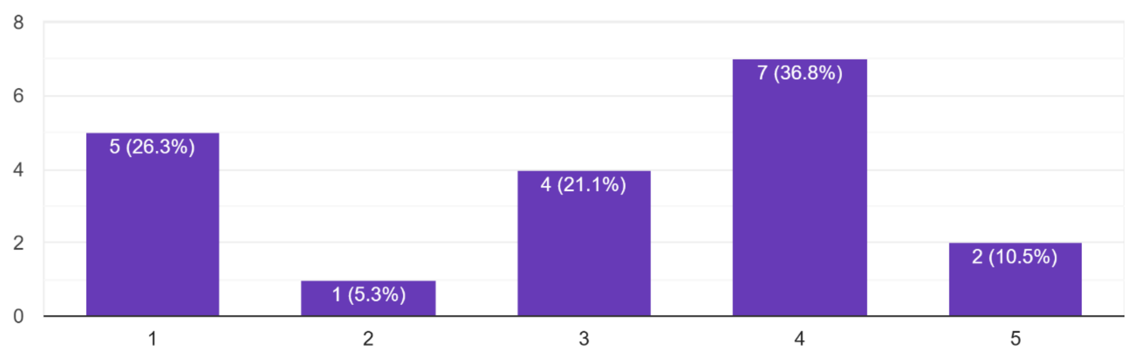
15. How much do you trust the information provided by this app? (1 = Do not trust at all, 5 = Trust completely)
19 responses



16. What would increase your trust in this app? (Select all that apply)
19 responses



17. How likely are you to use this app in real life? (1 = Very unlikely, 5 = Very likely)
19 responses



18. How likely are you to recommend this app to others? (0 = Not at all likely, 10 = Extremely likely)
19 responses